

Lab Manual

CYC-205

Introduction to Cyber Security



CUI

**Department of Computer Science
Islamabad Campus**

Page **1**

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Lab 01

Cyber Security Fundamentals

Objective:

This module aims to provide a foundational understanding of Cyber Security Fundamentals, emphasizing the criticality of browser, email, and social media security. Participants will learn practical techniques for securing browsers, implementing safe searching practices, and hardening Windows systems. By the module's end, participants will possess essential skills to navigate the digital landscape safely and protect against online threats.

Activity Outcomes:

Students will be able to identify common threats to browser, email, and social media security. Students will demonstrate proficiency in implementing secure browsing practices and safe searching techniques.

Students will understand the significance of Windows hardening in enhancing overall system security.

Instructor Note:

Ensure students engage actively with the lab task on securing browser cookies and passwords to reinforce theoretical concepts with practical application. Encourage discussion on real-world scenarios to deepen understanding.

1. Useful Concepts

Cybersecurity fundamentals encompass a range of practices and technologies designed to protect computer systems, networks, and data from unauthorized access, damage, or theft. At the core, it involves implementing robust security measures such as firewalls, antivirus software, and intrusion detection systems to safeguard against external threats. Critical components include browser security, which entails configuring settings to block malicious sites and managing cookies and passwords securely to prevent unauthorized access. Email security is equally essential, focusing on encryption and recognizing phishing attempts to protect sensitive information during communication. Additionally, Windows hardening involves applying security patches, configuring firewalls, and using built-in tools like BitLocker and Windows Defender to enhance the operating system's resilience against attacks. These measures collectively form the foundation of a strong cybersecurity posture, ensuring that digital assets are protected against a constantly evolving landscape of cyber threats.

Browser Security

Browser security refers to the measures and practices implemented to protect web browsers and their users from cyber threats like malware, phishing, and data breaches. This includes features like secure browsing modes, regular software updates, and the use of extensions or settings that enhance privacy and block malicious activities. Proper browser security is essential for safeguarding personal information and ensuring a safe online experience.

Email Security

Email security involves protecting email accounts and communications from unauthorized access, phishing attacks, and malware. Key practices include using strong passwords, enabling two-factor authentication, and deploying encryption to safeguard the content of emails. These measures are crucial for maintaining the confidentiality and integrity of sensitive information shared via email.

Windows Hardening

Windows hardening refers to the process of securing a Windows operating system by implementing various security measures to reduce the potential attack surface and enhance the overall security posture of the system. This is especially important for systems connected to networks or the internet, as they are susceptible to various forms of cyber-attacks.

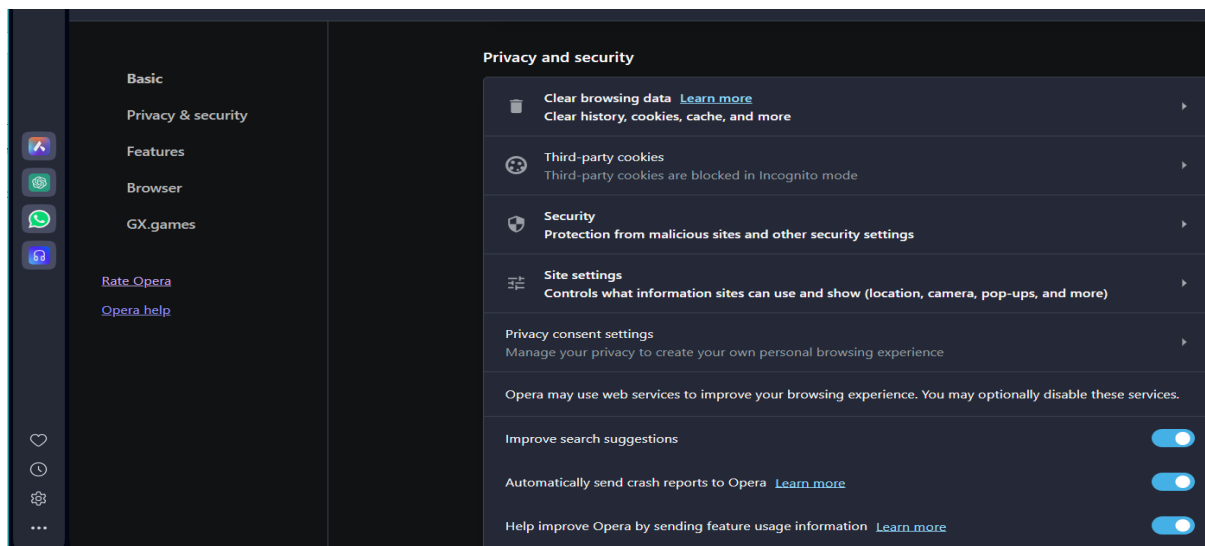
2. Solved Lab Activities (Allocated Time 3 Hr.)

<i>Sr.No</i>	<i>Allocated Time</i>	<i>Level of Complexity</i>	<i>CLO Mapping</i>
Activity 1	45 mins	Medium	CLO-3
Activity 2	45 mins	Medium	CLO-3
Activity 3	45 mins	Medium	CLO-3
Activity 4	45 mins	Medium	CLO-3
Activity 5	45 mins	Medium	CLO-3

Activity1:

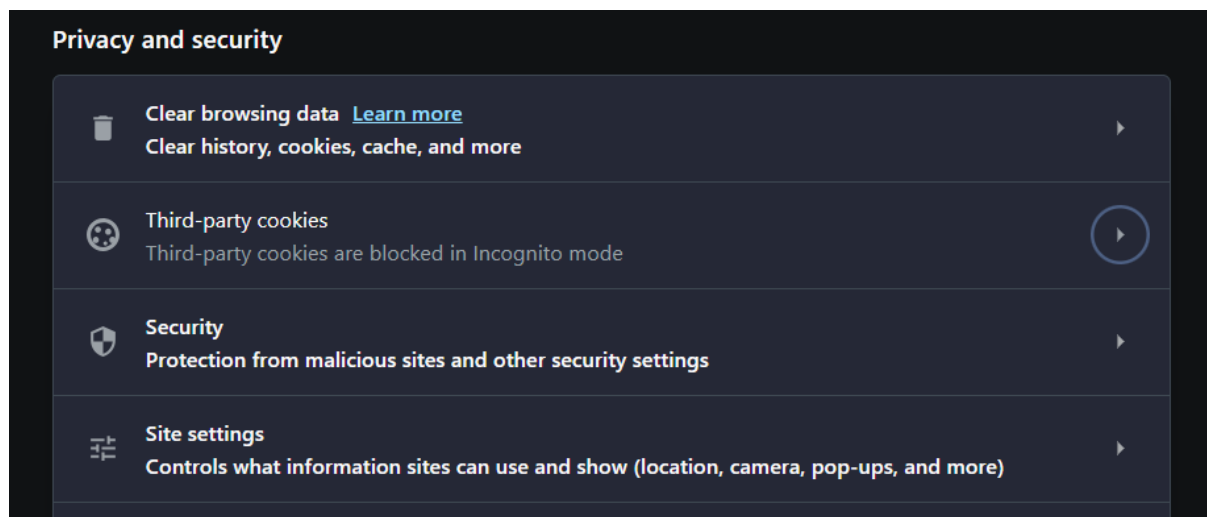
Browser Security:

Step#1: Blocking Third Party Cookies

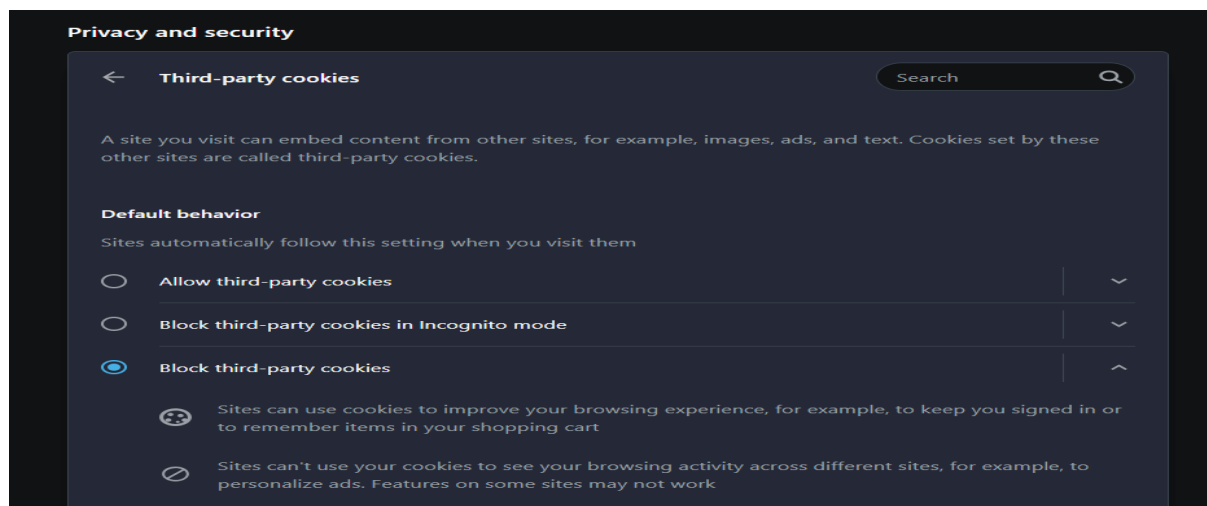


Step a: Click on settings in the bottom left corner and then click on privacy and security:

Step b: Click on third party cookies:

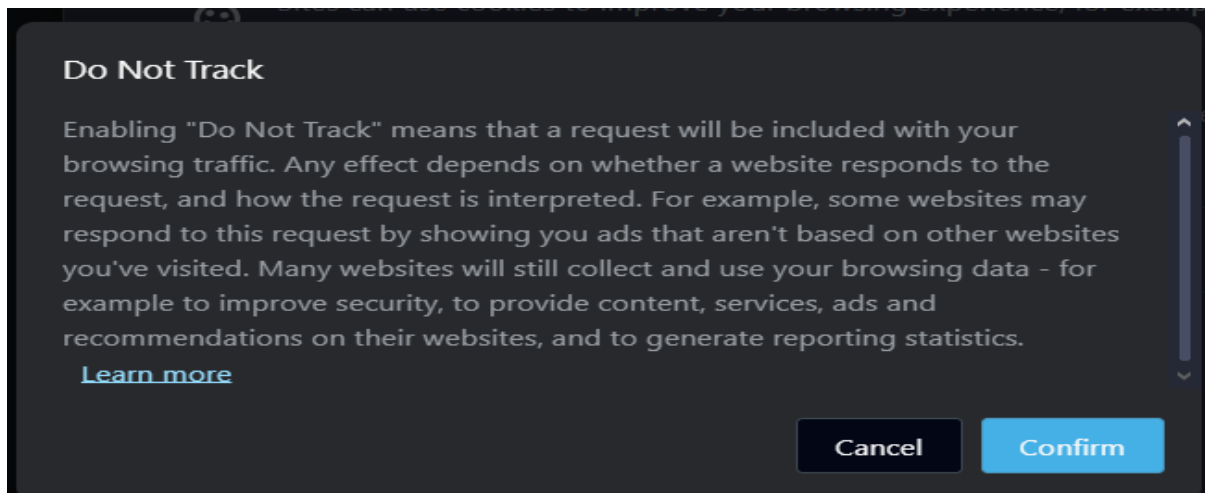


Step c: Click on “block third party cookies”:



Step#2: Enabling do not track

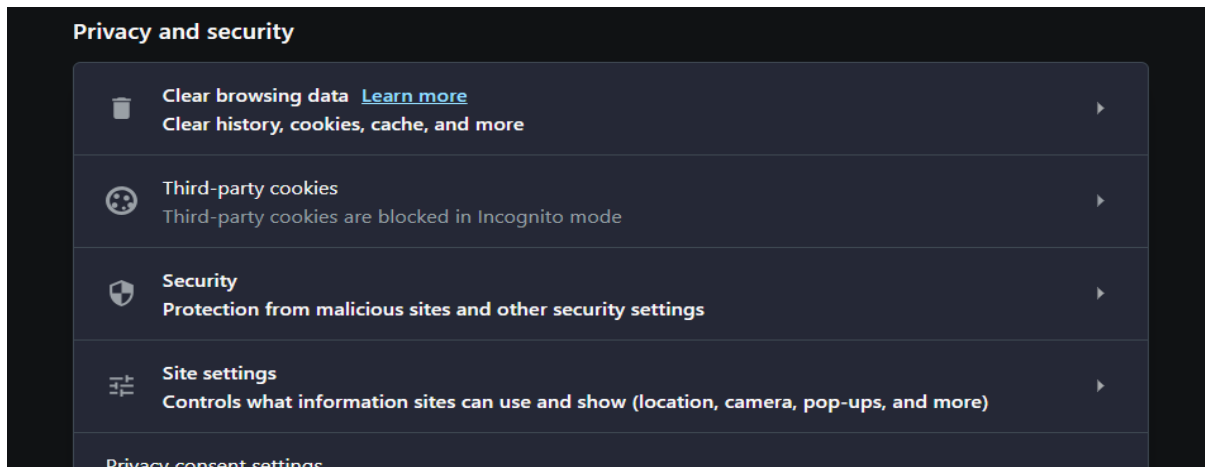
In the same settings, scroll down and enable do not track option:



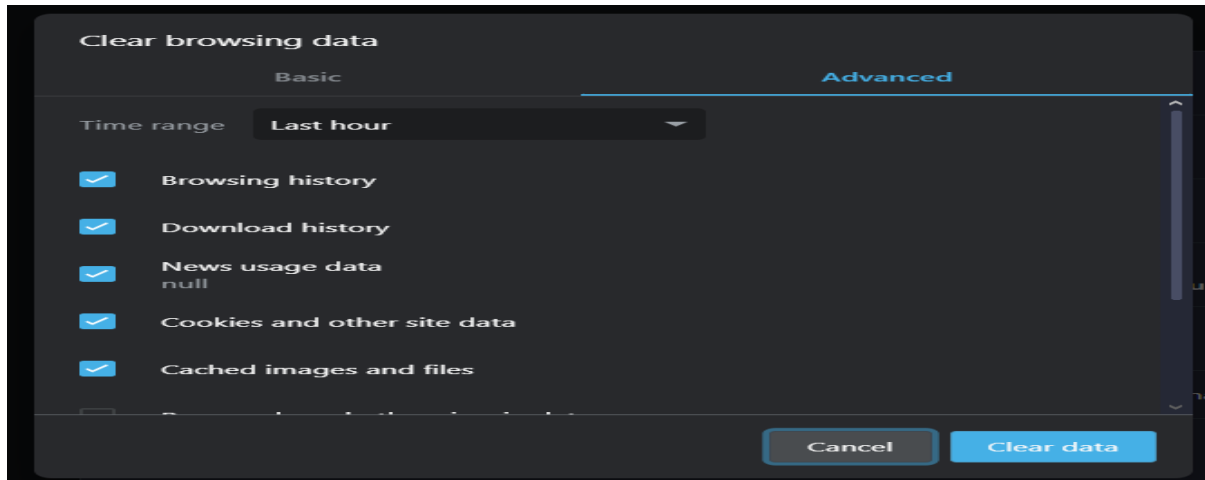
Step#3: Site data and cached files

Step a: Clearing browsing data

Go to privacy and security settings and click on “clear browsing data”:

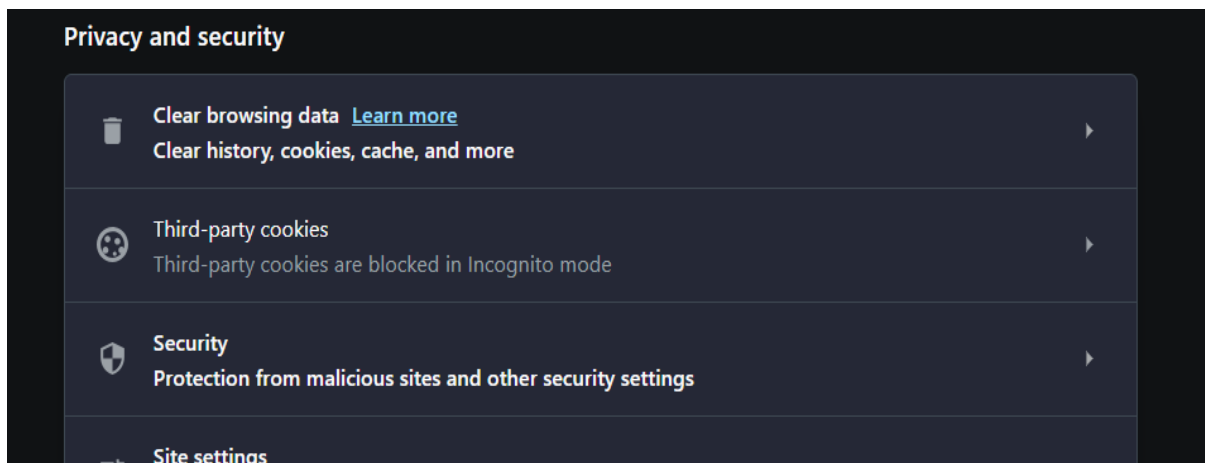


Step b: Click on advance and then select clear data:

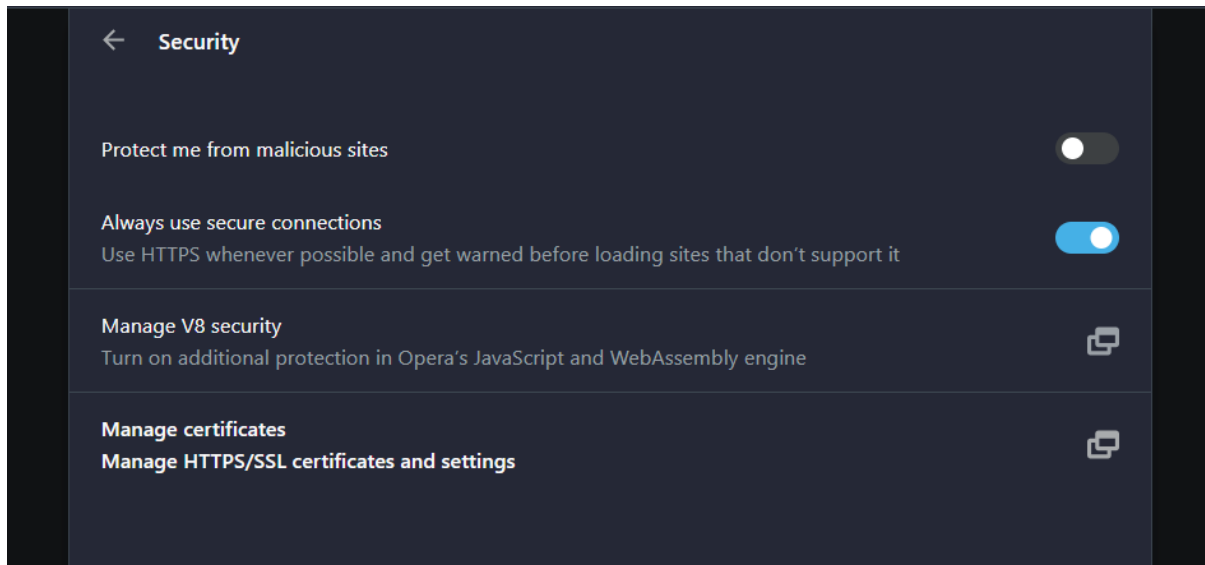


Step#4: Security certificates and https

Step a: Click on security settings in privacy and settings:



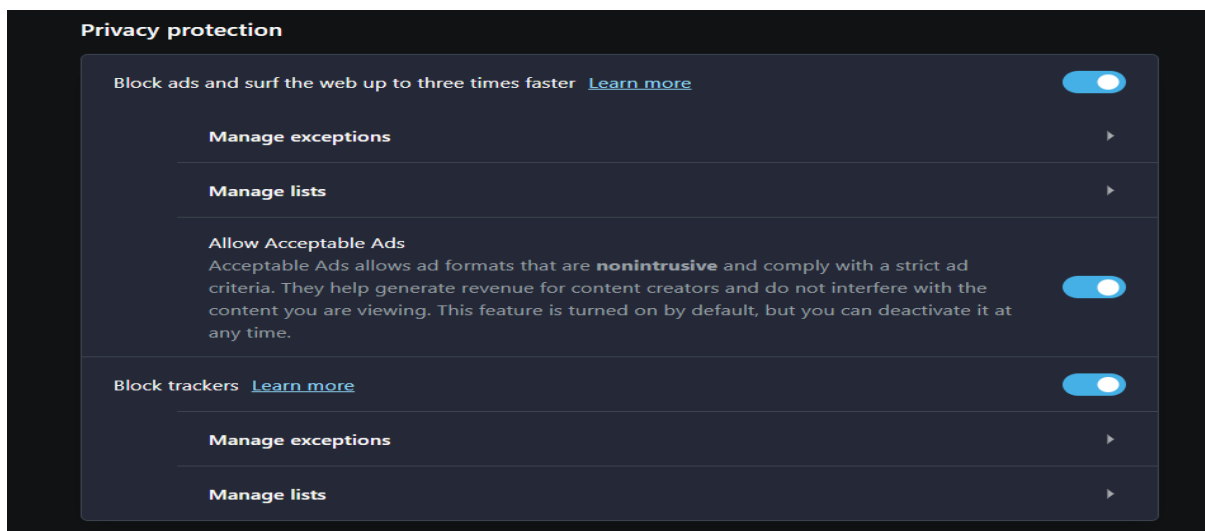
Step b: Enable always secure option:



Manage certificates if you want (remove or certificates if you want).

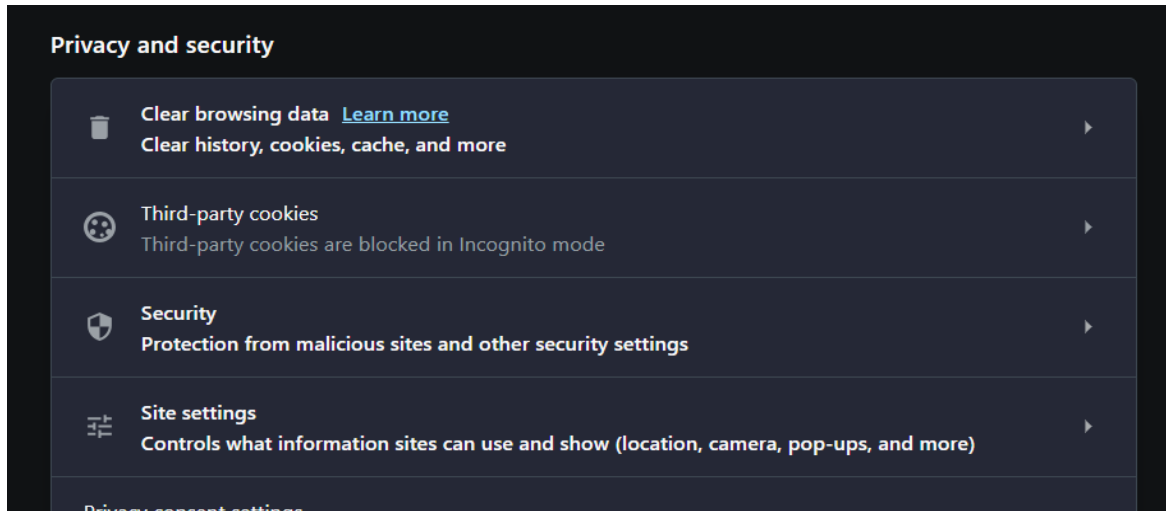
Step#5: Pop-up and ad blockers

Step a: Click on privacy and security and there click on block-ads option:



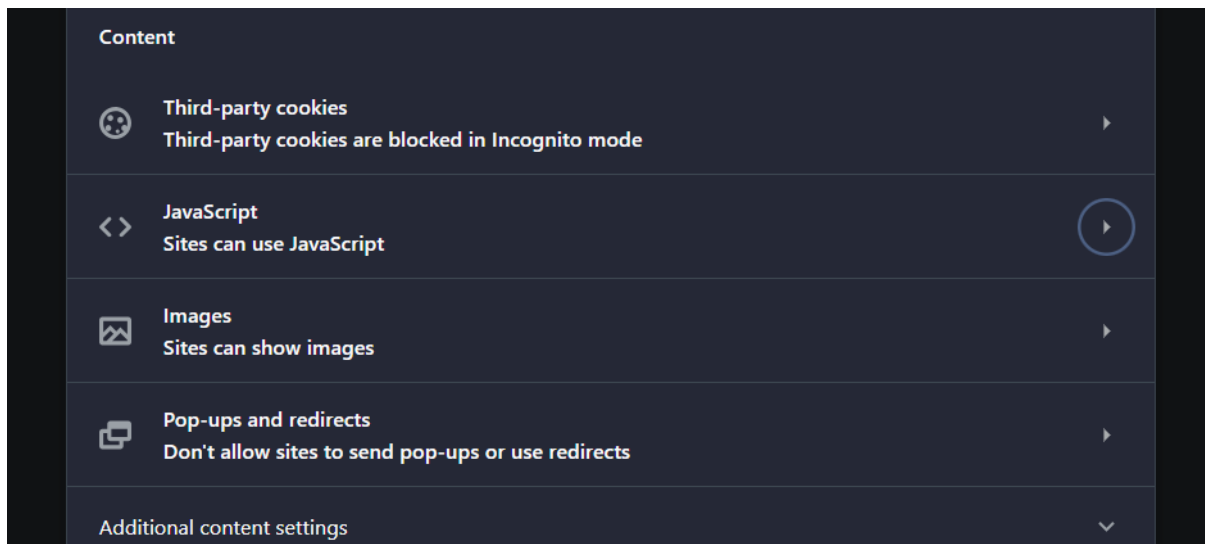
Also block website trackers if you want.

Step#6: Site settings

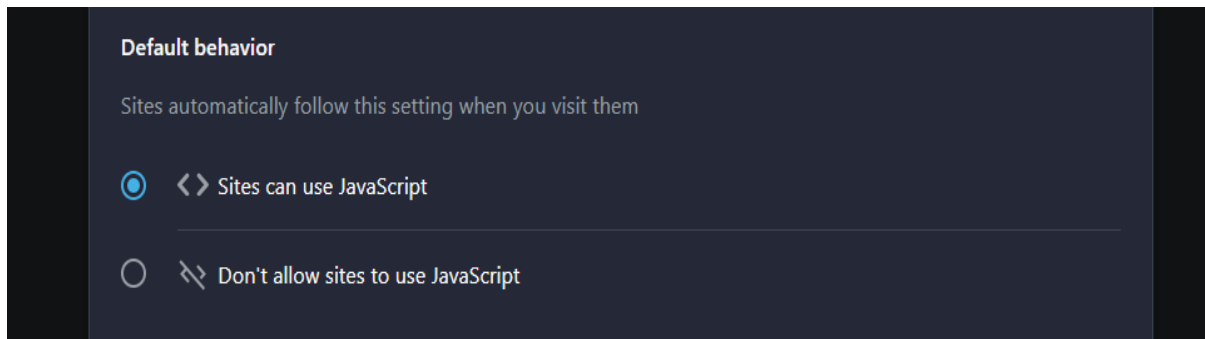


Step a: Click on site settings:

Step b: Click on java script:

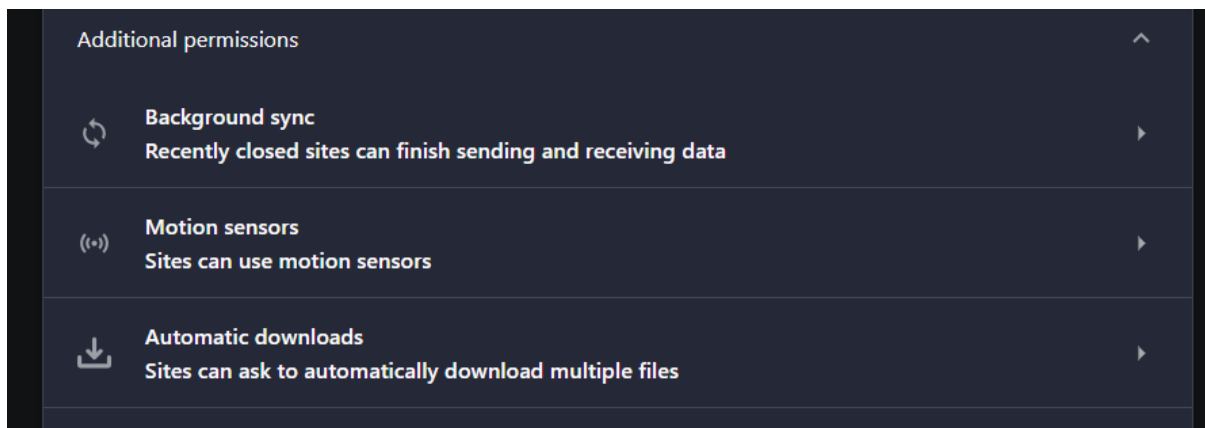


Step#3: Click on do not allow java script:

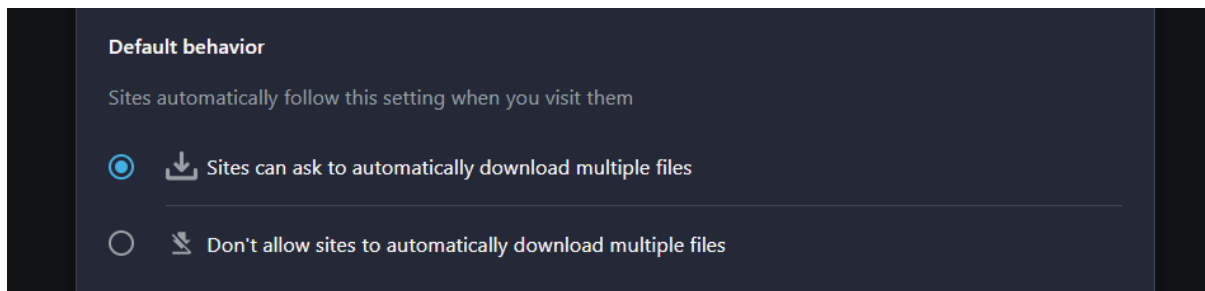


Step#7: Download protection

Step a: Go to privacy and security and go to site settings and then click on additional permissions then click on automatic downloads:



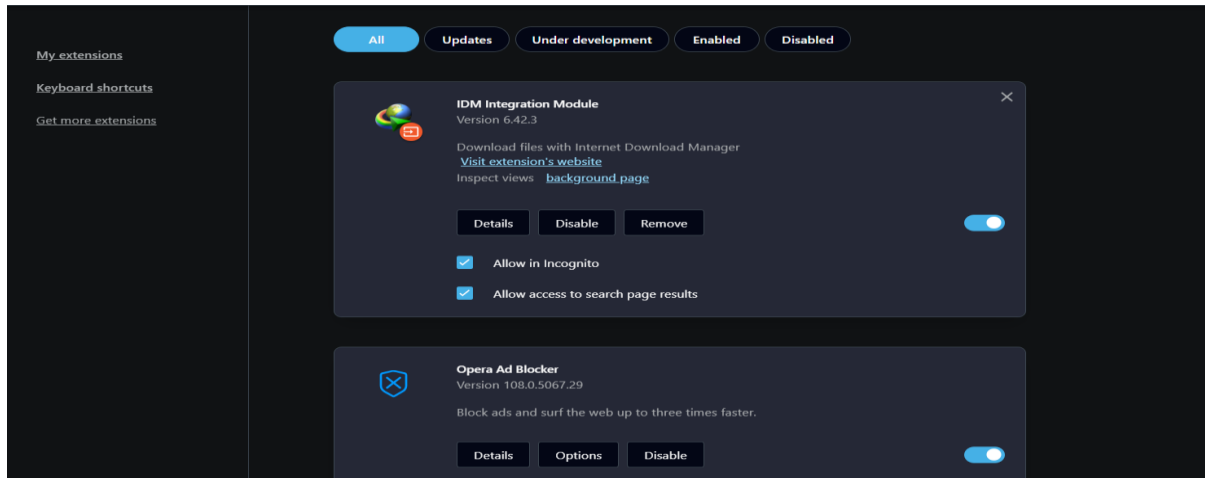
Step 2: Click on do not allow:



Step#8: Extension handling

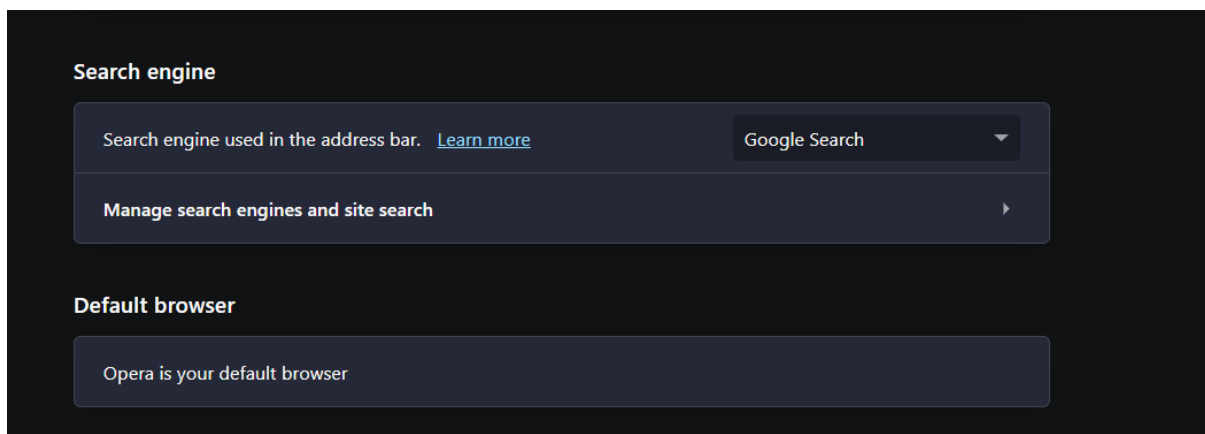
Step a: Click on the small puzzle icon or a cube icon in the bottom left or top right:

Step b: You can manage your entire extensions from there and also add other extensions if you want:



Step#9: Managing default browser

Step a: Go to basic settings and the scroll down to the following settings:



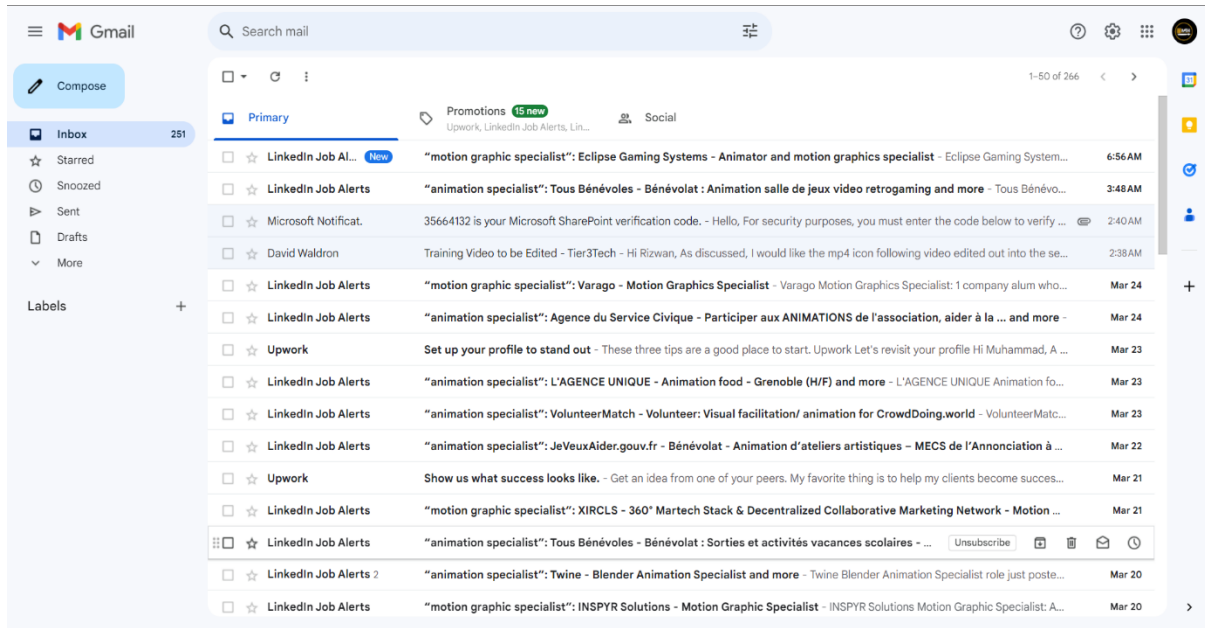
You can also change your search engines form here and manage default search engines.

Activity 2: Email Security

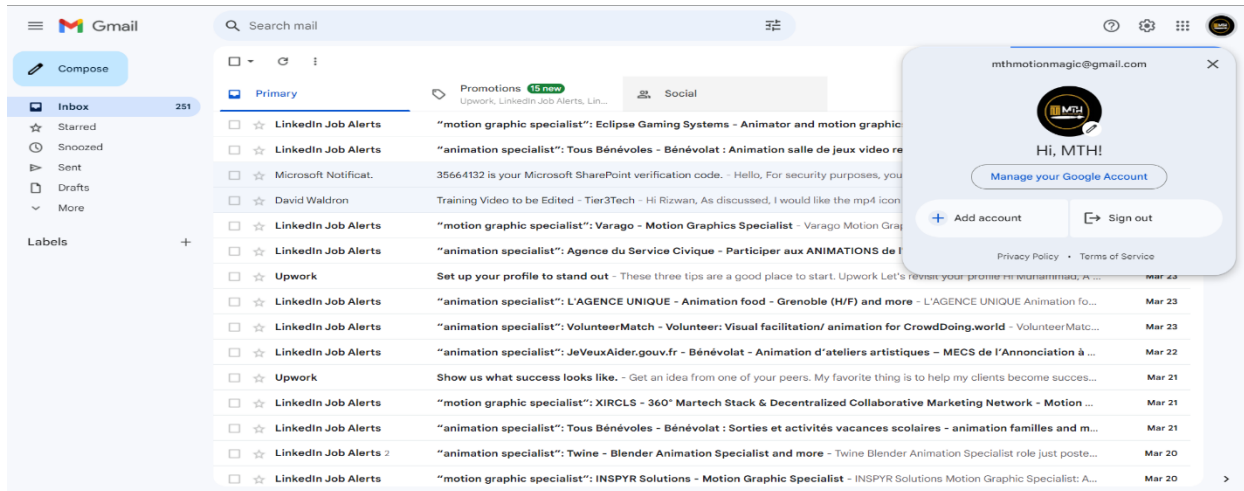
Step#1: Two-Factor Authentication (2FA)

Step#1: Log in to your Gmail account:

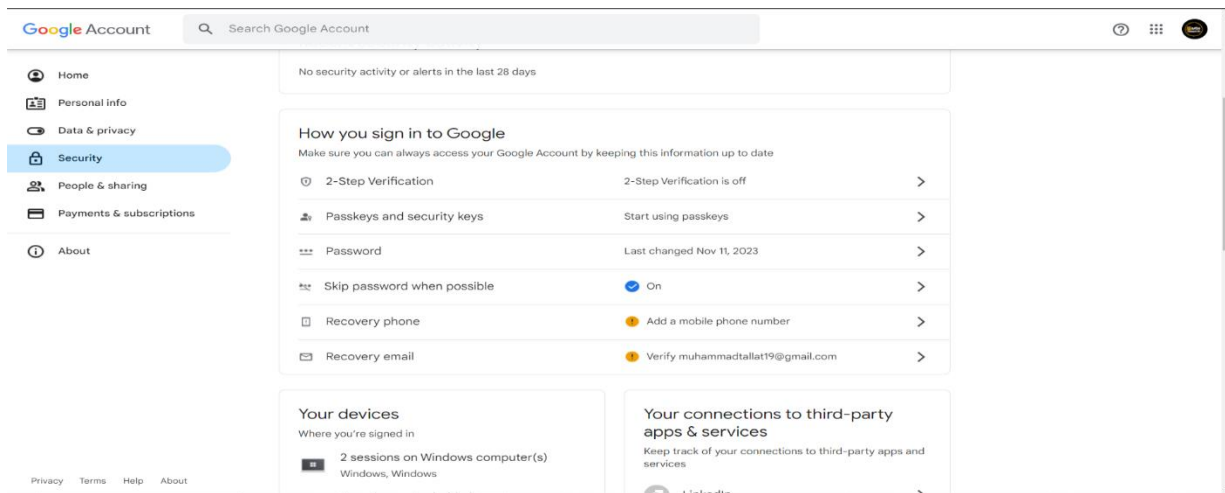
Step#2: Click on your profile picture in the upper-right corner:



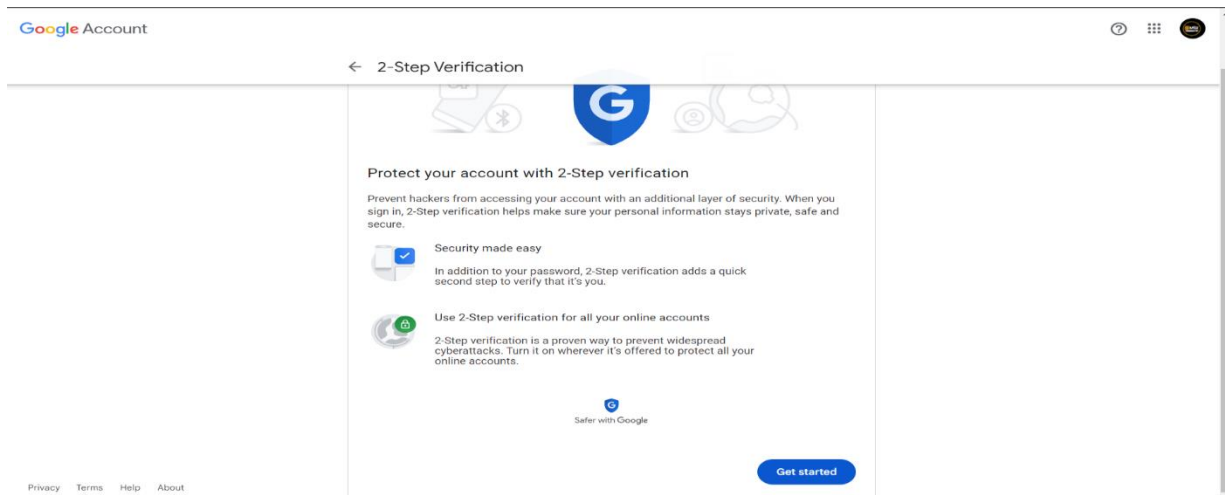
Step#3: Select "Manage your Google Account":



Step#4: In the left sidebar, click on "Security".



Step#5: Under "Signing in to Google," click on "2-Step Verification".



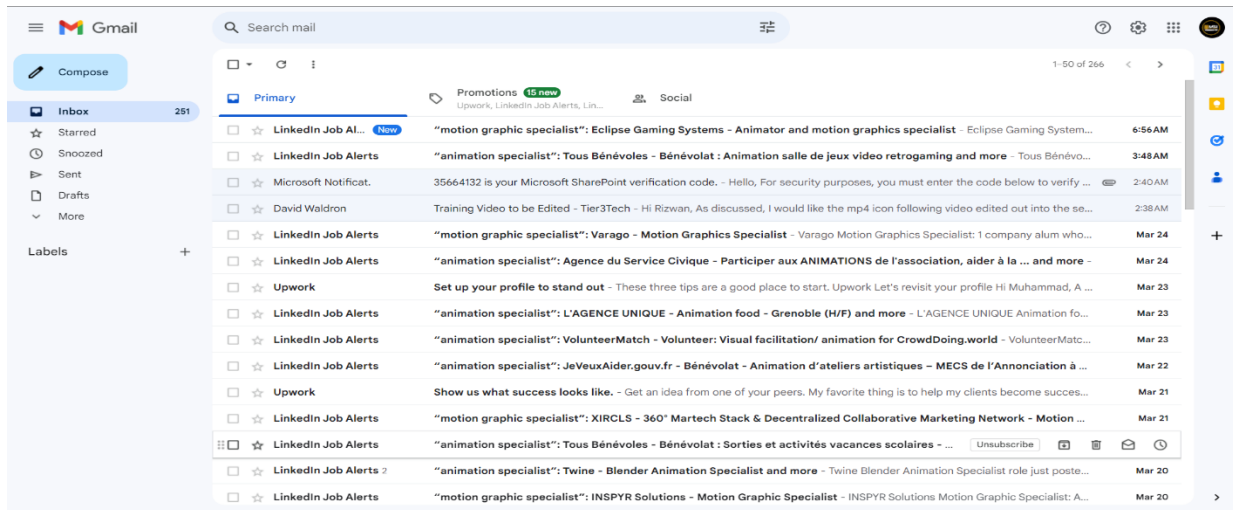
Step#6: Follow the on-screen prompts to enable 2FA and choose your preferred method (e.g., text message, Google Authenticator app).

Step#7: Verify your identity as prompted.

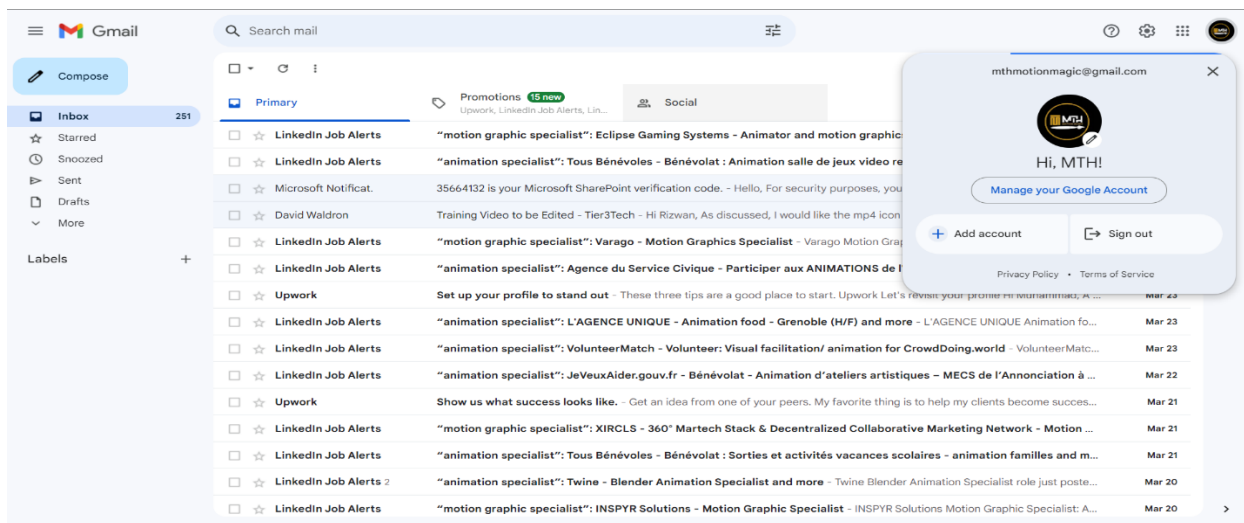
Step#2: Strong Password Policies

Step#1: Log in to your Gmail account:

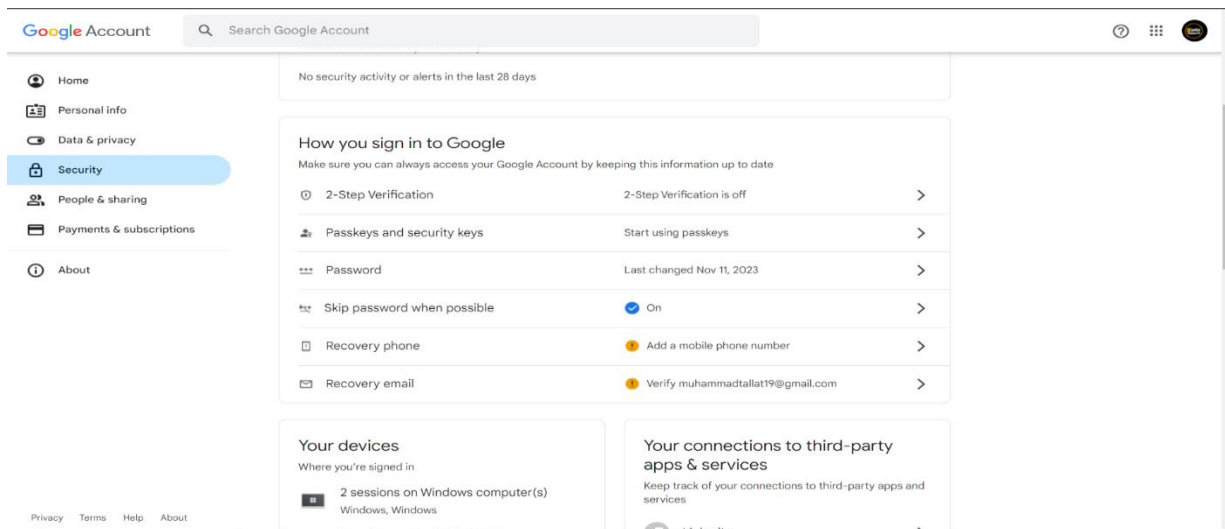
Step#2: Click on your profile picture in the upper-right corner:



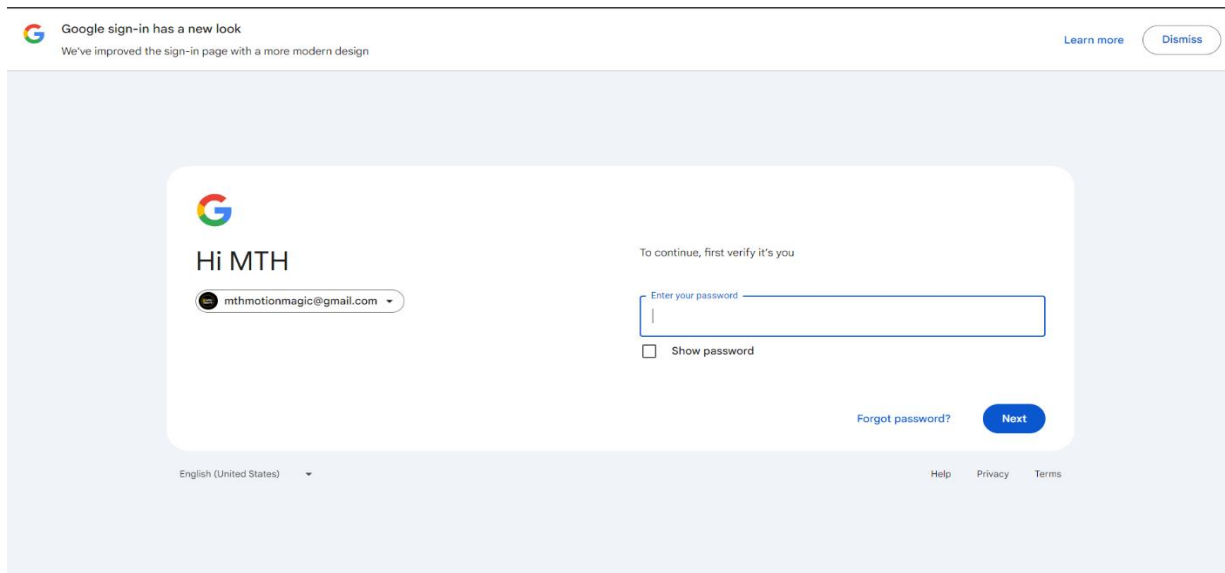
Step#3: Select "Manage your Google Account":



Step#4: In the left sidebar, click on "Security".



Step#5: Under "Signing in to Google," click on "Password."

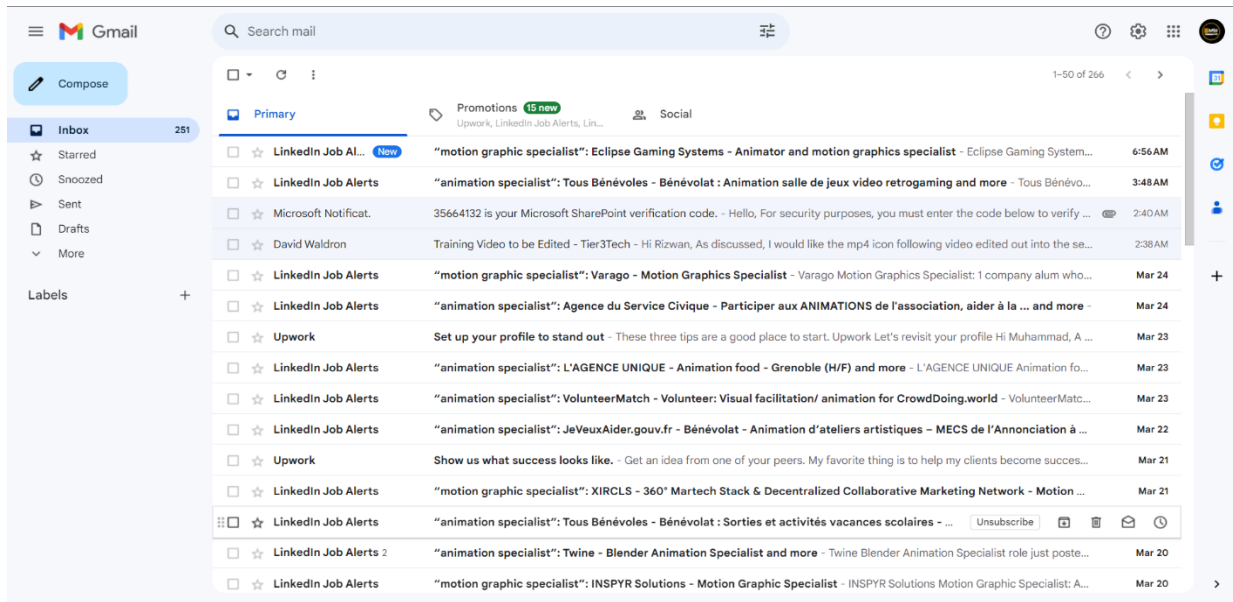


Step#6: Follow the on-screen instructions to change your password to a strong one.

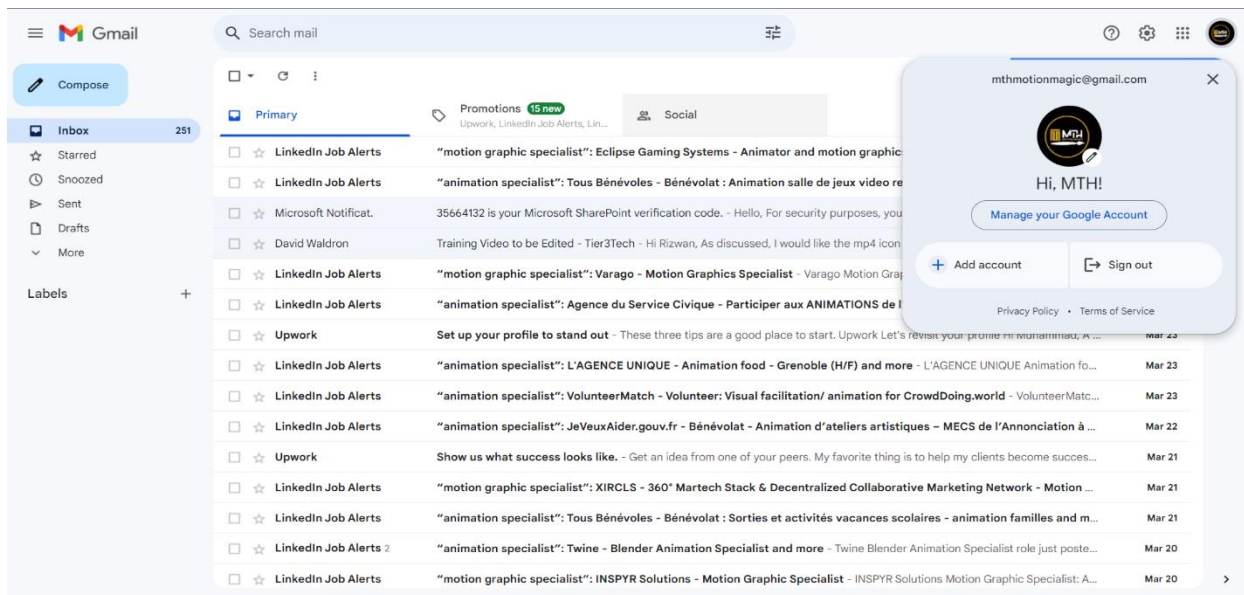
Step#3: Account Recovery Options

Step#1: Log in to your Gmail account:

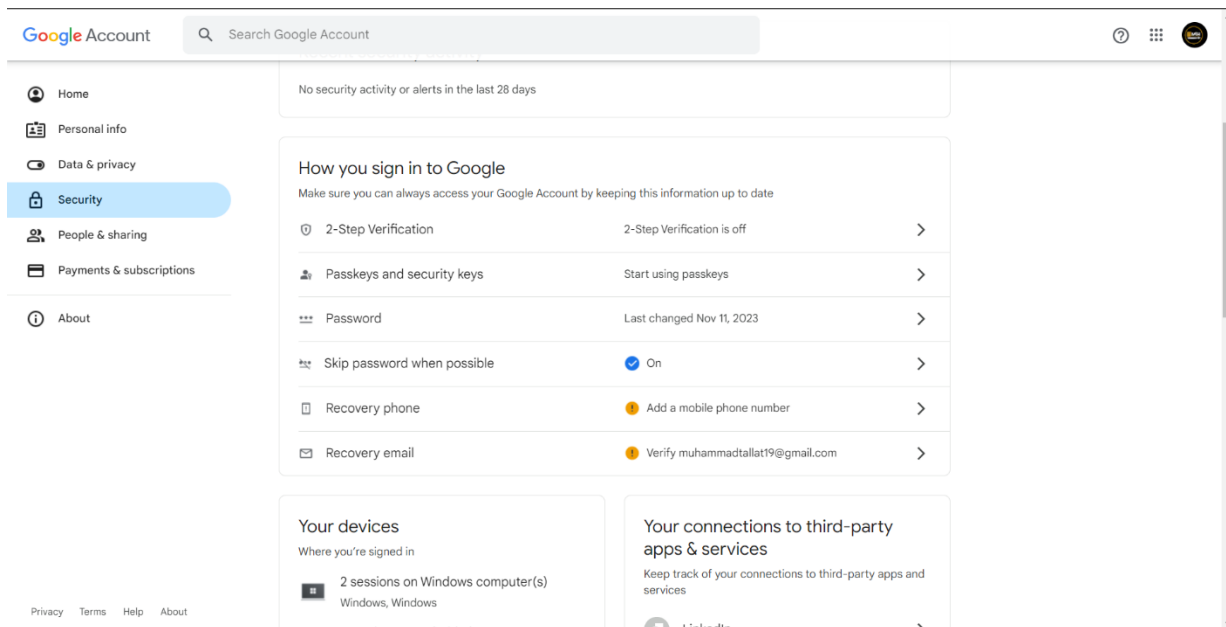
Step#2: Click on your profile picture in the upper-right corner:



Step#3: Select "Manage your Google Account":



Step#4: In the left sidebar, click on "Security".

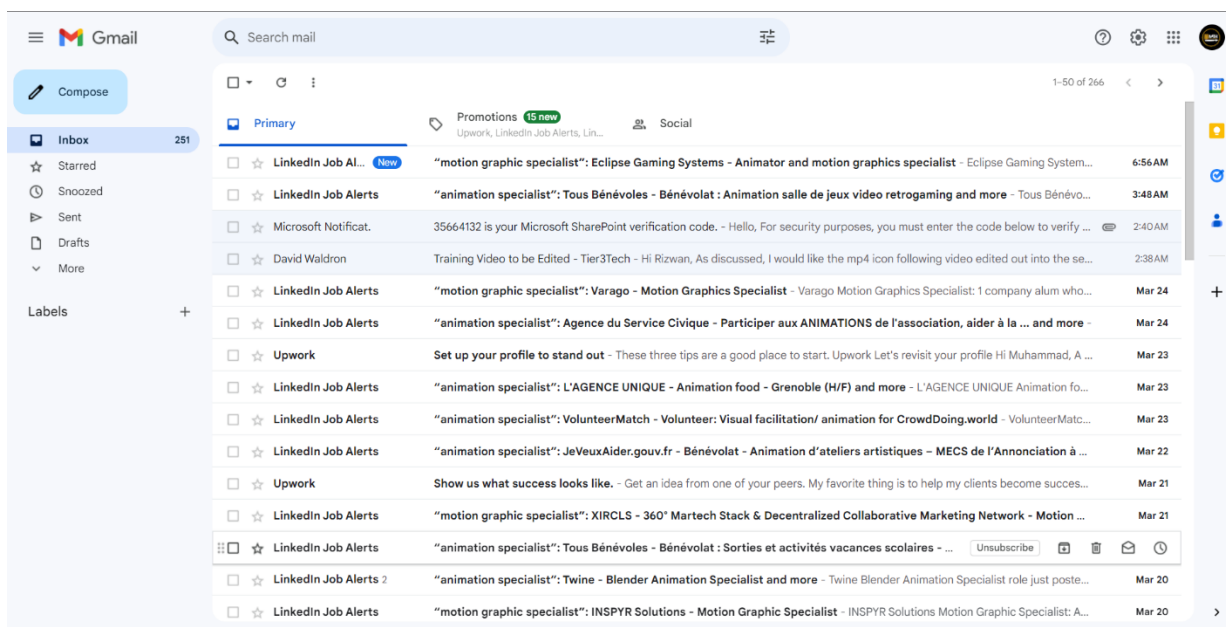


Step#5: Under "Ways we can verify it's you," click on "Recovery phone" and "Recovery email" to add or update recovery options.

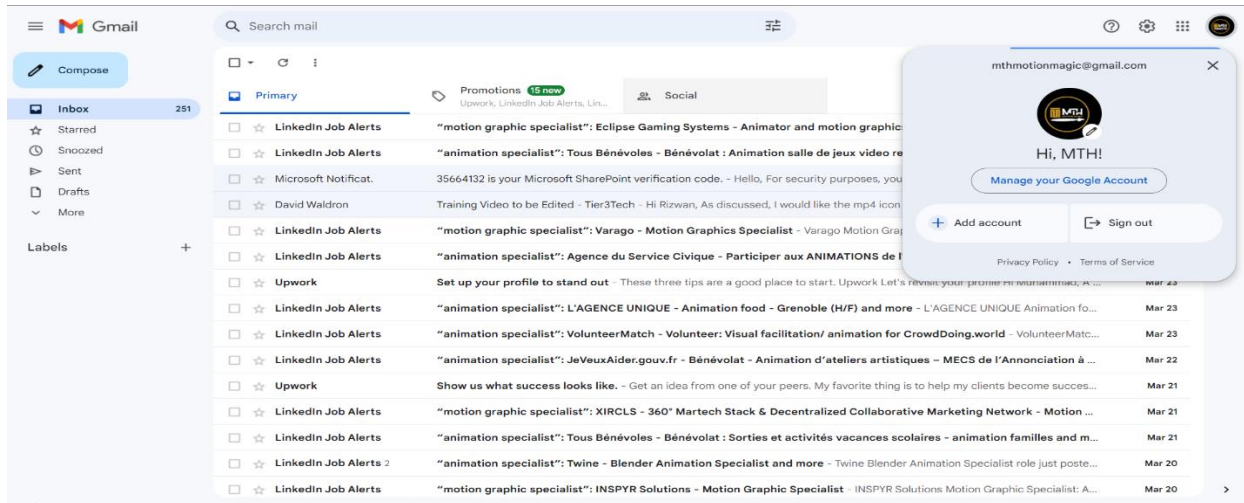
Step#4: Anti-Phishing Measures

Step#1: Log in to your Gmail account:

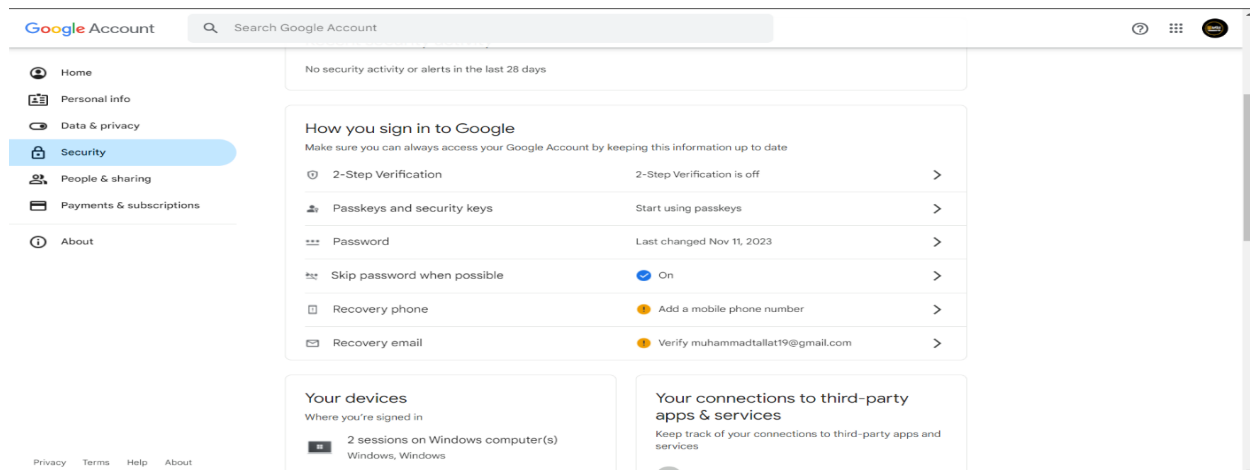
Step#2: Click on your profile picture in the upper-right corner:



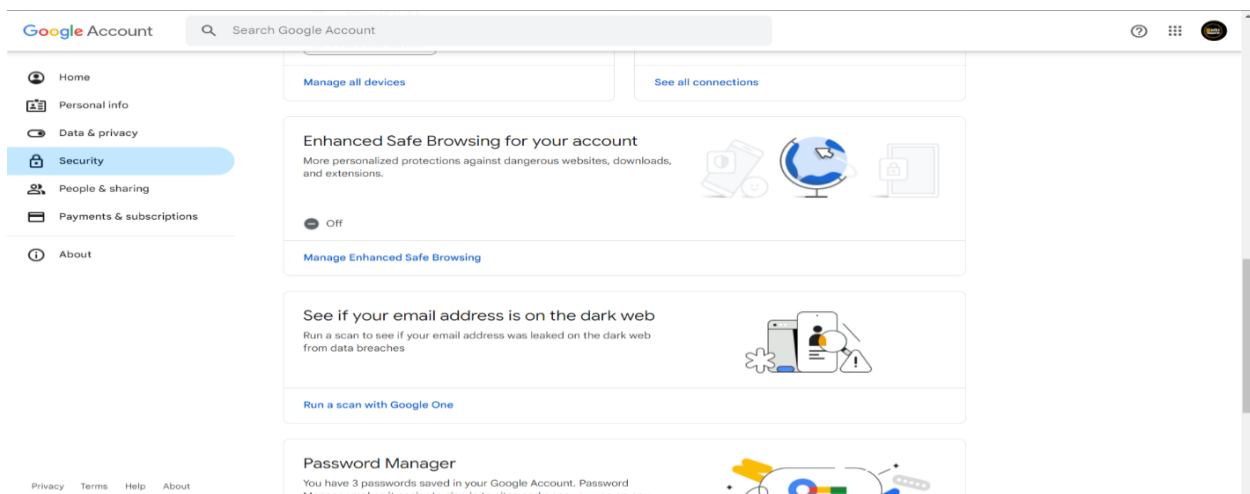
Step#3: Select "Manage your Google Account":



Step#4: In the left sidebar, click on "Security":



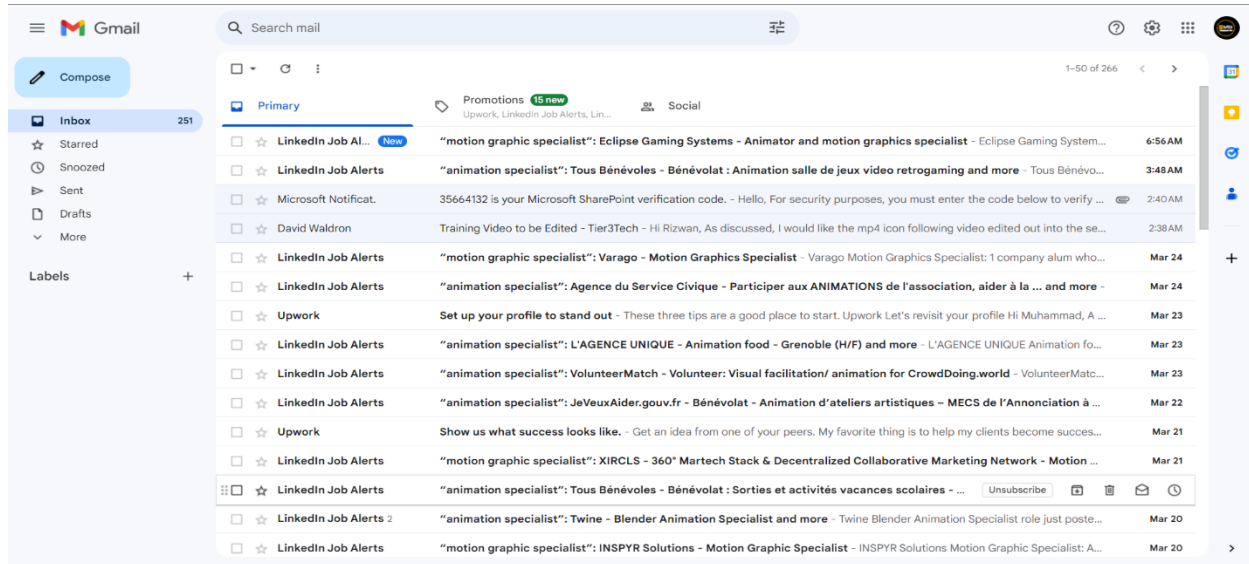
Step#5: Under "Security," enable "Enhanced Safe Browsing."



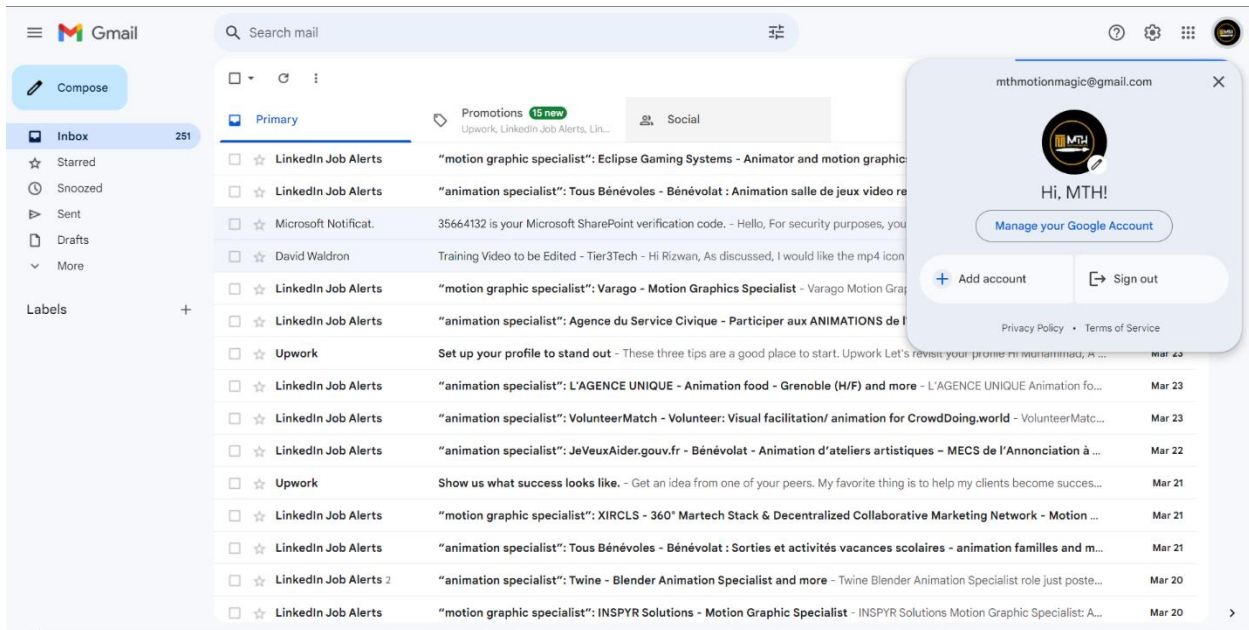
Step#5: Suspicious Activity Alerts

Step#1: Log in to your Gmail account:

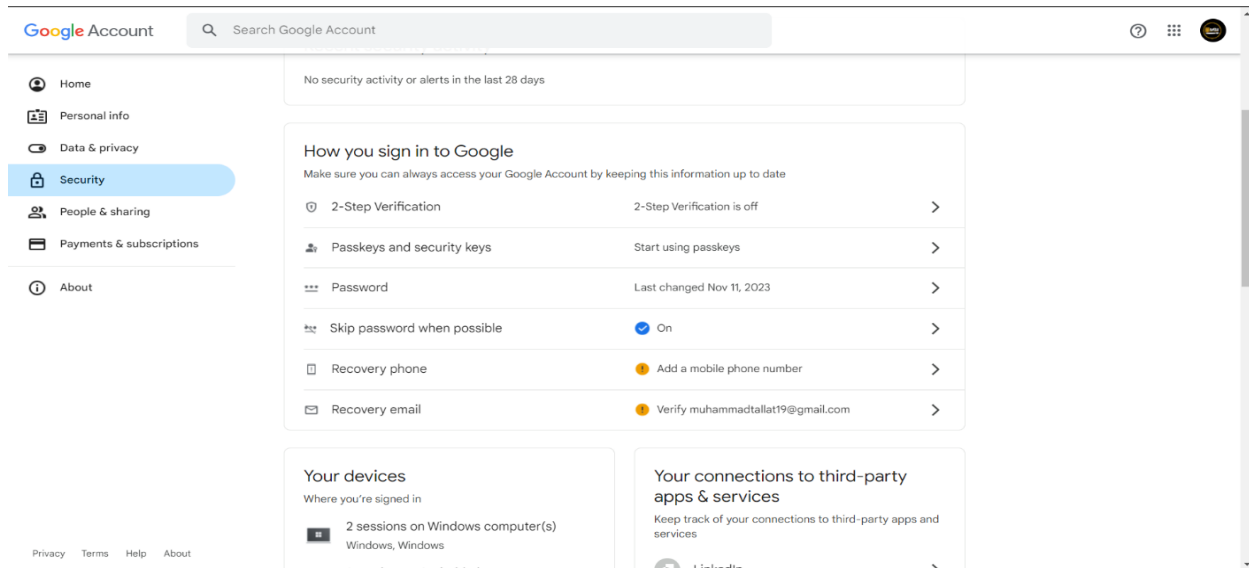
Step#2: Click on your profile picture in the upper-right corner:



Step#3: Select "Manage your Google Account":



Step#4: In the left sidebar, click on "Security".

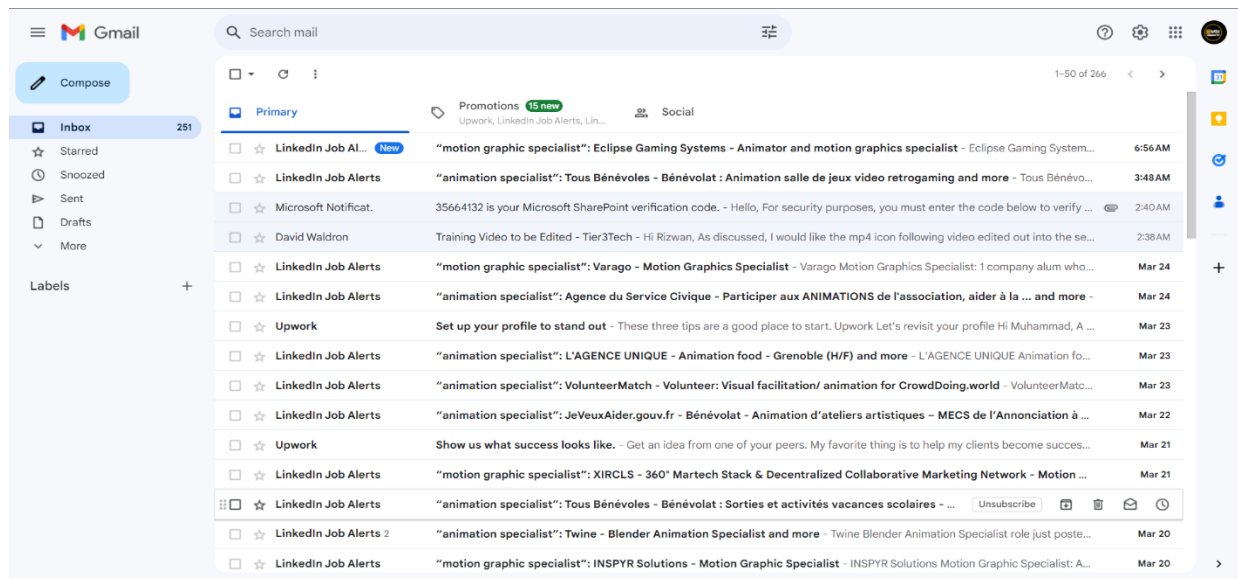


Step#5: Under "Notifications," turn on "Get notifications for important security events."

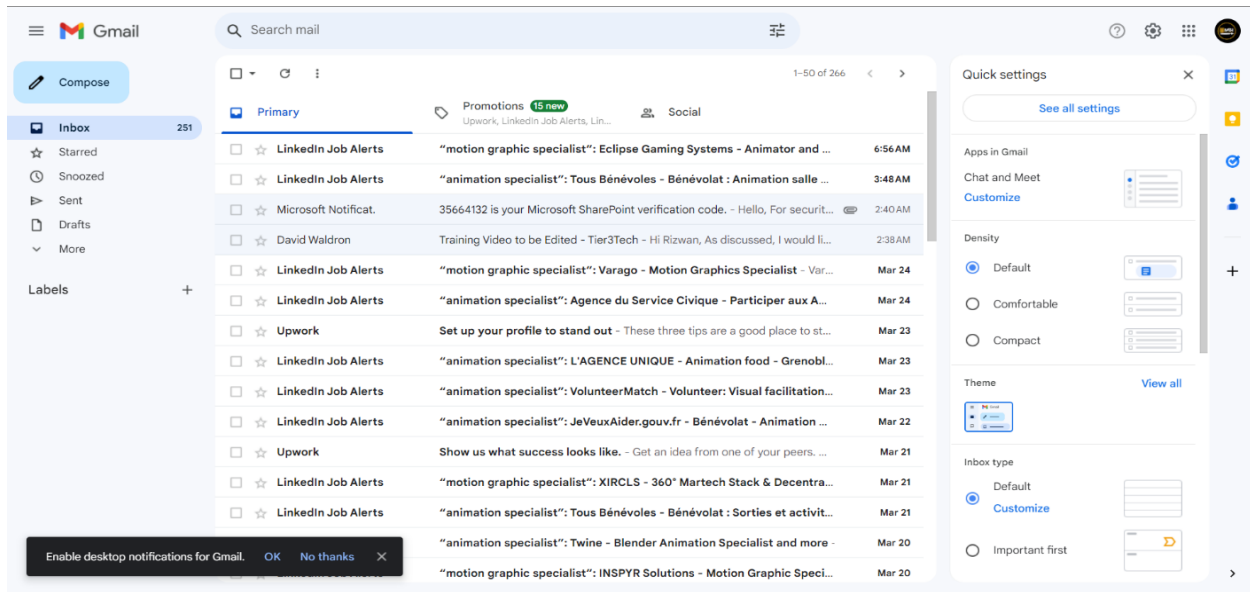
Step#6: Email Forwarding Restrictions

Step#1: Log in to your Gmail account:

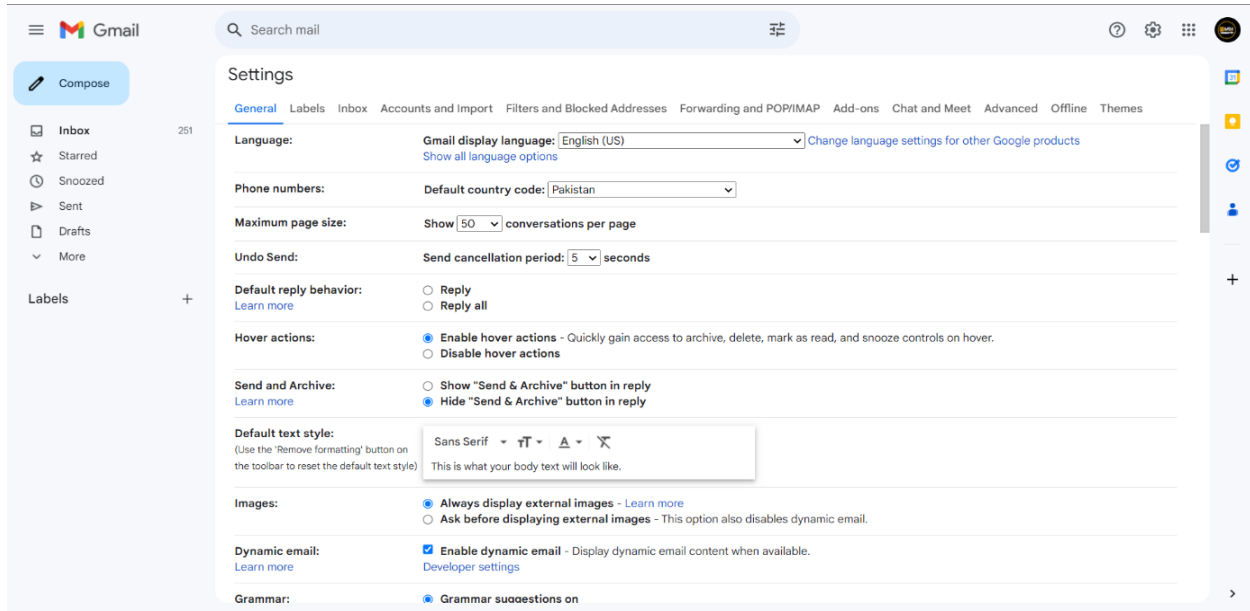
Step#2: Click on the gear icon in the upper-right corner:



Step#3: Select "See all settings."



Step#4: Navigate to the "General" tab.

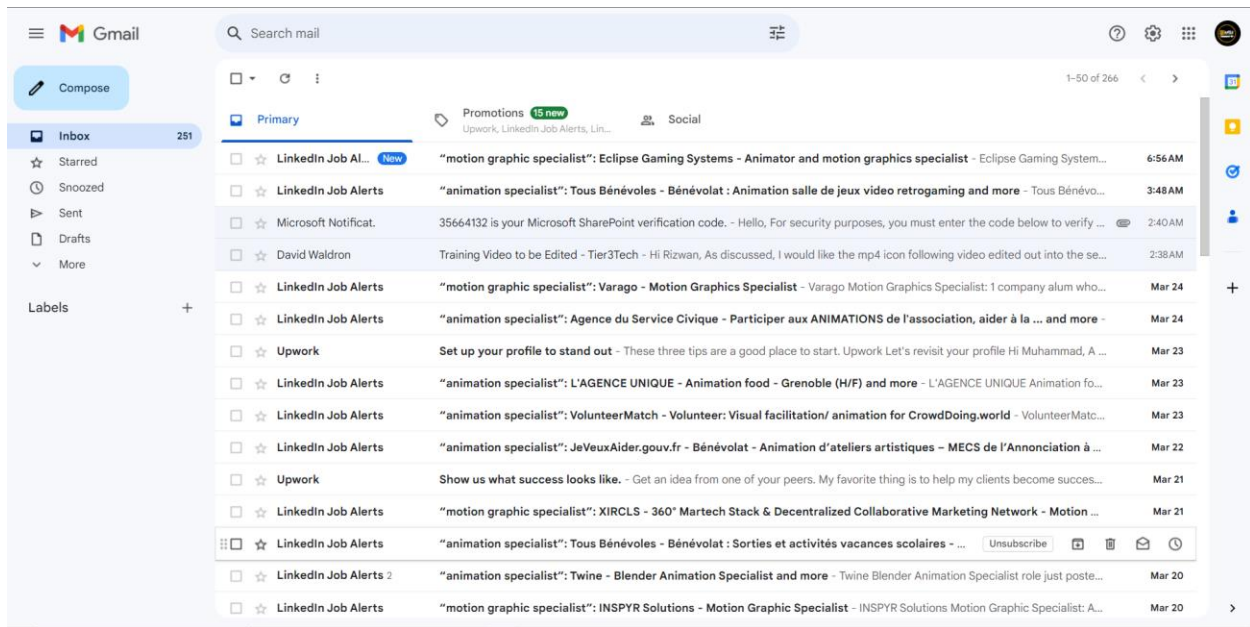


Step#5: Adjust privacy settings like email visibility and read receipt requests as needed.

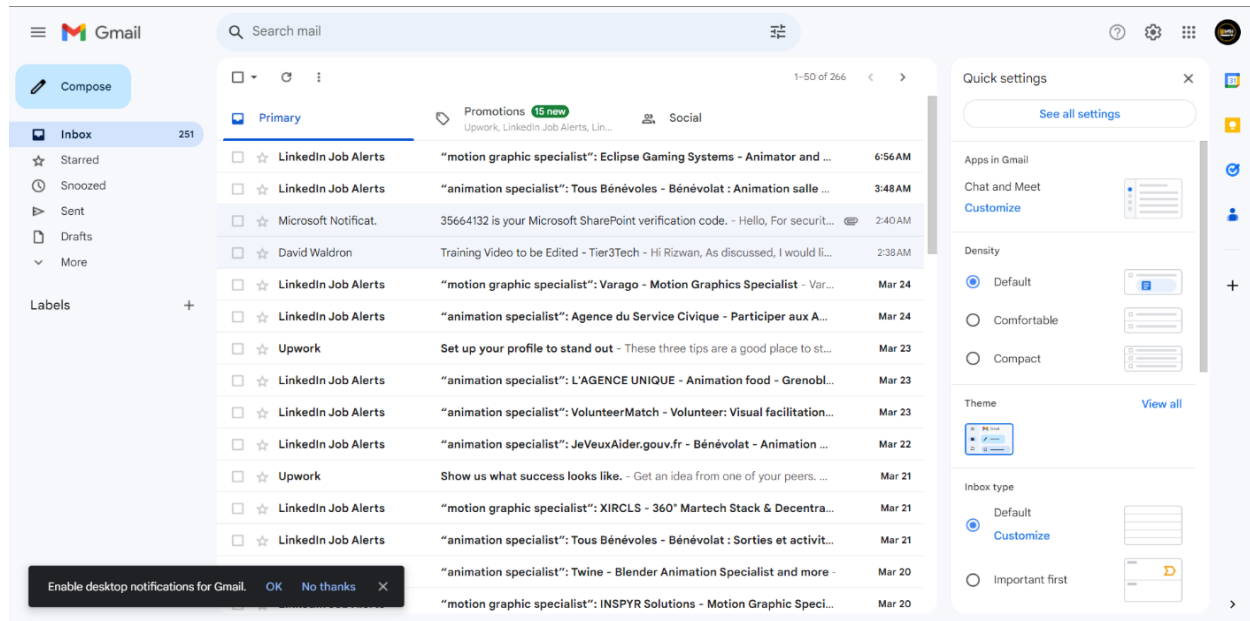
Step#7: Email Tracking Protection

Step#1: Log in to your Gmail account:

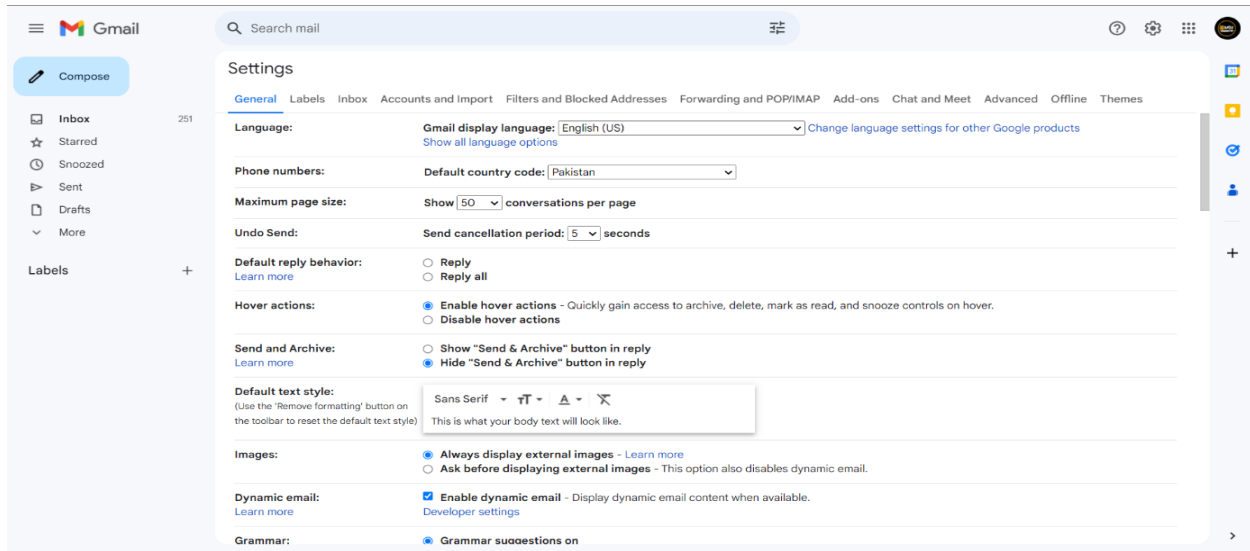
Step#2: Click on the gear icon in the upper-right corner:



Step#3: Select "See all settings."



Step#4: Navigate to the "General" tab.

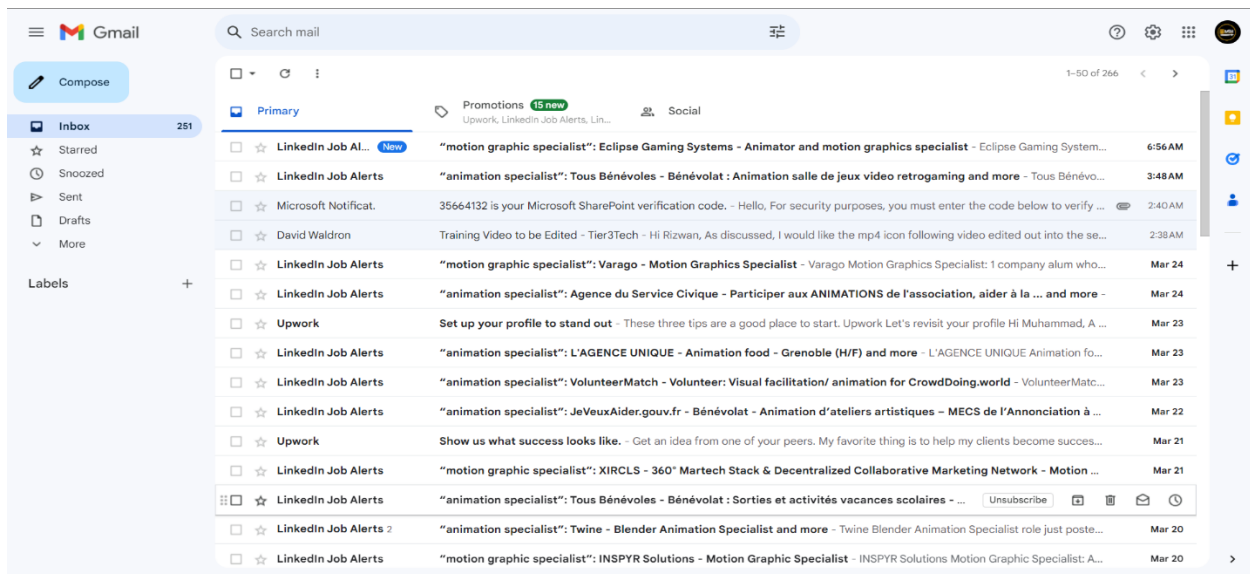


Step#5: Under "Images," choose "Ask before displaying external images."

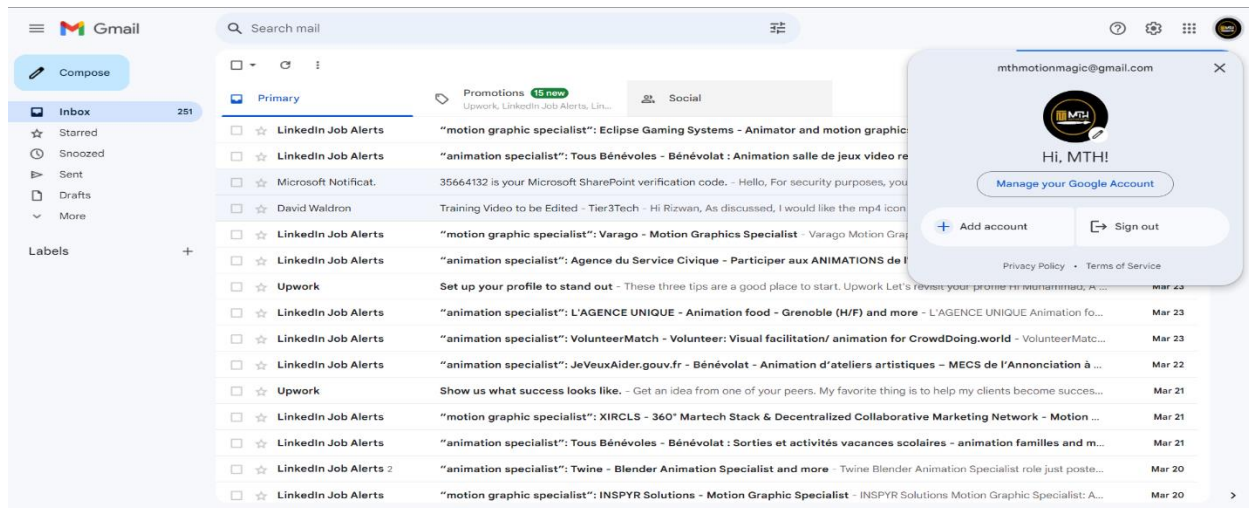
Step#8: Device Management

Step#1: Log in to your Gmail account:

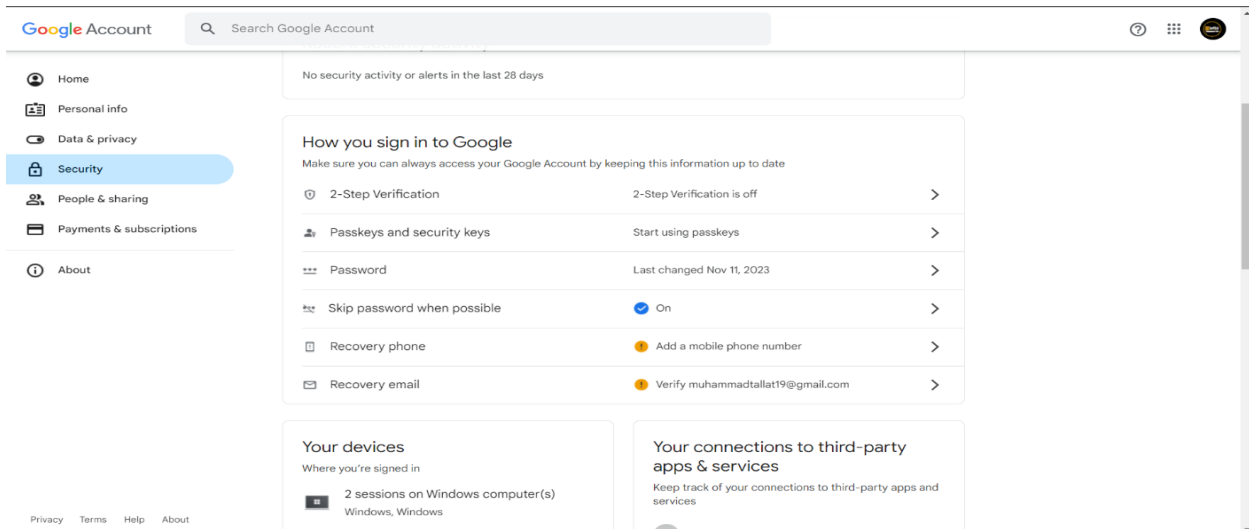
Step#2: Click on your profile picture in the upper-right corner:



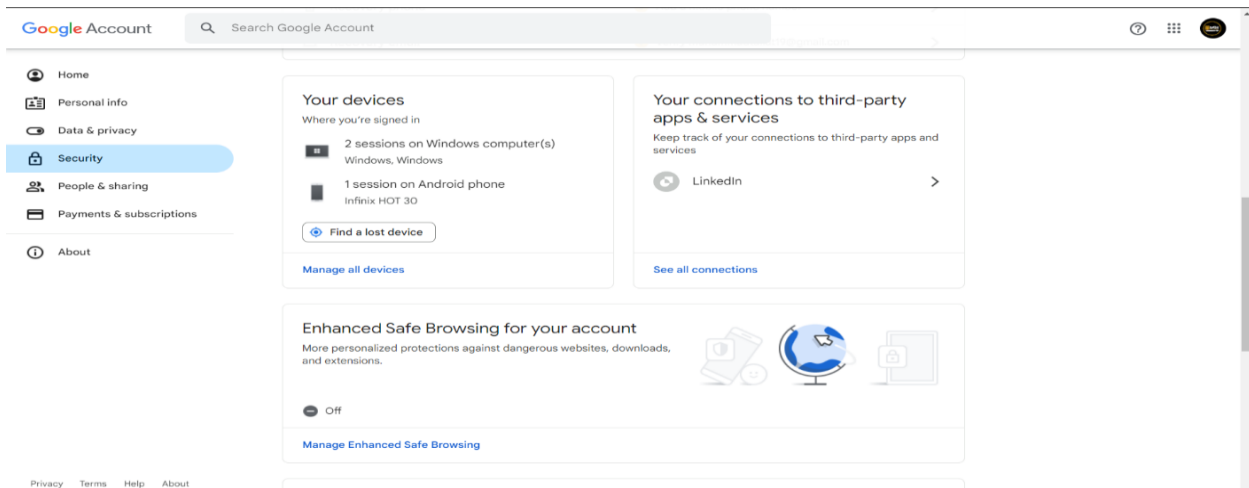
Step#3: Select "Manage your Google Account":



Step#4: In the left sidebar, click on "Security".



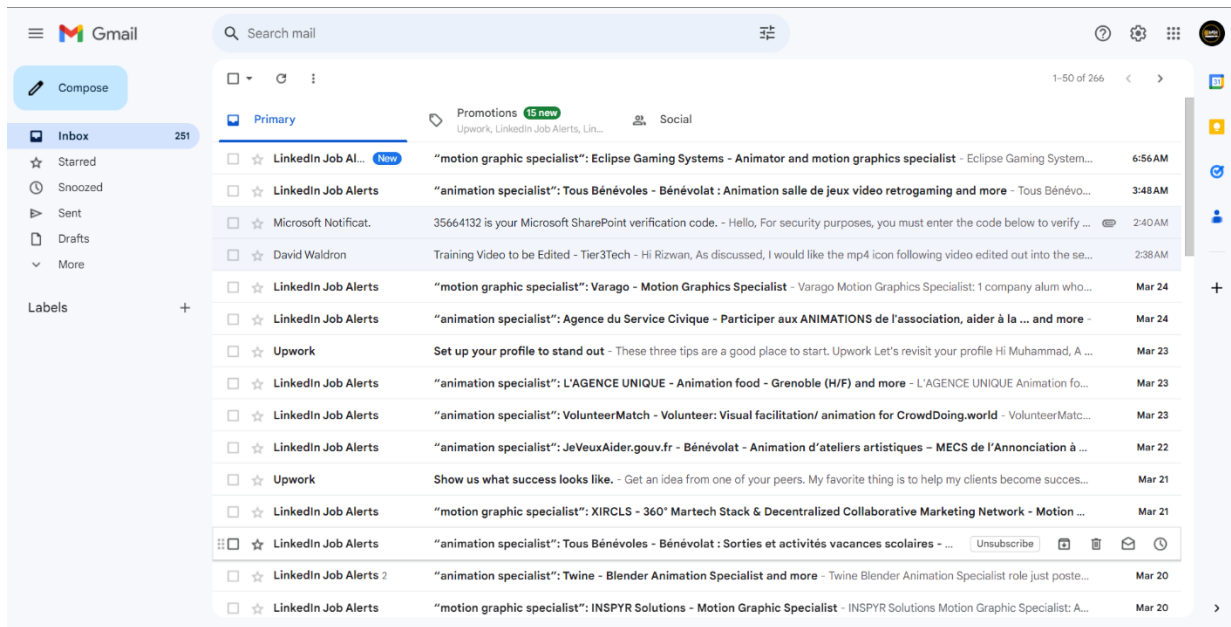
Step#5: Under "Your devices," review and manage the devices connected to your Google account.



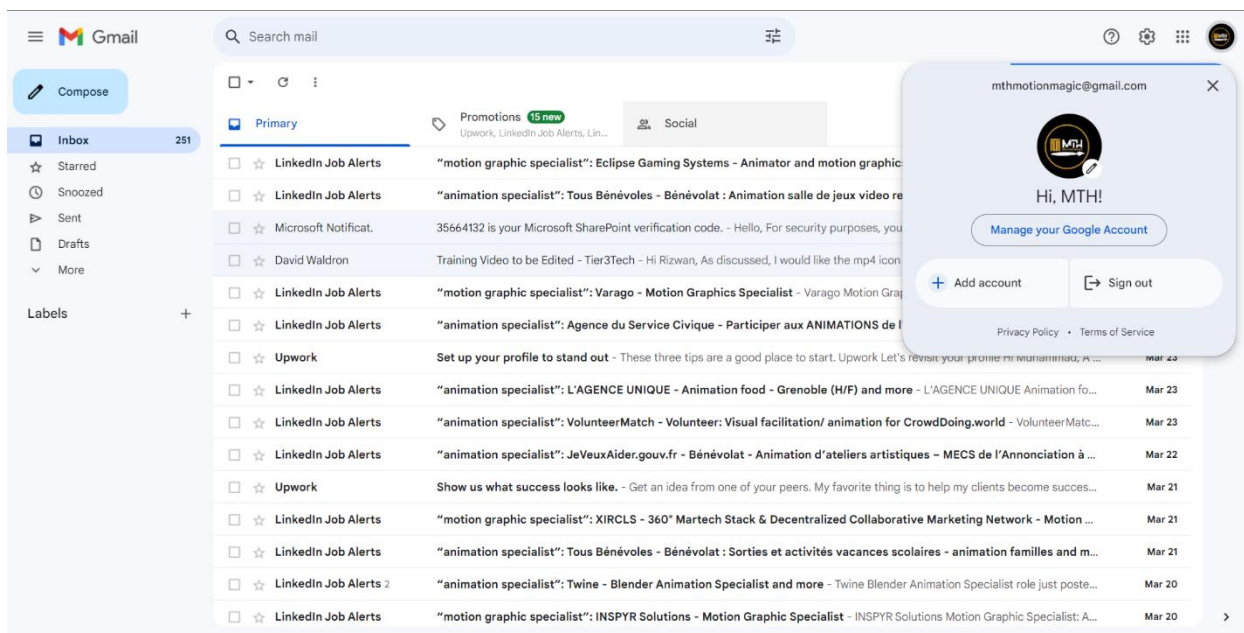
Step#9: Alerts for Unusual Activity

Step#1: Log in to your Gmail account:

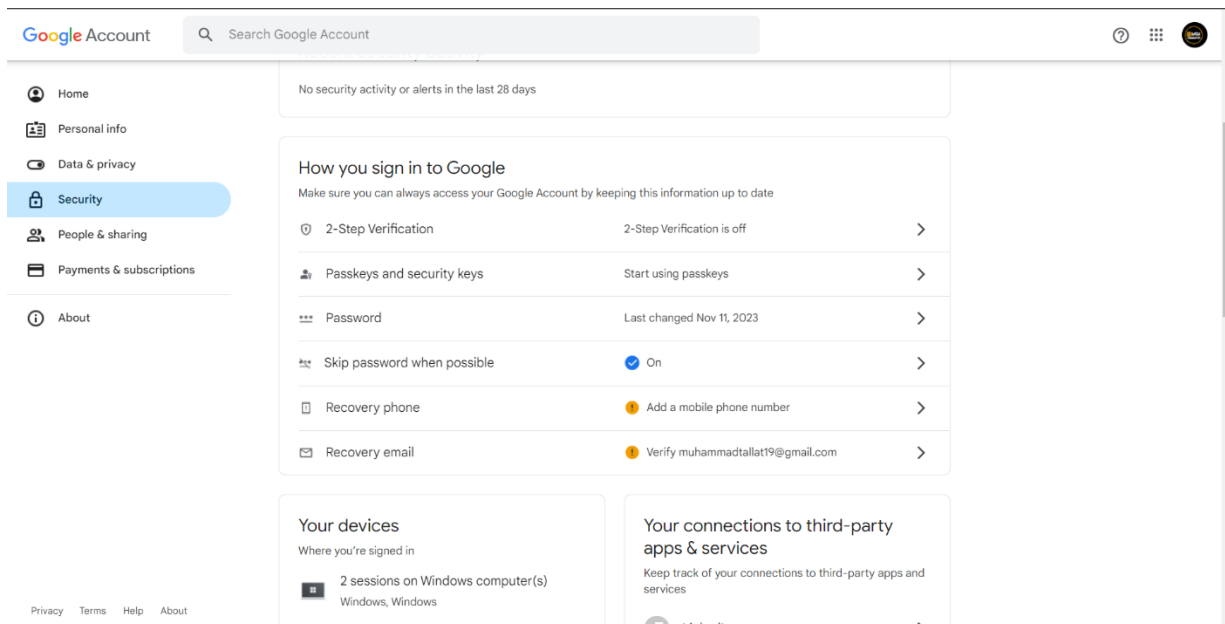
Step#2: Click on your profile picture in the upper-right corner:



Step#3: Select "Manage your Google Account":



Step#4: In the left sidebar, click on "Security."



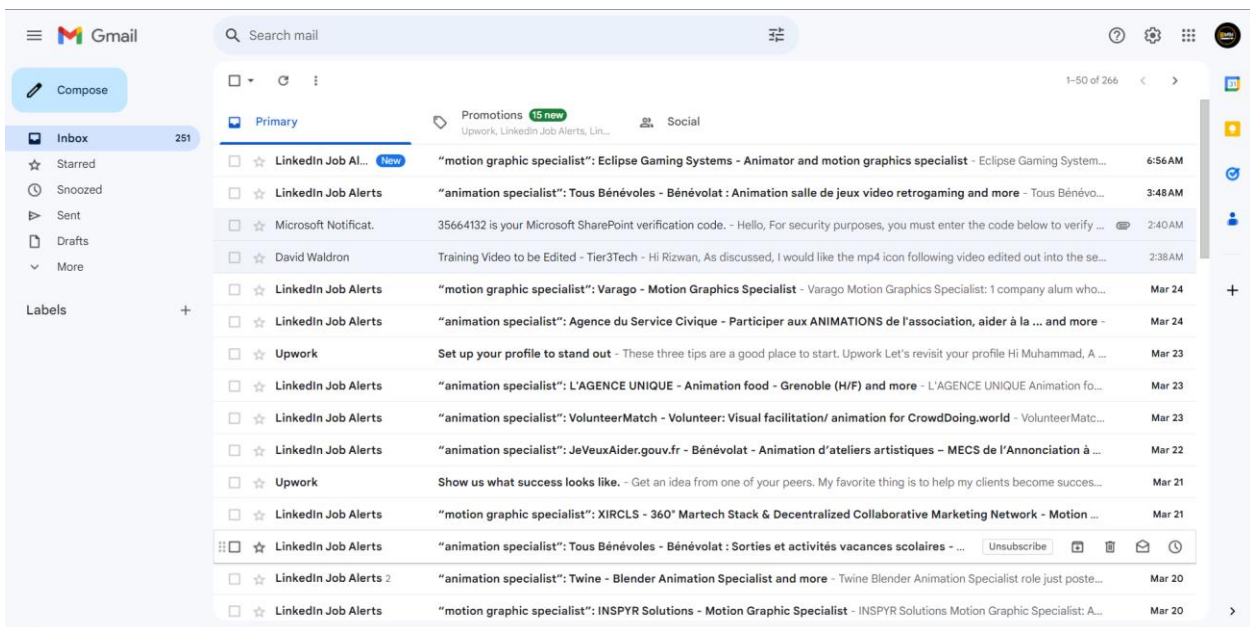
Step#5: Under "Signing in to Google," click on "Alerts."

Step#6: Configure alerts for unusual activity, such as suspicious sign-ins or changes to your account recovery information. You can receive alerts via email or text message.

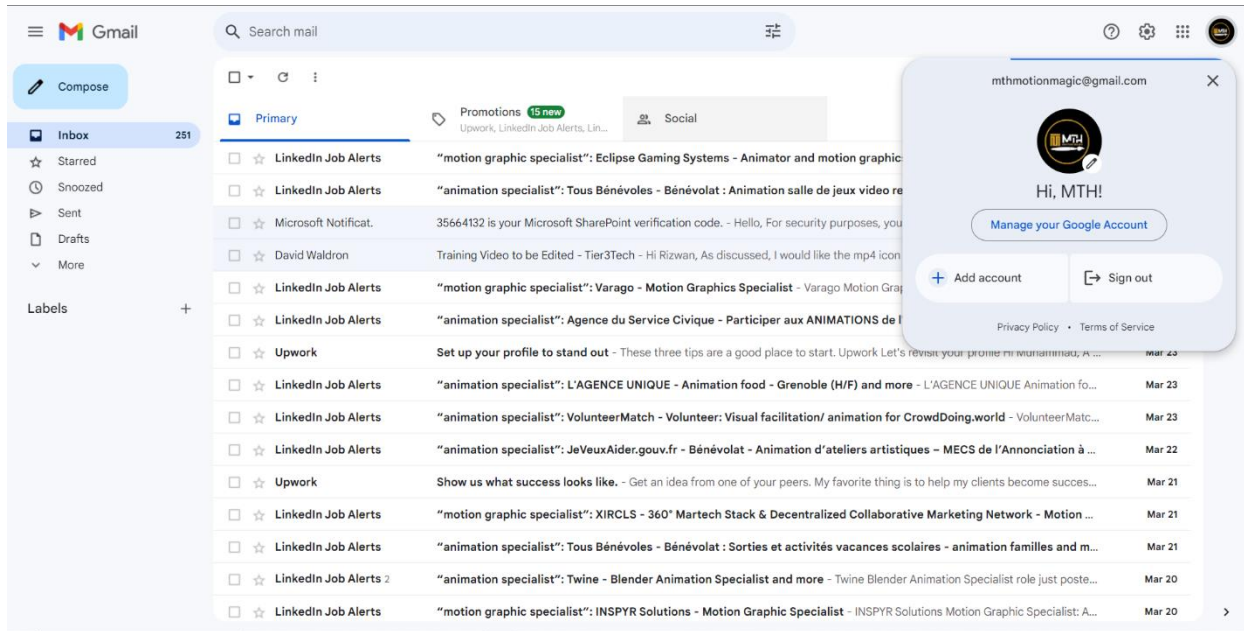
Step#10: Third-party App Access

Step#1: Log in to your Gmail account:

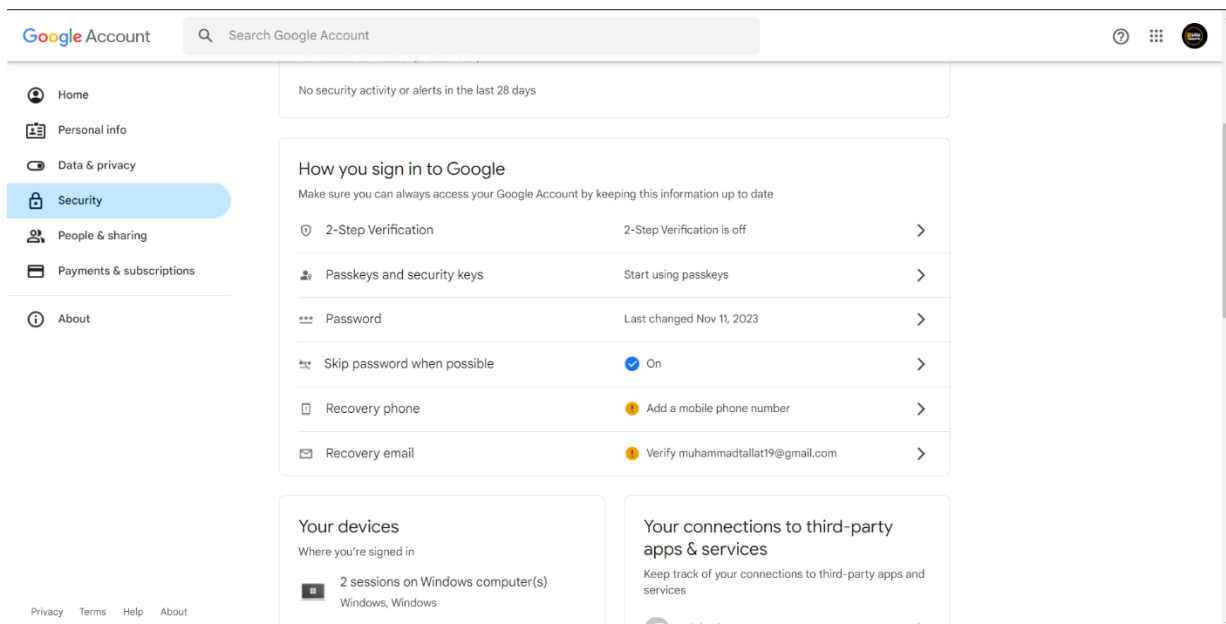
Step#2: Click on your profile picture in the upper-right corner:



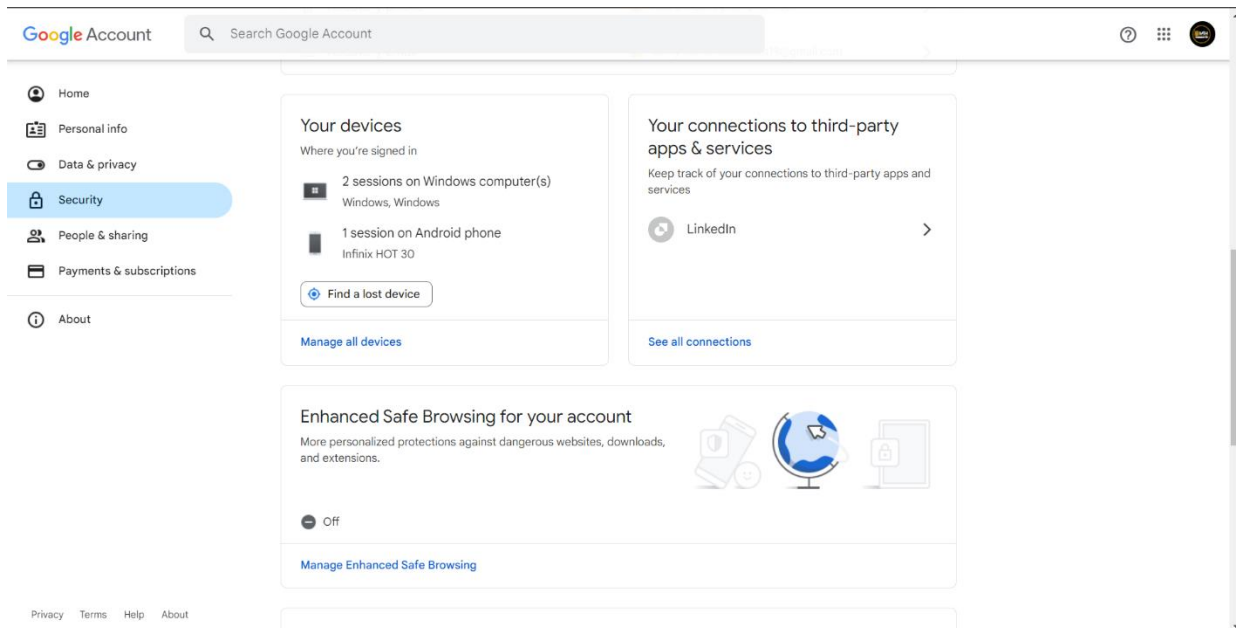
Step#3: Select "Manage your Google Account":



Step#4: In the left sidebar, click on "Security.":



Step#5: Under "Third-party apps with account access," review and manage the apps that have access to your Gmail account. Remove any that you no longer use or trust.



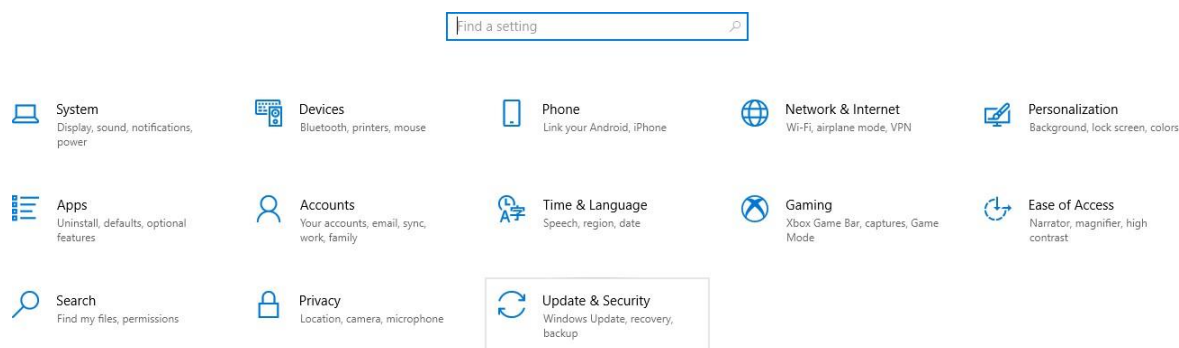
Activity 3: Windows Hardening

Windows Security Settings

Keep the Windows Updated:

Enable automatic Windows updates or regularly check for updates and install them.

Open Settings (type “Settings” into the search bar) > Update & Security > Windows Update. Here you’ll find several options for controlling Windows updates, but the most important one to set up is “active hours,” which ensures that Windows won’t reboot in the middle of your workday.





1. Use Strong Authentication:

Enforce strong password policies: Require complex passwords with a combination of uppercase, lowercase, numbers, and symbols.

Go to Settings > Accounts > Sign-in options.

Sign-in options

Manage how you sign in to your device

Select a sign-in option to add, change, or remove it.



Windows Hello Face

This option is currently unavailable—click to learn more



Windows Hello Fingerprint

This option is currently unavailable—click to learn more



Windows Hello PIN

Sign in with a PIN (Recommended)



Security Key

Sign in with a physical security key



Password

Sign in with your account's password



Picture Password

Swipe and tap your favorite photo to unlock your device

Require sign-in

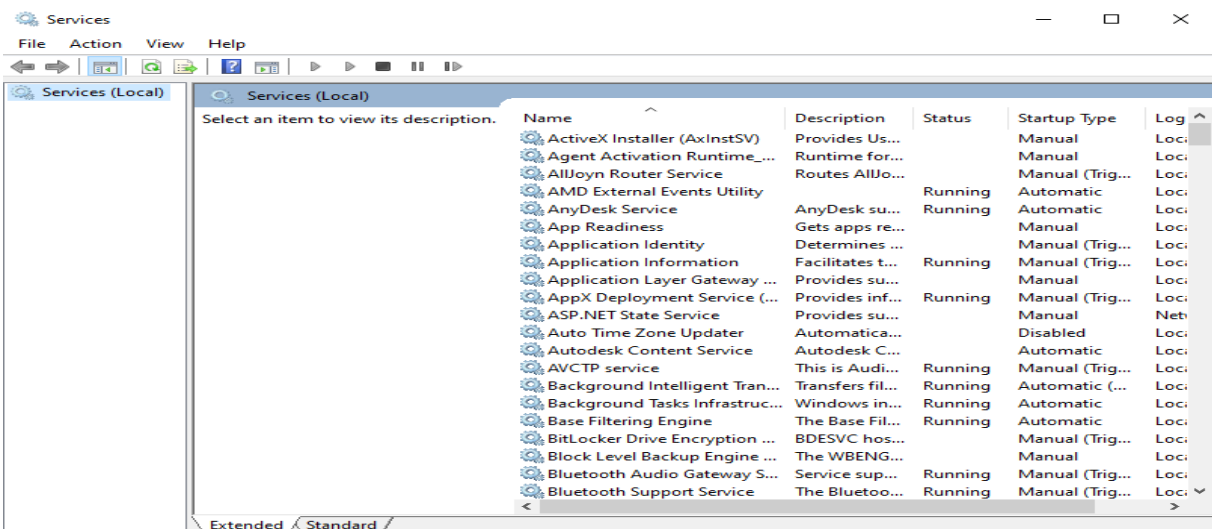
If you've been away, when should Windows require you to sign in again?

When PC wakes up from sleep ▾

2. Disable Unnecessary Services:

Use the "Services" application (services.msc) to disable or set unnecessary services to "Manual" or "Disabled."

Open Search in the Toolbar and type Services or open Run from Search menu, type services.msc and press enter.

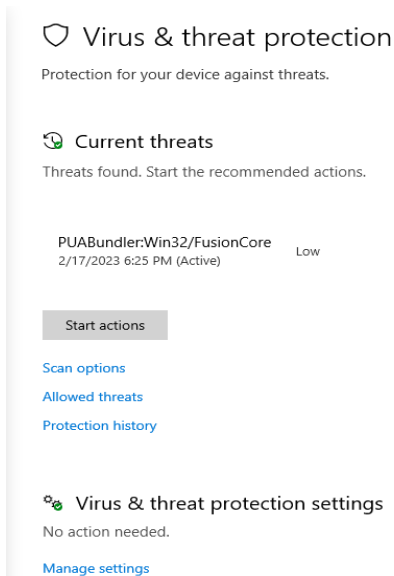


3. Set up Microsoft Defender properly

Open Settings > Update & Security (or Privacy & Security in Windows 11) > Windows Security.



On the “Virus & threat protection” page, you can start a manual scan if you want to check your computer right now. While you’re here, look under “Virus & threat protection settings” and click **Manage settings**. Make sure that every feature on the page is enabled.



On that same “Virus & threat protection” page, scroll down and click **Ransomware protection**. Here, you can optionally enable a feature called controlled folder access, which prevents programs from manipulating certain folders, including your Documents, Desktop, and Pictures folders. If you’re not comfortable working with folder hierarchies or messing around with access permissions, know that this feature will prevent you from using just about all software you download unless you go back and approve it (for details, see Bleeping Computer’s walk-through of the process). In addition, this feature protects only those folders, not your whole computer, from ransomware. Consistent backups are a far better safeguard against ransomware.

Ransomware protection

Set up OneDrive for file recovery options in case of a ransomware attack.

Set up OneDrive

[Manage ransomware protection](#)

[Dismiss](#)

Click **Firewall & network protection** and make sure the firewall is enabled for all three network options. If you are comfortable with more complex software, you may also want to check out third-party firewall tools like Simple wall.

Firewall & network protection

Who and what can access your networks.

Domain network

Firewall is on.

Private network

Firewall is on.

Public network (active)

Firewall is on.

Select the App & browser control tab and enable Reputation-based protection. This step will block the installation of unapproved applications entirely. It might also block apps that you intend to install, so be sure to remember where this option is if you need to disable it to install software you know to be safe.

App & browser control

App protection and online security.

Reputation-based protection

These settings protect your device from malicious or potentially unwanted apps, files, and websites.

Potentially unwanted app found. Your device may perform poorly.

[Review](#)

[Reputation-based protection settings](#)

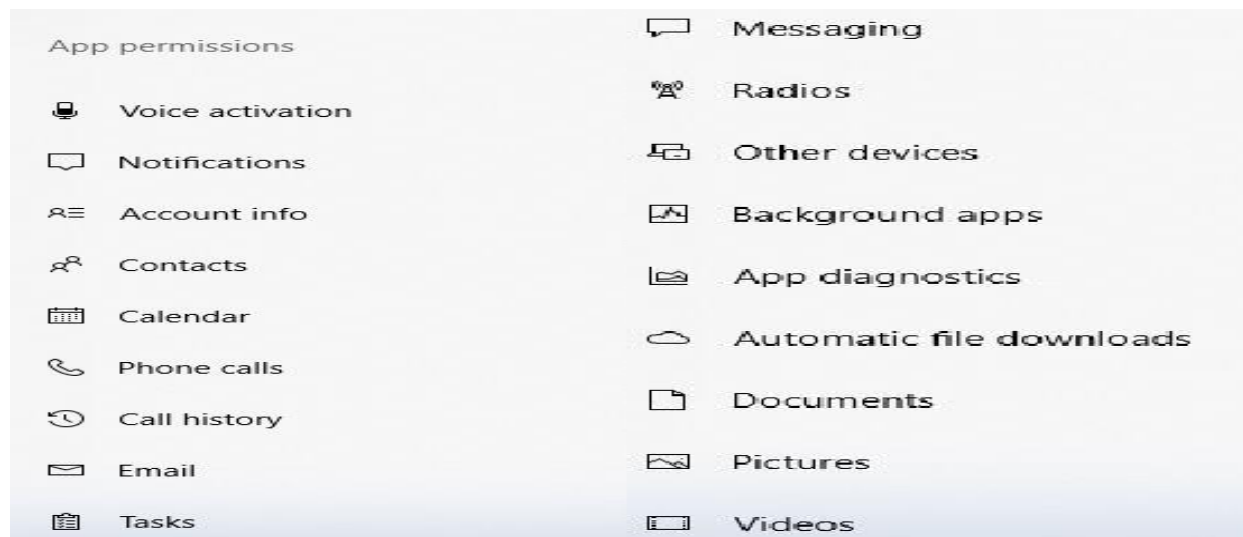
[Dismiss](#)

4. Permissions audit

Like your phone, your laptop has all sorts of permissions settings to prevent applications from accessing certain data, such as your contacts, your location, the camera, and more. Pop into this settings page occasionally to make sure no app is overreaching for permissions it doesn't need.

Open Settings > Privacy (or Privacy & security in Windows 11) and then scroll down to the "App permissions" section.

Go through each permission and ensure that apps aren't accessing data they don't need to function. Pay attention in particular to the "Location" section, which is one of the more personal permissions, and disable access for any app that doesn't make sense on that list.



5. Check your privacy settings

Windows tends to collect a lot of data about you, and there isn't always much you can do about it. But you can at least opt out of a few of the more egregious aspects:

Open Settings > Privacy (or Privacy & security on Windows 11) > General.

Consider disabling anything on this page you don't find helpful, especially the Let apps show me personalized ads by using my advertising ID option. If you do so, you'll still see ads, but instead of using details like your browser-app usage or your usage history to show you customize ads, apps will show you generalized ad.

General

Change privacy options

Let apps use advertising ID to make ads more interesting to you based on your app activity (Turning this off will reset your ID.)

☒ On

Let websites provide locally relevant content by accessing my language list

☒ On

Let Windows track app launches to improve Start and search results

☒ On

Show me suggested content in the Settings app

☒ On

6. User Account Control (UAC):

User Account Control (UAC) is a security feature in Windows that helps prevent unauthorized changes to your computer. It notifies you before changes are made to your PC that require administrator-level permission. Below are detailed steps for configuring and working with UAC in Windows 10. The process is similar in other versions of Windows.

Working with UAC:

When UAC is active, you'll encounter UAC prompts when performing actions that require administrator-level permission. Here's how to respond to these prompts:

Standard User Action:

If you're logged in as a standard user and you attempt to perform an action that requires administrator permissions (e.g., installing software), a UAC prompt will appear.

Admin Approval:

To proceed, click "Yes." This will require you to enter your administrator password or use another form of administrator authentication if you're not already logged in as an admin.

Credentials Prompt:

If you're not logged in as an administrator, you'll see a prompt asking for administrator credentials (username and password). Enter the necessary information.

Elevated Access:

Once you've provided the correct credentials, the action will be performed with elevated privileges, and the program will be allowed to make changes to your system.

Unauthorized Actions:

If you didn't initiate the action or if you're unsure about its legitimacy, click "No" or "Cancel" to prevent the action from proceeding.

UAC is a crucial security feature in Windows, as it helps prevent unauthorized changes to your system, reducing the risk of malware or other malicious software making harmful alterations. It's generally recommended to keep UAC enabled at the default or a higher level for optimal security.

Configuring User Account Control (UAC):

Access UAC Settings:

Press the Windows key on your keyboard or click the Start button.

Type "UAC" and press Enter. You'll see "Change User Account Control settings" in the search results. Click on it to open the UAC settings.

UAC Slider:

The UAC settings window has a slider with four levels:

Always notify: This is the most secure setting. It notifies you every time a program tries to make changes to your computer, even if it's a trusted action.

Notify me only when apps try to make changes to my computer (default): This is the default setting and provides a balance between security and usability. It notifies you when apps try to make changes but not for Windows settings changes.

Notify me only when apps try to make changes to my computer (do not dim my desktop): This is the same as the default setting but doesn't dim your desktop when UAC prompts appear.

Never notify: This setting turns off UAC. It is not recommended for security reasons.

Choose the UAC Level:

Slide the UAC slider to your desired level of notification. For most users, the default setting is appropriate.

Confirm Changes:

You may be prompted to enter your administrator password or provide confirmation if you're changing the UAC level to a more secure setting.

Apply and Restart:

Click "OK" to apply the changes and close the UAC settings window.

Depending on the changes you made, you may need to restart your computer for the new UAC settings to take effect.

UAC is a crucial security feature in Windows, as it helps prevent unauthorized changes to your system, reducing the risk of malware or other malicious software making harmful alterations. It's generally recommended to keep UAC enabled at the default or a higher level for optimal security.

Password and Account Policies:

- Press **Win + R**, type **secpol.msc**, and press Enter.
- In the Local Security Policy console, navigate to Security Settings > Account Policies.
- Set password policies such as "Password must meet complexity requirements" and "Maximum password age" to your desired values.

Firewall Configuration:

- Go to "Settings" > "Update & Security" > "Windows Security" > "Firewall & network protection."
- Click on "Advanced settings" to configure incoming and outgoing rules.
- Create rules to allow only necessary network communications.

AutoPlay

In Windows 10, AutoPlay is a feature that automatically launches an appropriate program to play or handle media files, open folders to view files, or perform other actions when you insert a removable media device (e.g., USB drive, CD/DVD) into your computer. Configuring AutoPlay security settings is important to prevent potential security risks associated with automatically running programs from external devices. Here's how to manage AutoPlay security settings:

1. Access AutoPlay Settings:

- Click on the Windows icon in the taskbar to open the Start menu.
- Type "AutoPlay" in the search bar, and select "AutoPlay settings" from the search results.

2. Configure AutoPlay Options:

- In the AutoPlay settings window, you'll find a series of drop-down menus next to different types of media and devices. These menus allow you to specify what Windows should do when each type of media or device is connected.
- For each media type (e.g., "Removable drive," "Memory card"), click the drop-down menu to select an action:

Take no action:

This option disables AutoPlay for the selected media type. Windows will do nothing when you insert a device of this type.

Open folder to view files (File Explorer):

Windows will open File Explorer to view the files on the device.

Ask me every time:

Windows will prompt you to choose an action every time you insert a device of this type.

Import photos and videos (Photos):

Windows will open the Photos app for photo and video media.

Play (default media player):

Windows will open the default media player for audio and video media.

3. Save Changes:

- After configuring the AutoPlay options for each media type, make sure to click the "Save" button to save your settings.

4. Additional AutoPlay Settings:

- Below the media type settings, you may find additional settings related to AutoPlay. For example, there might be a setting to "Use AutoPlay for all media and devices." You can toggle this setting on or off based on your preferences.
- By configuring AutoPlay settings, you can enhance the security of your Windows system by preventing potentially harmful programs from automatically running when you insert removable media devices. It's advisable to select the "Take no action" option or "Ask me every time" for media types that you don't regularly use to minimize security risks.

Activity 4:**Browser save cookies and password lab task****1. Managing Cookies:****Clearing Cookies:**

- Open your web browser.
- Access the browser's settings or preferences. This is usually found in the menu or three-dot icon at the top right corner.
- Navigate to the privacy or security settings.
- Look for an option to clear browsing data or cookies.
- Select the option to clear cookies.
- Choose the time range for which you want to clear cookies (e.g., last hour, last day, all time).
- Confirm and execute the action.

Blocking Cookies:

- Access your browser's settings or preferences.
- Navigate to the privacy or security settings.
- Look for an option related to cookies.
- Depending on the browser, you may have options to block all cookies, block third-party cookies, or customize cookie settings.
- Choose the desired option to block cookies.
- Adjust any additional settings according to your preferences.
- Save changes.

2. Managing Saved Passwords:**Viewing Saved Passwords:**

- Open your web browser.
- Access the browser's settings or preferences.

- Look for the section related to passwords or security.
- Find the option to manage saved passwords or view saved passwords.
- You may need to enter your system password or confirm your identity.
- Once accessed, you should see a list of websites with saved passwords.

Deleting Saved Passwords:

- Follow the steps to view saved passwords as mentioned above.
- Identify the website for which you want to delete the saved password.
- Select the website entry.
- Look for an option to delete or remove the saved password.
- Confirm the action.
- Repeat the process for any other passwords you wish to delete.

Activity 5

Install and Set Up GoPhish

Download GoPhish: Visit the *GoPhish website* and download the latest version suitable for your operating system.

Install GoPhish: Unzip the downloaded file.

Navigate to the GoPhish directory and run the executable file.

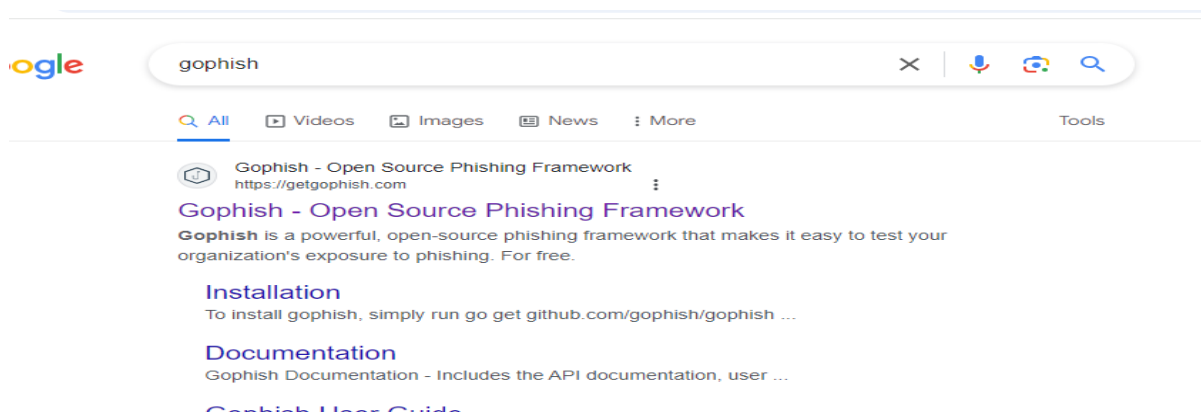
GoPhish will launch a web server, typically accessible via <http://localhost:3333>.

Access GoPhish Dashboard: Open a web browser and navigate to <http://localhost:3333>.

Log in using the default credentials (you should change these for security).

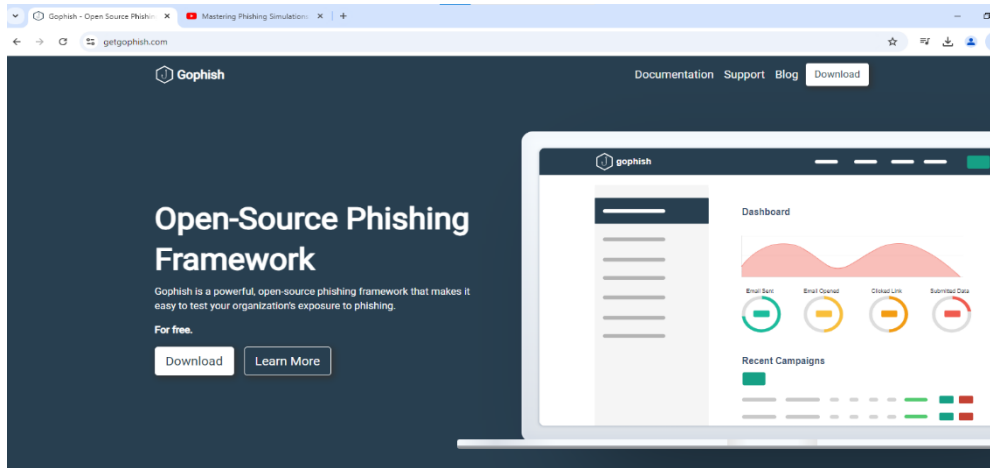
Step 1:

Open your web browser and then type 'gophish'.



Step 2:

Open the first link and then click download to download it.



Step 3:

After it is downloaded, scroll down and then select windows 64 file.

Enjoy

SHA256 Hash	Filename
9ed2f88d6582b798a3448fb51080ff782b40cec8fb4855895720e9f782f68511	gophish-v0.12.1-linux-32bit.zip
44f598c1eeb72c3b08fa73d57049022d96cea2872283b87a73d21af78a2c6d47	gophish-v0.12.1-linux-64bit.zip
f80786dcb2c4037d7f9d6c9e8e8655c5eeaa8e46ef4032bda704b9ce18e23491	gophish-v0.12.1-osx-64bit.zip
e6936b8a472c730dcb0da64024d82341806869af666fad10f8639e7f85b1b7e6	gophish-v0.12.1-windows-64bit.zip

Contributors



mcab and 29vivek

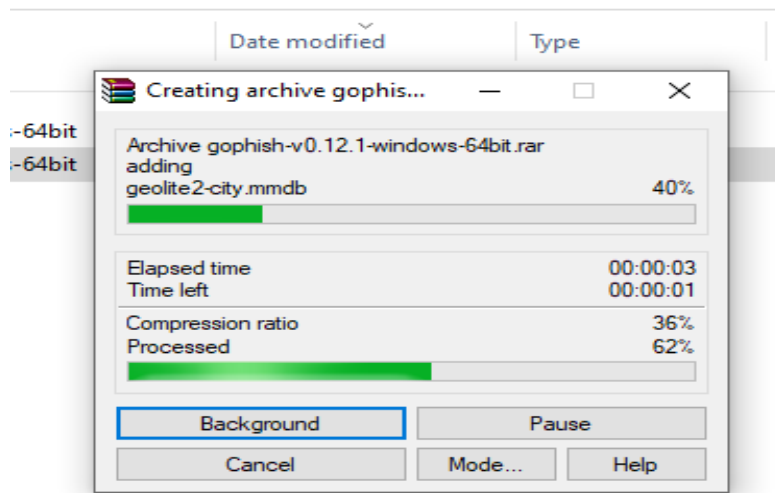
▼ Assets 6

gophish-v0.12.1-linux-32bit.zip	31.4 MB	Sep 14, 2022
gophish-v0.12.1-linux-64bit.zip	31.8 MB	Sep 14, 2022
gophish-v0.12.1-osx-64bit.zip	33.2 MB	Sep 14, 2022
gophish-v0.12.1-windows-64bit.zip	32.1 MB	Sep 14, 2022
Source code (zip)		Sep 14, 2022
Source code (tar.gz)		Sep 14, 2022

Be patient, it will take some time to download.

Step 4:

After it is downloaded, open it's location in folders and then extract files.



Step 5:

After files are extracted, click on the windows 64-bit folder.

▼ Today (2)

	gophish-v0.12.1-windows-64bit	9/18/2024 9:53 PM	WinRAR ZIP archive	32,900 KB
	gophish-v0.12.1-windows-64bit	9/18/2024 9:54 PM	File folder	

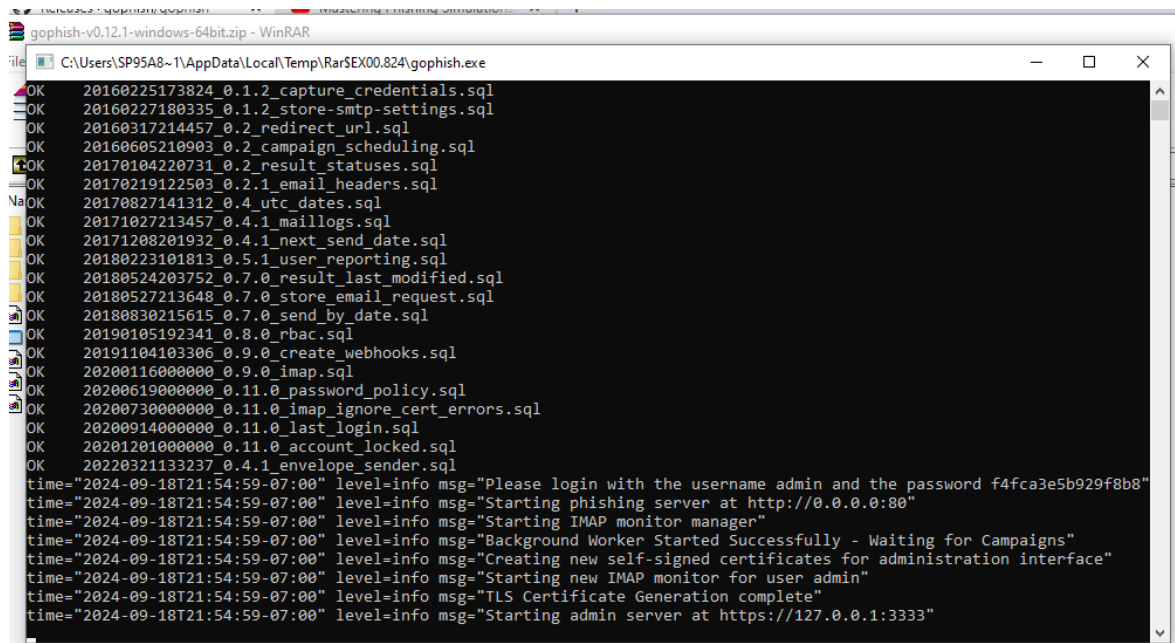
Step 6:

After opening it click the go phish file

	db		Folder	9/14/2022 10:4...	
	static		Folder		
	templates		Folder	9/14/2022 10:4...	
	config.json	460	230	File json	9/14/2022 10:4... 80248C26
	gophish.exe	34,590,314	11,712,887	Application	9/14/2022 10:4... 7D4F62E0
	LICENSE	1,115	661	File	9/14/2022 10:4... 24A2A343
	README.md	3,285	1,641	File md	9/14/2022 10:4... A3522FD4
	VERSION	7	7	File	9/14/2022 10:4... DA913A30

Step 7:

After that the command line interface will appear:
It will show the username= admin and password



Step 8:

```
ion complete"
t https://127.0.0.1:3333"
```

Copy the website link and then open it in a new window.

Step 9:

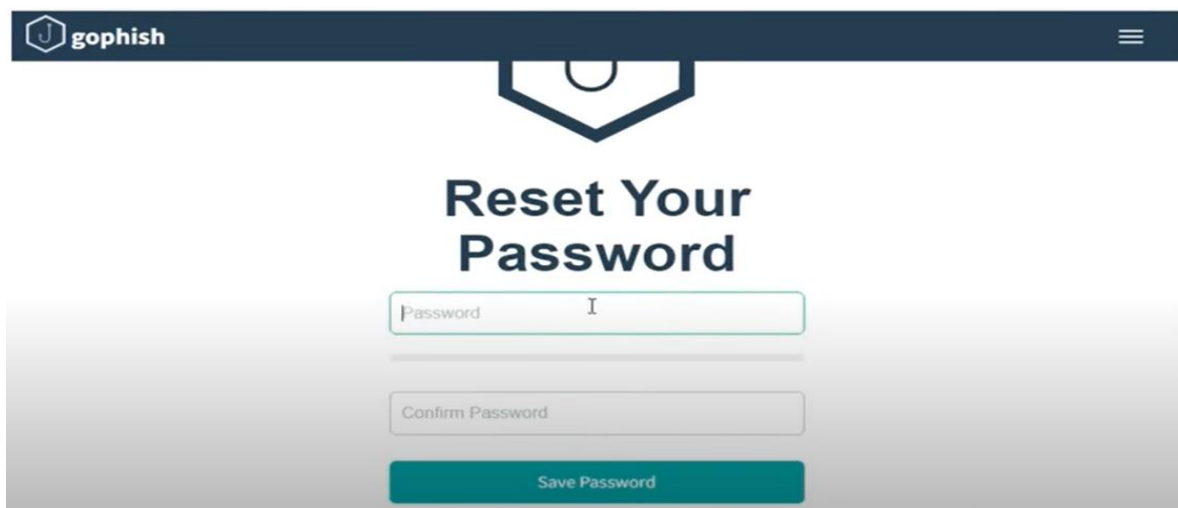
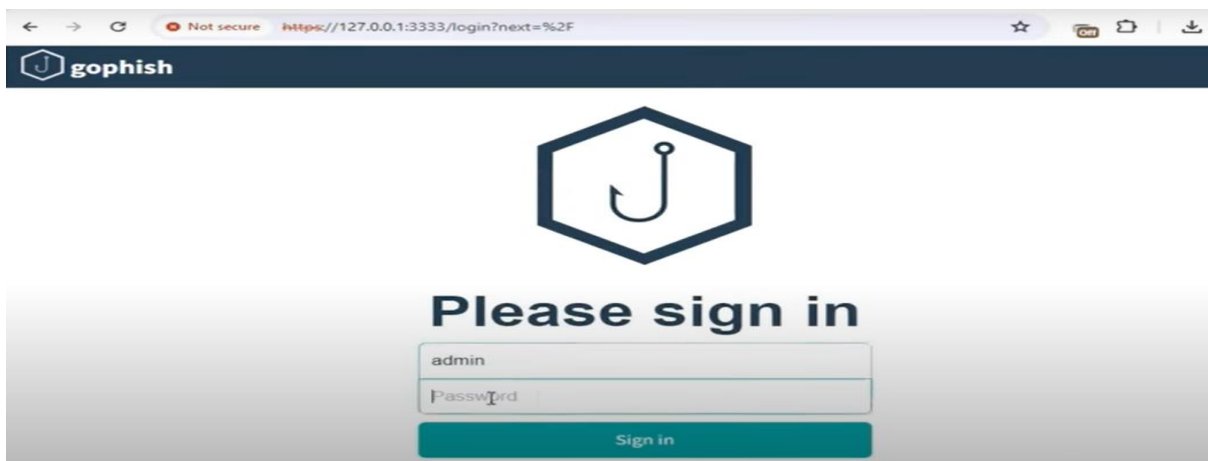
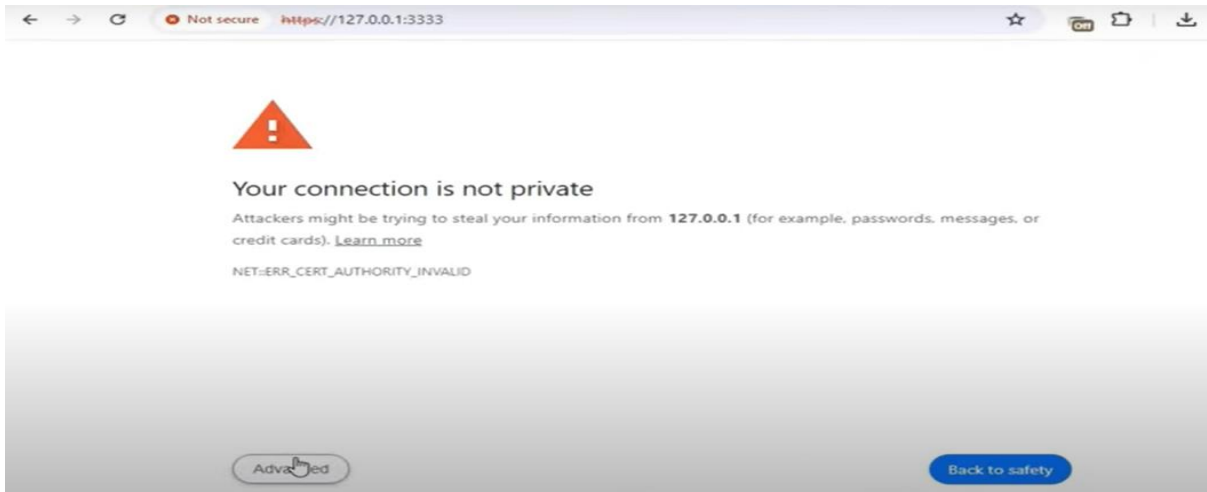
Copy the password

```
e password f4fca3e5b929f8b8"
80"
```

Step 10:

- Enter the website link into google.com and click on advanced to proceed.
- Go to gophish site, and after entering username and password, it will asked you to reset your password.
- Open the dashboard, and then go to sending profiles page enter the required entries then save it.

- After that send the test email to the person you want to send the email to check it configured correctly

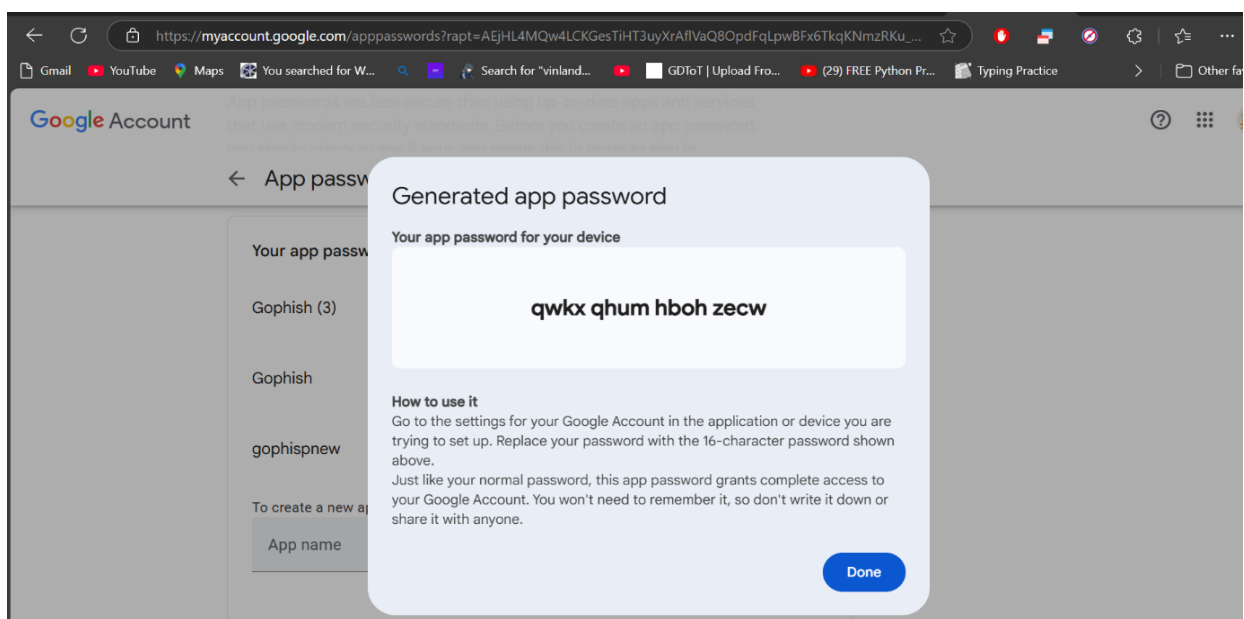
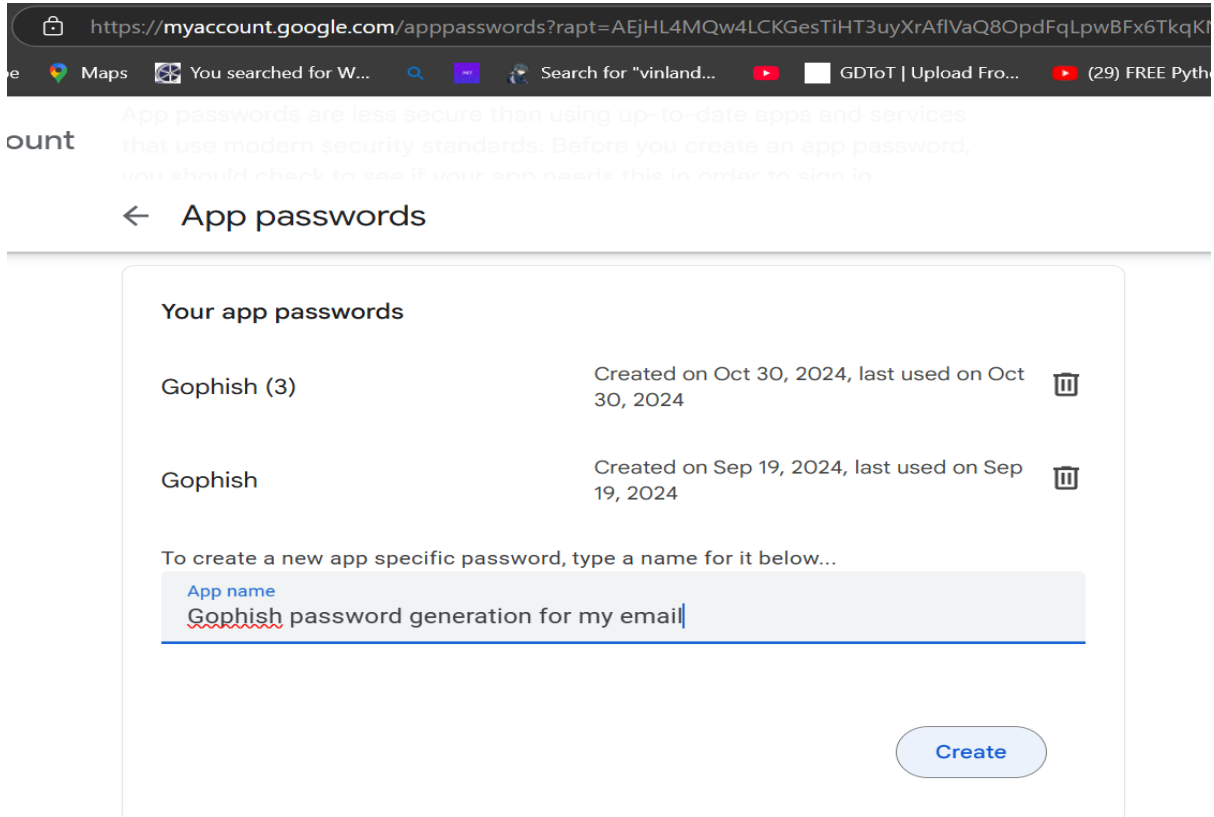


Step10.1

Go to this link for create your own password for your email which you enter into the gophish.

Link: myaccount.google.com/apppasswords

After you click this link, create a session and it will give you your own password which will only be applicable for Gophish. Remember it as you will not be able to view it again.



Step10.2

Now go to Google and enter smtp server gmail. For TLS port, it is 587. For SSL port, it is 465. You have to enter it in front of gmail(**smtp.gmail.com:587**)

Configuring Gmail SMTP Server

Once you have the app password, you can configure your email client or application to use Gmail's SMTP server. Here are the necessary details:

- **SMTP Server Address:** `smtp.gmail.com`
- **Port:** 587 for TLS 465 for SSL
- **Authentication:** Yes
- **Username:** Your full Gmail address (e.g., `yourname@gmail.com`)
- **Password:** The 16-digit app password you generated ².

Example Configuration in Outlook

Step 10.3 Now fill the form accordingly.

Edit Sending Profile

Name:

Interface Type:

SMTP From:

Host:

Username:

Password:

gophish

Dashboard

Campaigns

Users & Groups

Email Templates

Landing Pages

Sending Profiles

Account Settings

User Management

Webhooks

Admin

Admin

Sending Profiles

+ New Profile

Show 10 entries

Search:

Name	Interface Type	Last Modified Date	
Sells Manager	SMTP	October 15th 2024, 7:47:50 am	
test-1	SMTP	October 15th 2024, 7:34:25 am	

Showing 1 to 2 of 2 entries

Previous

1

Next

Step 11:

Create a profile and request to send a test email.

No data available in table

Showing 0 to 0 of 0 entries

Previous

Next

Send Test Email

Cancel

Save Profile

Step 12:

Send it to other user this phish email to test it, its work and then Confirm it!

Send Test Email



✓ Email Sent!

Send Test Email to:

Shoaib

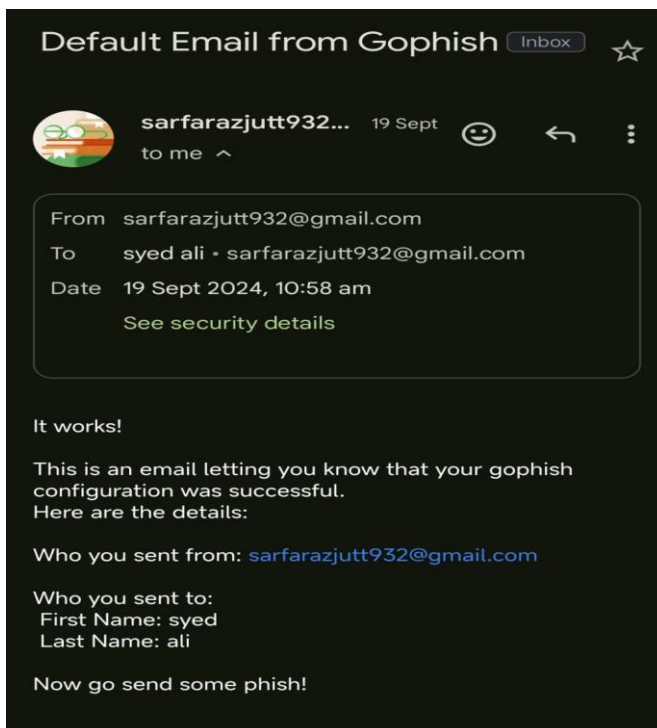
Shafique

shoaibshafique621@gmail.c

student

Cancel

Send



Step 13:

Navigate to the landing page and import the website.

Landing Pages

✔ Page added successfully!

+ New Page

Show entries

Name ▲

Last Modified Date ▼

google.com

September 21st 2024, 9:46:26 pm

Step 14:

Create an Email Template.

Email Templates

+ New Template

Show entries

Name ▲

Modified Date ▼

temp1

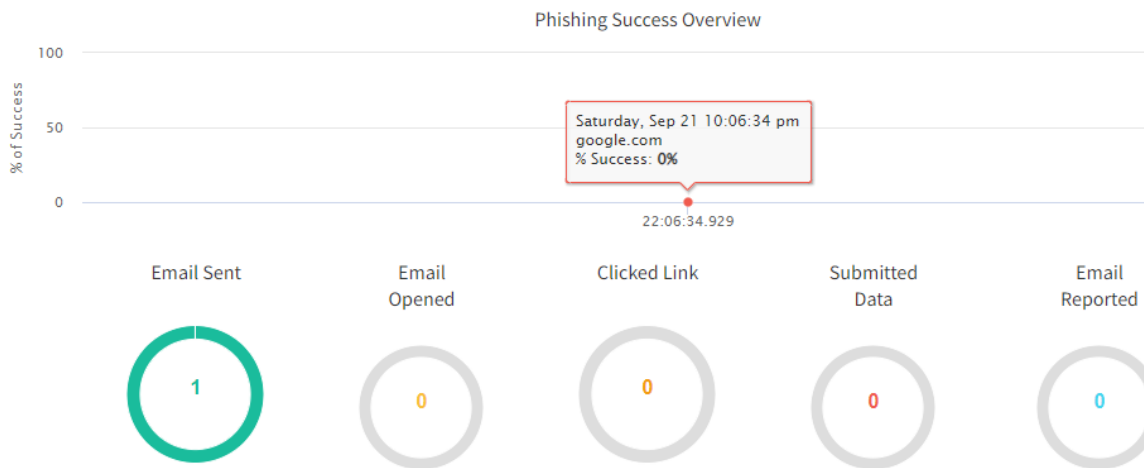
September 21st 2024, 9:58:33 pm

Step 15:

- Create user group and launch campaign.
- Then send that phishing email to desired or specific user to gain credentials for testing purposes (social engineering attacks on your organization member) to protect your systems and sensitive data.

Step 16:

Dashboard



- Open that link.
- It will ask your username and password.
- Enter it and confirm it from the dashboard.
- There you will see the results.

Step 3: Craft a Phishing Email

Create an Email Template: Navigate to the “Email Templates” section. Click “New Template” to start crafting your phishing email.

Design the Email:

Subject: Choose a subject that will entice the recipient to open the email. For example, “Urgent: Account Verification Needed” or “Your Invoice is ready”.

Body: Write a convincing email body. It should include:

Greeting: Start with a generic or personalized greeting (e.g., “Dear User”).

Message: Create a sense of urgency or curiosity. For instance, “We detected unusual activity in your account. Please verify your identity by clicking the link below.”

Call to Action: Include a phishing link (this will be a placeholder for your actual campaign URL). For example, “Click here to verify your account.”

Footer: Close the email with a signature that matches the organization you are impersonating.

Insert Variables: GoPhish allows you to personalize emails using variables like {{.First Name}} or {{.Last Name}}, which will be replaced with actual data from your target list.

Include a Phishing Link: Add a hyperlink that will direct users to a fake landing page or a cloned website (this can be created using GoPhish's landing page feature). Ensure the link text appears legitimate, like "Login to Your Account" or "View Invoice."

Preview and Test: Preview your email to see how it will look to recipients. Send a test email to ensure everything appears as intended.

Graded Lab Tasks

Note: The instructor can design graded lab activities according to the level of difficulty and complexity of the solved lab activities.

Lab Task 1

To enhance browser security by configuring privacy settings, managing cookies, and securing stored passwords, Students will learn to configure browser settings for enhanced security and manage passwords effectively to protect against online threats.

Lab Task 2

To secure email communications by implementing encryption, recognizing phishing emails, and enabling two-factor authentication (2FA), Students will enhance their ability to secure email communications, recognize phishing attempts, and understand the importance of 2FA. Moreover, try some online email phishing tool which crafts a simple email message and send it to one or several recipients using a specified mail server.

Lab Task 3

Assume that you receive an email which you suspect to be generated by some fake source. Provide an analysis of a phishing email with an explanation of why it is suspicious.

Lab Task 4

To secure a Windows operating system by configuring security settings, applying updates, and using built-in security tools, Students will learn to harden a Windows operating system by applying security updates, configuring the firewall, and using built-in security tools to protect against various threats.

Lab Task 5

Configure security settings using Group Policy or Local Security Policy. Apply Windows security baselines if available. Provide screenshots of Group Policy or Local Security Policy settings. Include a summary of the applied security baselines.

Lab 02

Introduction to Basic Cryptography

Objective:

The objective of this module is to introduce participants to the foundational concepts of cryptography and its critical role in data security. By understanding and applying basic cryptographic techniques, participants will be equipped to protect sensitive information and recognize the significance of secure communication protocols. This module aims to build a solid foundation in cryptography, preparing participants to handle security challenges in various IT and cybersecurity roles.

Activity Outcomes:

Students will understand the fundamental concepts of cryptography. We will perform a password security activity that is the ability to create and manage strong, resilient passwords that effectively protect against unauthorized access. We will check password security in KeePass which is a free, open-source password manager that securely stores and manages passwords using strong encryption. We will learn how to crack password hashes using Hashcat

Instructor Note:

Ensure participants have cryptographic tools and software. Provide clear explanations and practical demonstrations. Encourage hands-on practice and support understanding of cryptographic principles and data security.

1) Useful Concepts:

Cryptography is the science of securing communication and data through techniques like encryption and decryption to protect information from unauthorized access. Password security and cryptography are vital for protecting sensitive data by converting it into unreadable formats using algorithms like hashing and encryption. Strong password policies, combined with cryptographic techniques such as salting and key derivation functions, prevent unauthorized access and enhance data security. These practices are essential for safeguarding user credentials and maintaining secure systems.

Strength of different passwords:

Welcome to the world of cryptography, where secrets are kept safe through the power of mathematics and clever algorithms. In this manual, we'll introduce you to the fundamentals of cryptography, focusing specifically on understanding the strength of different passwords. Let's dive in.

Understanding Cryptography:

Cryptography is the science of secure communication in the presence of adversaries. It involves creating codes and ciphers to protect sensitive information from unauthorized access.

Cryptography relies on various techniques such as encryption, decryption, hashing, and digital signatures to achieve its goals.

Importance of Strong Passwords:

Passwords are the first line of defence in securing your digital accounts.

Strong passwords are essential to prevent unauthorized access to your accounts and protect your personal information.

Strength of Passwords:

The strength of a password is determined by its complexity and length.

A strong password should be:

Long: The longer the password, the harder it is to crack. Aim for passwords that are at least 12 characters long.

Complex: Include a mix of uppercase and lowercase letters, numbers, and special characters (!, @, #, etc.).

Unpredictable: Avoid using easily guessable information such as common words, birthdays, or sequential patterns.

Password Creation Tips:

Avoid using dictionary words or common phrases. Consider using passphrase instead of a single word. Passphrases are longer and easier to remember. Use a combination of letters, numbers, and special characters. Don't use the same password for multiple accounts. Each account should have a unique password. Regularly update your passwords to maintain security.

Password Strength Assessment:

There are online tools and password strength meters available that can help you assess the strength of your passwords.

These tools analyses factors such as length, complexity, and predictability to provide an estimation of how secure your password is.

Best Practices:

Enable two-factor authentication (2FA) whenever possible. 2FA adds an extra layer of security by requiring a second form of verification, such as a code sent to your phone. Use a reputable password manager to securely store and manage your passwords. Be cautious of phishing attempts and never share your passwords with anyone.

Hashcat

Hashcat is a powerful and versatile password recovery tool widely used in cybersecurity for cracking hashed passwords. Known as the world's fastest password-cracking utility, Hashcat supports various attack modes, including brute-force, dictionary, and hybrid attacks. It can handle a wide range of hash types, such as MD5, SHA-1, and bcrypt, making it essential for penetration testers and security professionals. Hashcat leverages the processing power of GPUs to accelerate the cracking process, providing both efficiency and scalability. Its open-source nature and robust community support make it a go-to tool for password auditing and recovery.

KeePass

KeePass is a free, open-source password manager that securely stores and organizes passwords using strong encryption. It offers a simple, user-friendly interface, making it easy to manage multiple accounts. With features like password generation and auto-fill, KeePass enhances both security and convenience. Trusted by individuals and organizations, it's a reliable tool for protecting sensitive credentials.

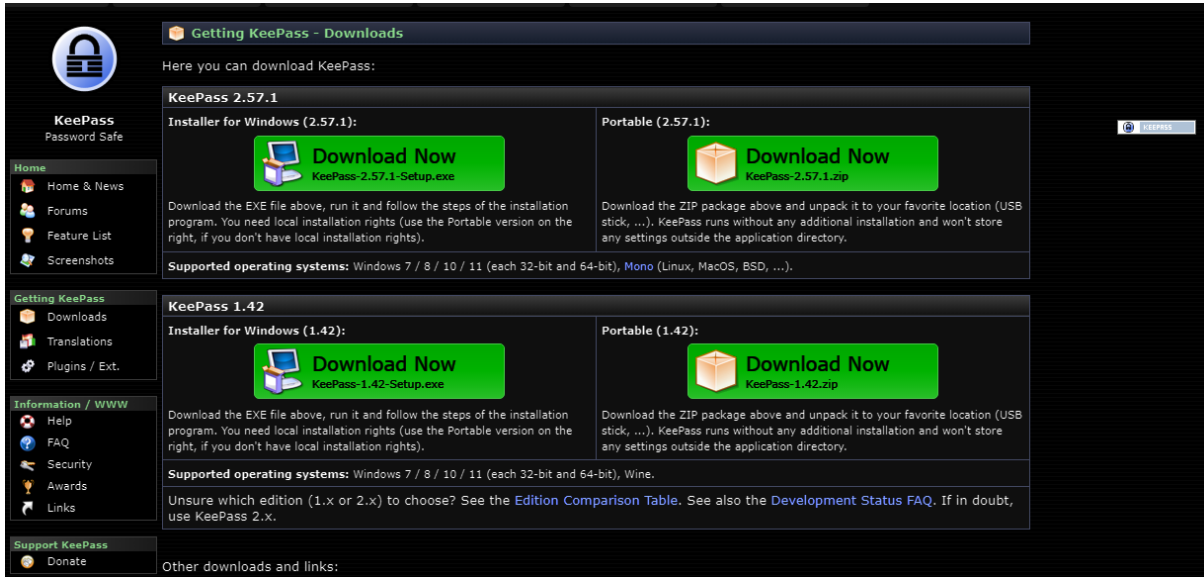
2) Solved Lab Activities (Allocated Time 3 Hr.)

<i>Sr.No</i>	<i>Allocated Time</i>	<i>Level of Complexity</i>	<i>CLO Mapping</i>
<i>Activity 1</i>	<i>20 mins</i>	<i>Easy</i>	<i>CLO-3</i>
<i>Activity2</i>	<i>20mins</i>	<i>Easy</i>	<i>CLO-3</i>
<i>Activity3</i>	<i>20mins</i>	<i>Medium</i>	<i>CLO-3</i>
<i>Activity4</i>	<i>20mins</i>	<i>Medium</i>	<i>CLO-3</i>
<i>Activity5</i>	<i>20mins</i>	<i>Medium</i>	<i>CLO-3</i>
<i>Activity6</i>	<i>20mins</i>	<i>Medium</i>	<i>CLO-3</i>
<i>Activity7</i>	<i>20mins</i>	<i>Medium</i>	<i>CLO-3</i>
<i>Activity8</i>	<i>20mins</i>	<i>Medium</i>	<i>CLO-3</i>
<i>Activity9</i>	<i>20mins</i>	<i>Medium</i>	<i>CLO-3</i>

KeePass

Installation:

Download KeePass from [here](#).



Master Key:

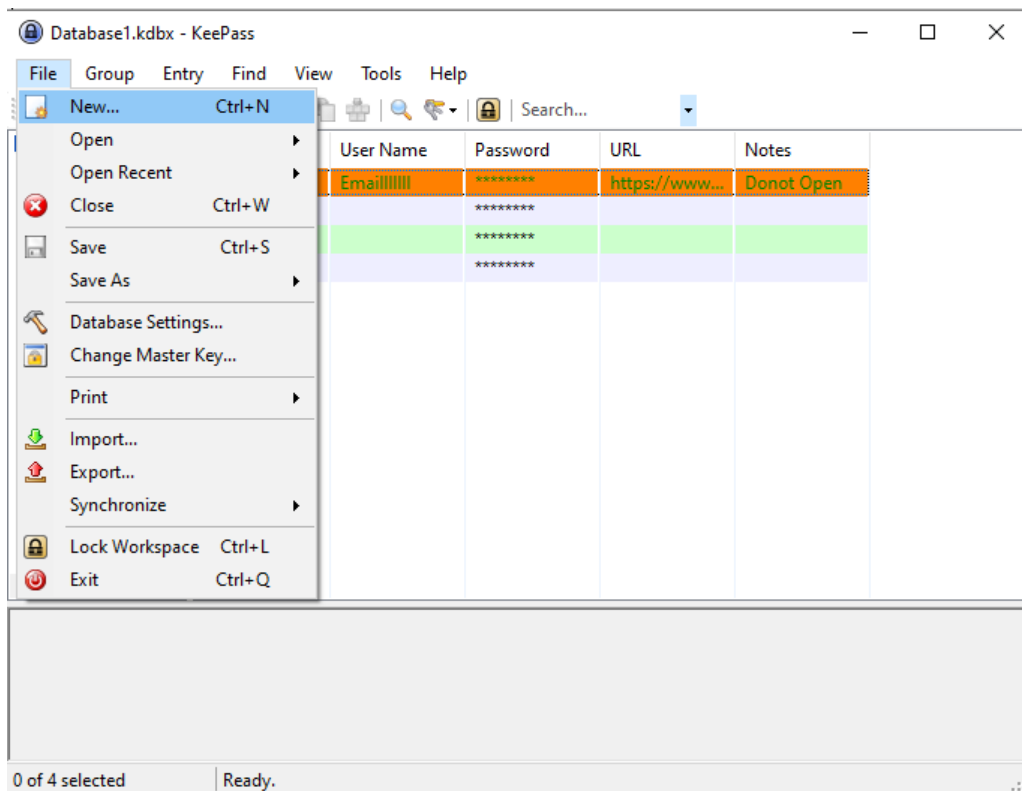
It is the Key through which the database gets encrypted or decrypted. It is derived from the Master Password or a Key File or both.

Master Password:

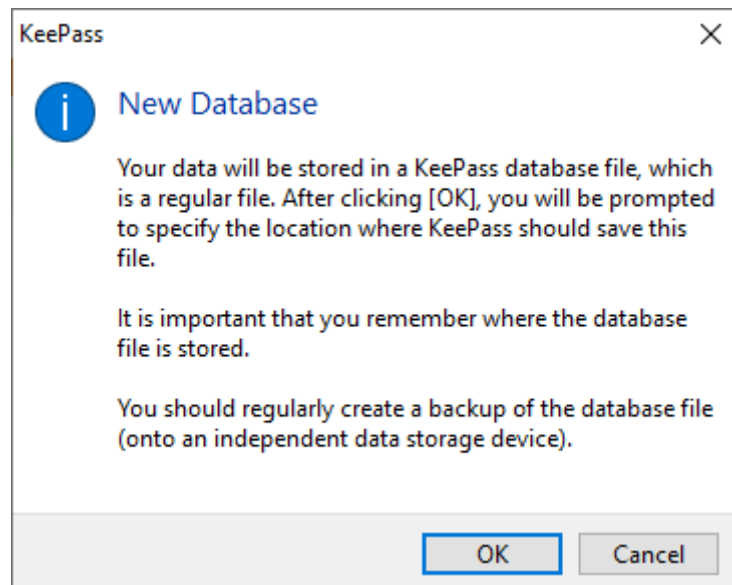
It is the fixed number of string that the user should remember and has to enter in order to access their database.

Activity 1: Generate Database

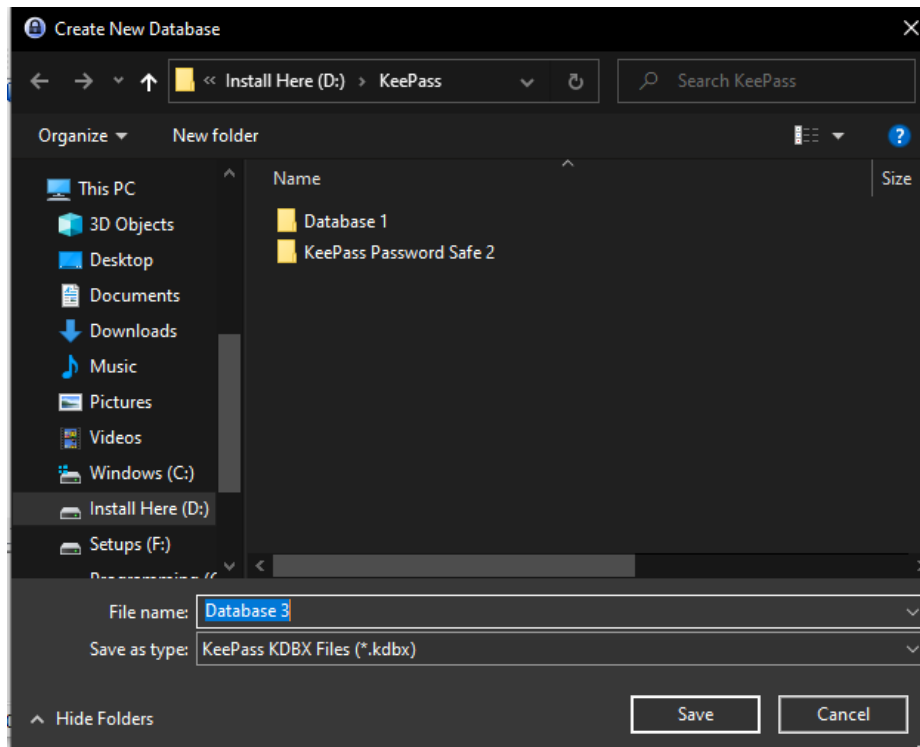
Step 1: Click on File and then on New. You can also press Ctrl+N.



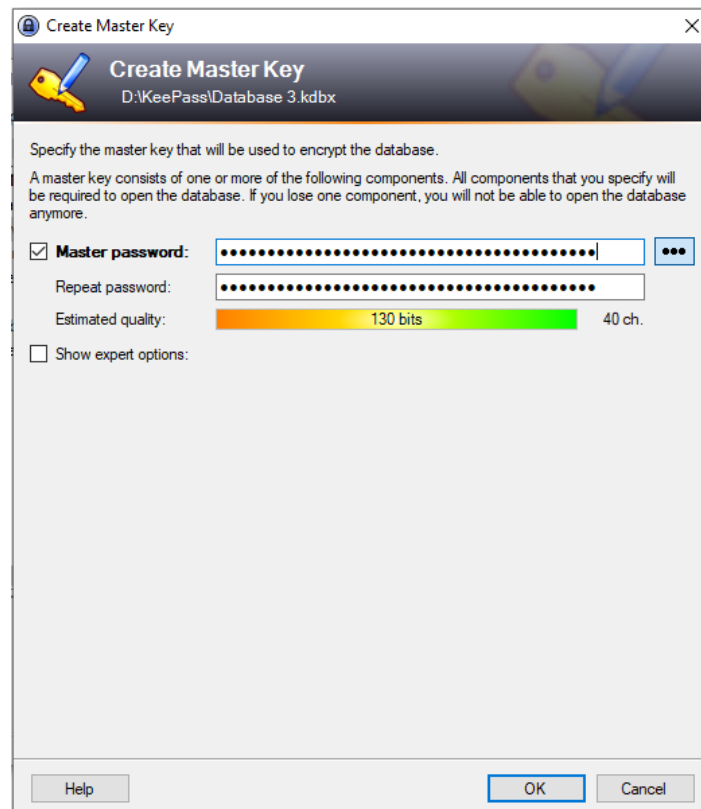
Step 2: Click Ok.



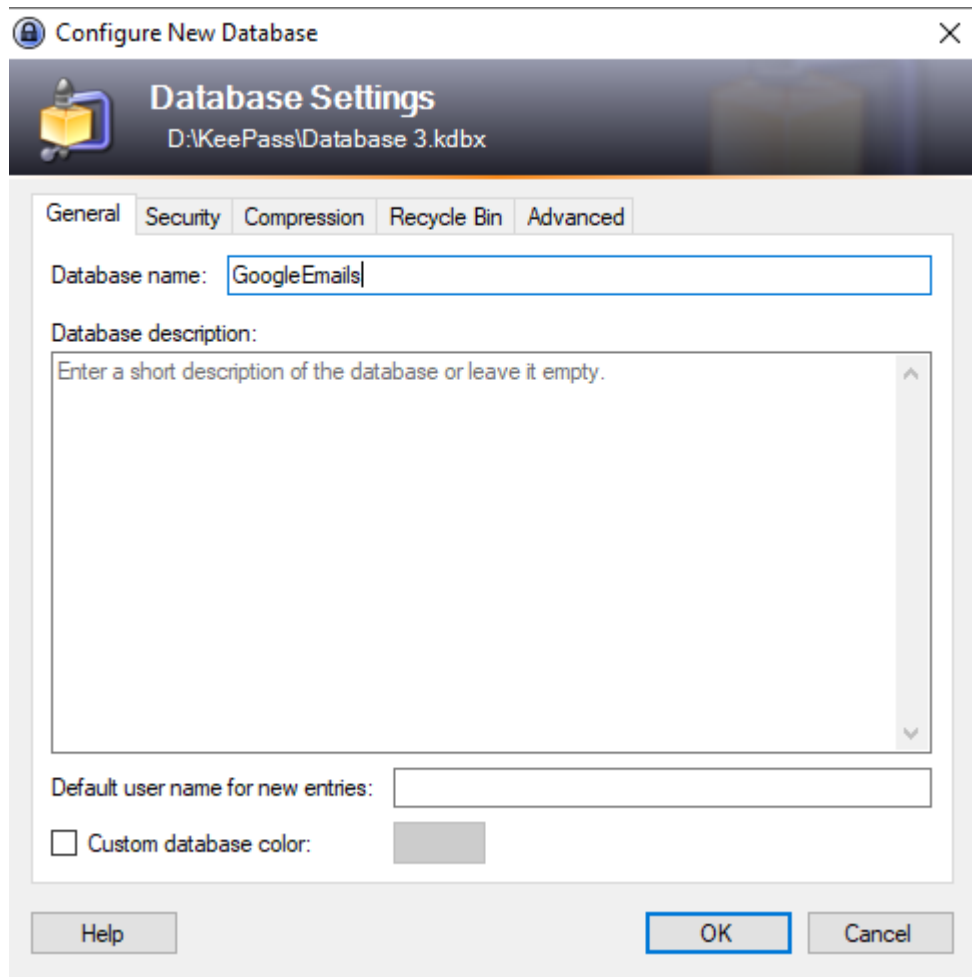
Step 3: Give it a Name and a Directory to store it.



Step 4: Now give it a Master Key. Best Practice is to have a Large Key that would be difficult to brute force.



Step 5: Now give it a Name.



Configure New Database

Database Settings
D:\KeePass\Database 3.kdbx

General Security Compression Recycle Bin Advanced

Database name:

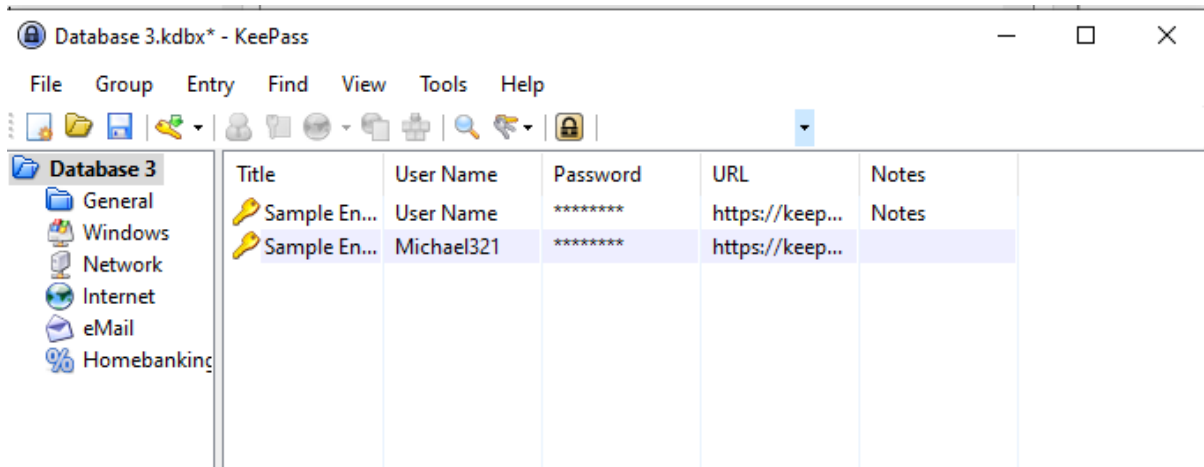
Database description:
Enter a short description of the database or leave it empty.

Default user name for new entries:

☐ Custom database color:

Help OK Cancel

Step 6: Your database will be created.



Database 3.kdbx* - KeePass

File Group Entry Find View Tools Help

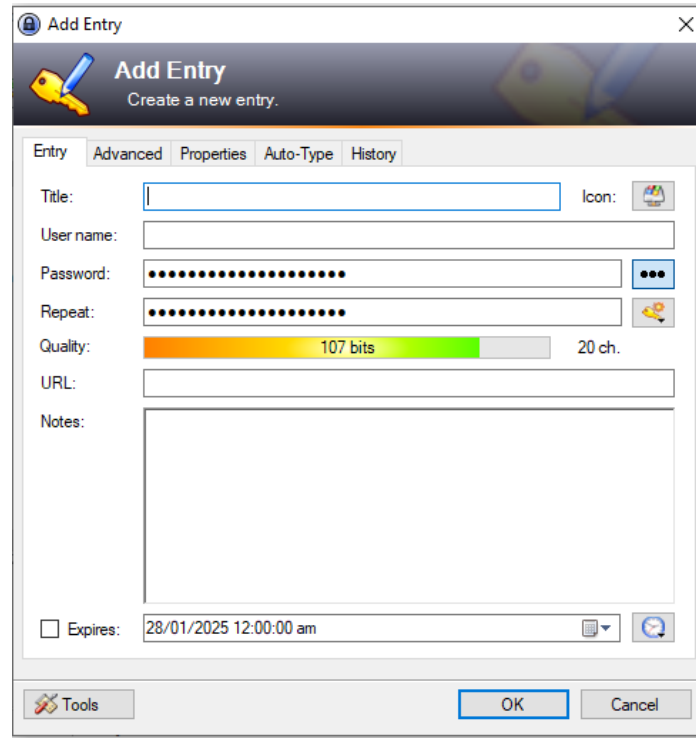
Database 3

- General
- Windows
- Network
- Internet
- eMail
- Homebanking

Title	User Name	Password	URL	Notes
Sample En...	User Name	*****	https://keep...	Notes
Sample En...	Michael321	*****	https://keep...	

Activity 2: Generate Keys

Step 1: Press Ctrl+I on your Keyboard while being in the Database and the group where you want to store the Key. This popup will appear.

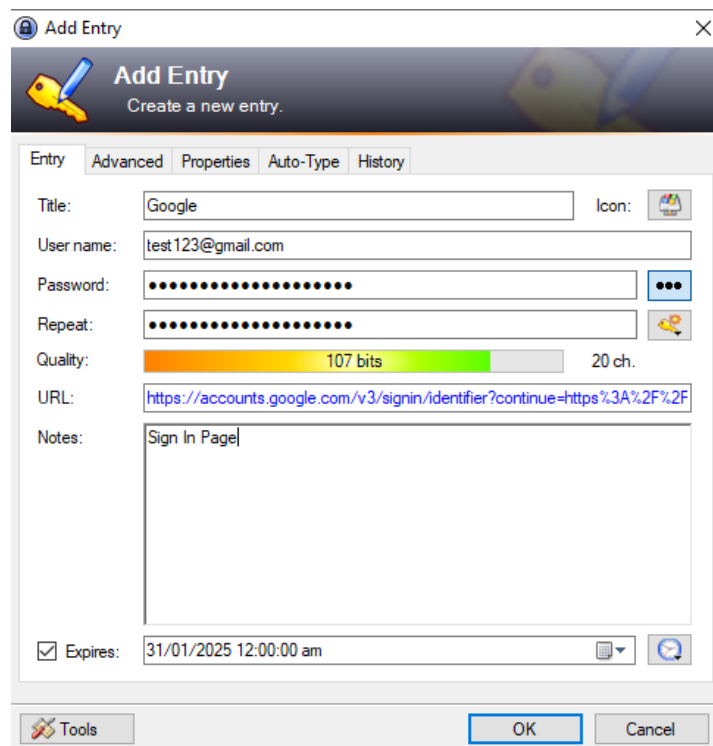


The 'Add Entry' dialog box is shown with the 'Entry' tab selected. The fields are as follows:

- Title: (empty)
- User name: (empty)
- Password: (masked with dots)
- Repeat: (masked with dots)
- Quality: 107 bits (indicated by a green bar)
- URL: (empty)
- Notes: (empty text area)
- Expires: ☐ 28/01/2025 12:00:00 am

Buttons at the bottom: Tools, OK, Cancel.

Step 2: Here you can add the Title, Username and Password of your choice. By Default, a strong password would be given to you.

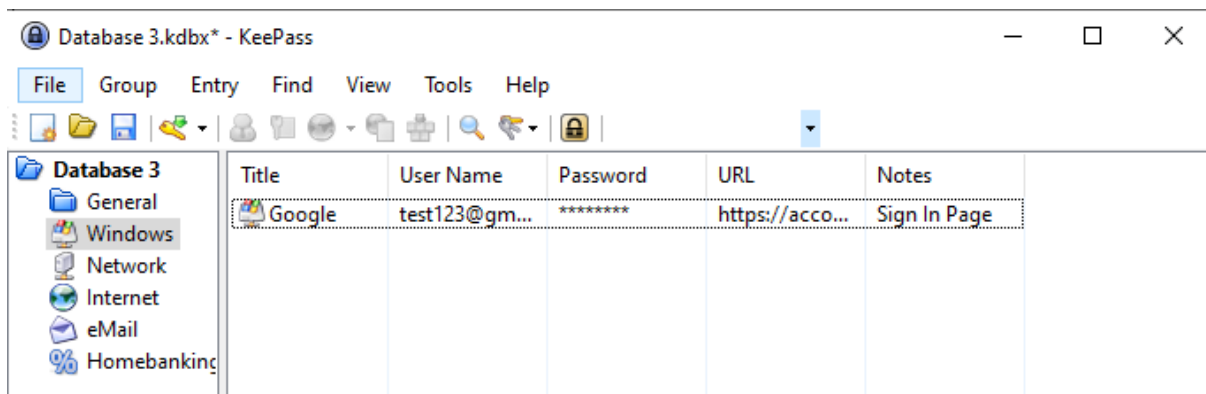


The 'Add Entry' dialog box is shown with the 'Entry' tab selected. The fields are filled with example data:

- Title: Google
- User name: test123@gmail.com
- Password: (masked with dots)
- Repeat: (masked with dots)
- Quality: 107 bits (indicated by a green bar)
- URL: <https://accounts.google.com/v3/signin/identifier?continue=https%3A%2F%2F>
- Notes: Sign In Page
- Expires: ☒ 31/01/2025 12:00:00 am

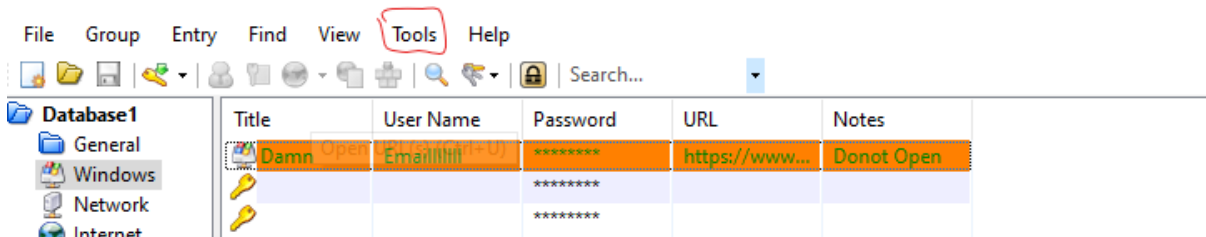
Buttons at the bottom: Tools, OK, Cancel.

Step 3: Your Entry would be made in the Database.

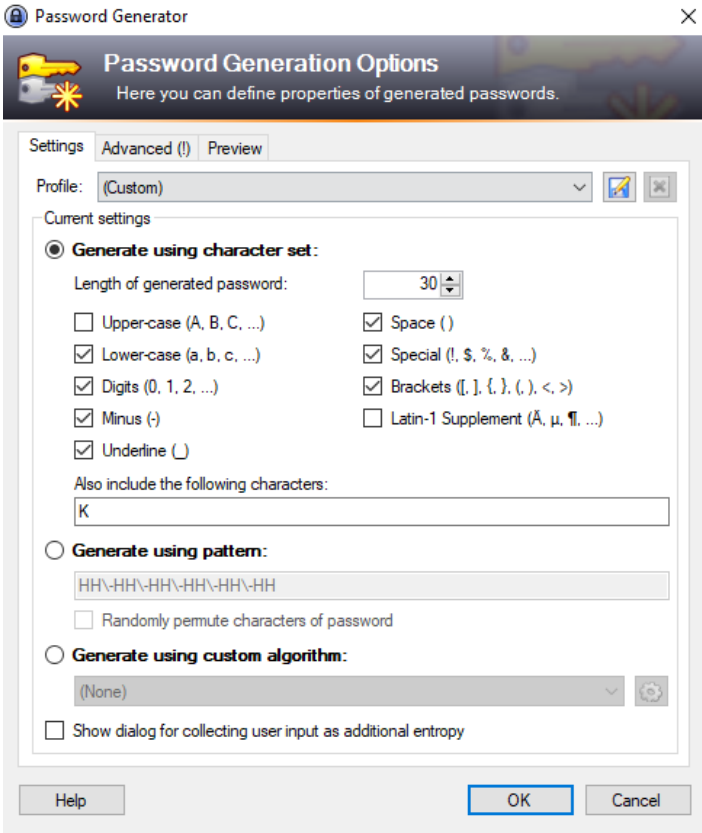


Activity 3: Password Generator

1. Go to Tools.

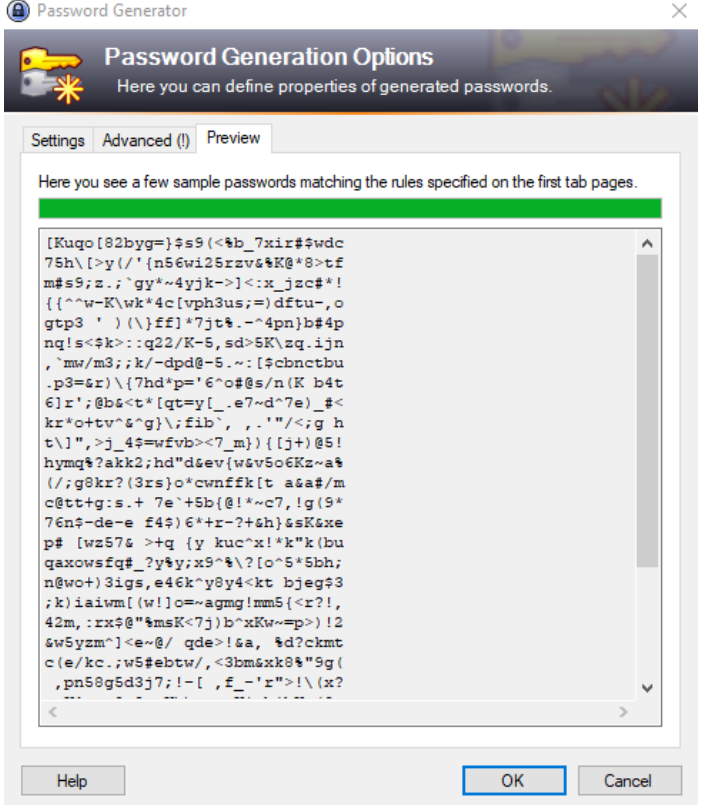


2. Then Generate Password and choose Password of your liking.



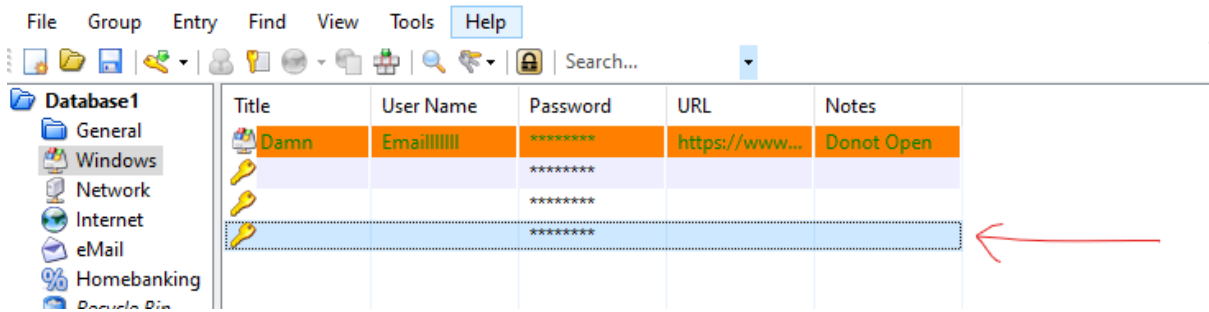
The screenshot shows the 'Advanced' tab of the Password Generator. The 'Profile' is set to '(Custom)'. Under 'Current settings', the 'Generate using character set' option is selected. The 'Length of generated password' is 30. The following character sets are checked: Upper-case (A, B, C, ...), Lower-case (a, b, c, ...), Digits (0, 1, 2, ...), Minus (-), Underline (_), Space (), Special (!, \$, %, &, ...), Brackets ([,], {, }, (,), <, >), and Latin-1 Supplement (Ä, µ, ¶, ...). The 'Also include the following characters' field contains 'K'. The 'Generate using pattern' and 'Generate using custom algorithm' options are not selected. The 'Show dialog for collecting user input as additional entropy' checkbox is also unchecked. The 'OK' button is highlighted.

3. You can also view what the password would look like in Preview.



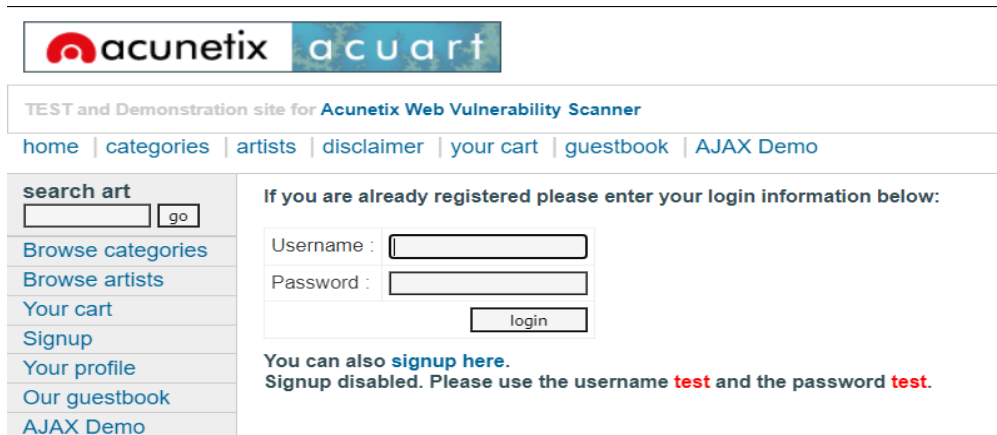
The screenshot shows the 'Preview' tab of the Password Generator. It displays a sample password: [Kuqo{82byg=\$s9(<%b_7xir#\$wdc75h\ [>y /' {n56wi25xv6%K@*8>tfm#s9; z.;`gy^~4yjk->]<:x\_jzc\*! { {^~w-K\wk\*4c[vph3us;=) dftu-, o gtp3 ' ' ) (\}ff] \*7jt%-^4pn}b#4p nq!s<\$k>.:q22/K-5, sd>5K\zq.ijn ,`mw/m3;;k/-dpd@-5.~: [\$cbnctbu .p3=6x)\{7hd*p='6^o#@s/n(K b4t 6)x';@b6<t* [qt=y[_ .e7~d^7e)_#< kr*o+tv^6^g)\;fib`, ,.'"/<;g h t\],">j_4\$=wfvb><7_m)} { [j+)@5! hymq?ak2;hd"d6ev{w6v5o6Kz~a% (/;g8kr? (3xs)o*cwnffk[t a6a#/m c@tt+g:s.+ 7e^+5b{!~*~c7,!g(9* 76n\$-de-e f4\$) 6*+r-?+6h}\$sK6xe p# [wz576 >+q {y kuc^x!*k" k(bu qaxowsfq#_?y\$y;x9^% \? [o^5^5bh; n@wo+) 3igs,e46k^y8y4<kt bjeg\$3 ;k)iaiw[m(w!)o~agmg!mm5{<x?! , 42m,.rx\$@ "%msK<7j)b^xKw~=>)!2 6w5yzm^]<e~/ qde>!6a, %d?ckmt c(e/kc.;w5#ebtw/, <3bm6xk8%"9g(,pn58g5d3j7;![,f_-'r">!\(x? < >]. The 'OK' button is highlighted.

4. Click Ok to Create the Password.

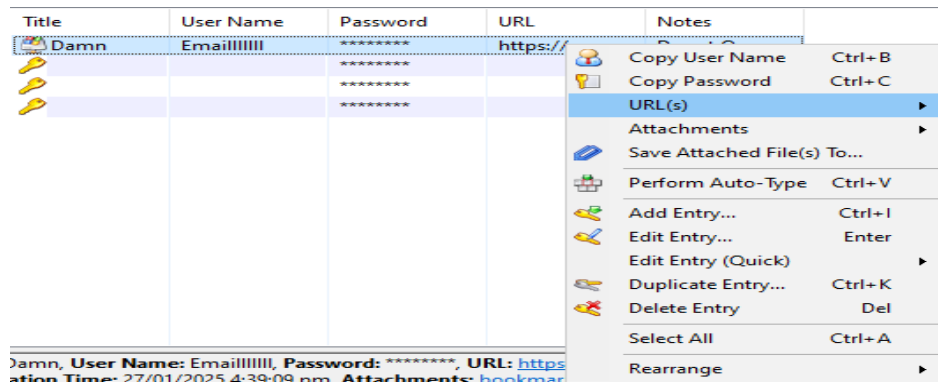


Activity 4: Auto Type

1. Click where you want to enter the Email and Password.



2. Now go to KeePass and Right-Click on the Key that you want to Enter.



3. Now Click on Perform Auto-Type. It will input the Keys itself into the Login.

TEST and Demonstration site for Acunetix Web Vulnerability Scanner

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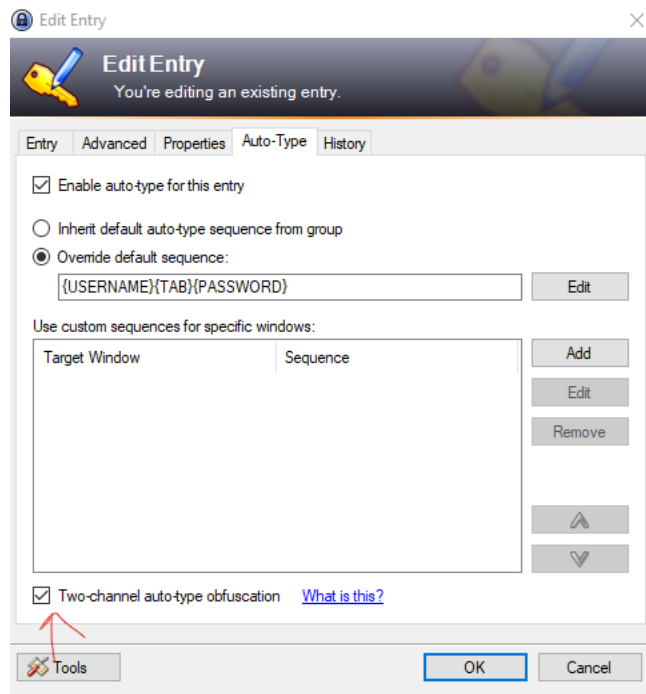
If you are already registered please enter your login information below:

Username :

Password :

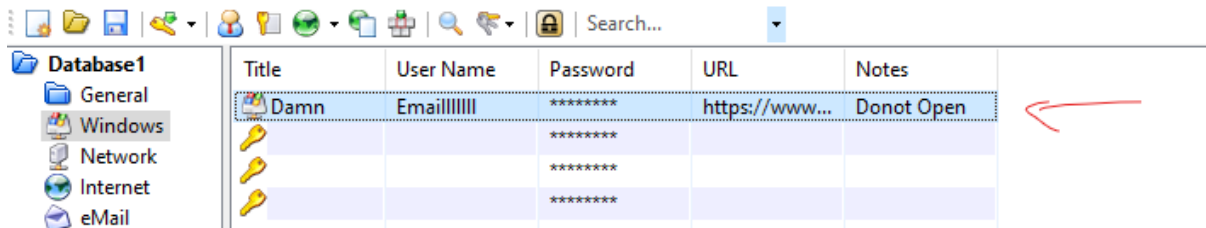
You can also [signup here](#).
 Signup disabled. Please use the username **test** and the password **test**.

Moreover, you can also edit what the AutoType should do in the Edit Entry and in AutoType tab and customize it for your own interest. One thing to note is that you should turn on Two-channel auto-type obfuscation as it renders keyloggers useless.

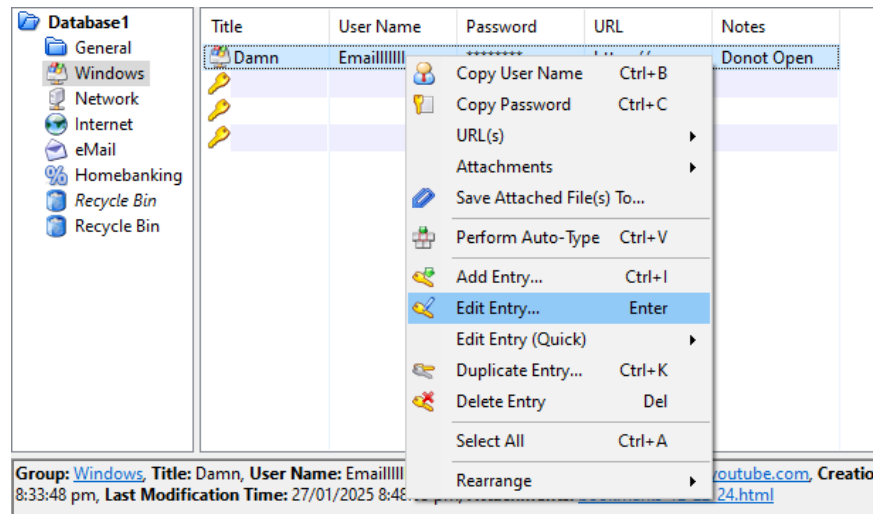


Activity 5: Storing Strings

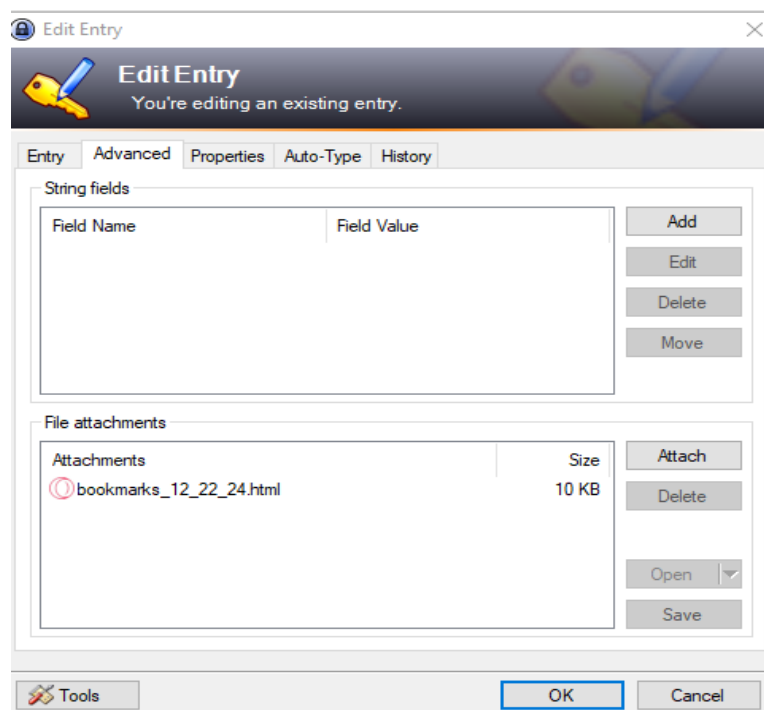
1. Click on the Email & Password where you want to store Strings.



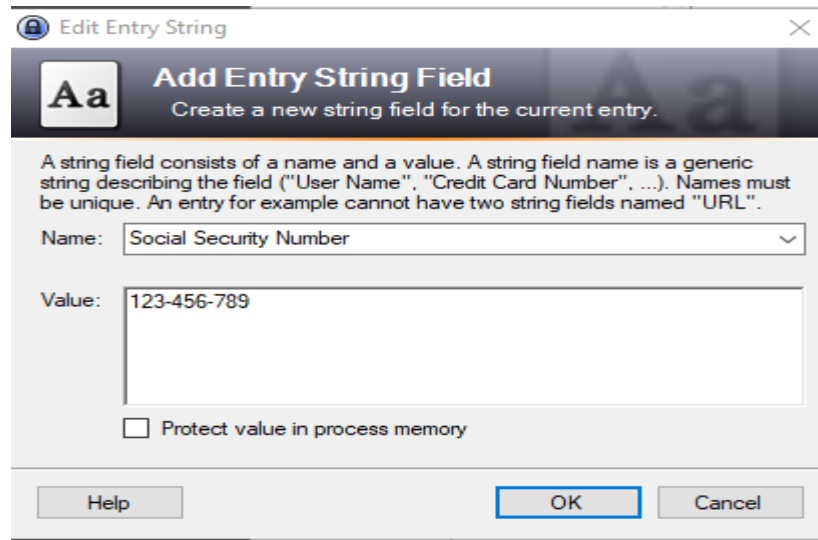
2. Right-Click on it and Click on Edit-Entry.



3. Now go to Advanced Tab and then to String Fields.



4. Now add any new String of Data that you want to add.



Edit Entry String

Add Entry String Field
Create a new string field for the current entry.

A string field consists of a name and a value. A string field name is a generic string describing the field ("User Name", "Credit Card Number", ...). Names must be unique. An entry for example cannot have two string fields named "URL".

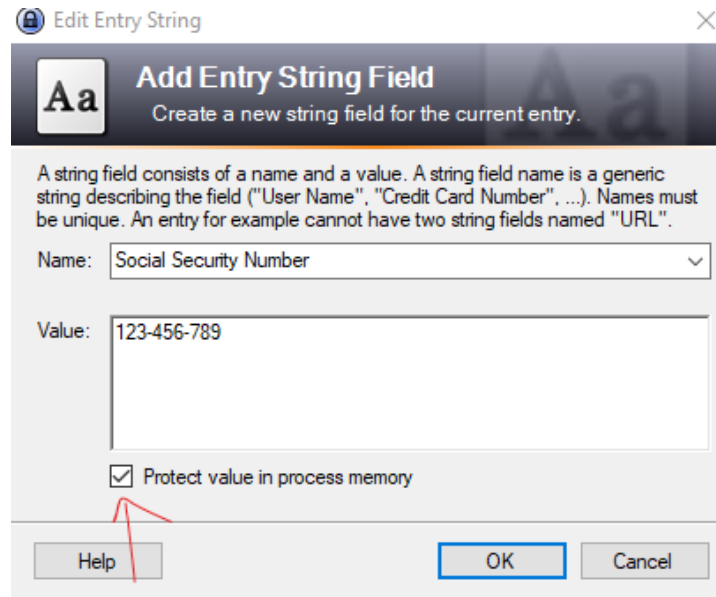
Name: Social Security Number

Value: 123-456-789

☐ Protect value in process memory

Help OK Cancel

5. Click on Protect value in process memory to avoid String field exposure. (This option would hide your Data Value from outside until you specifically access it).



Edit Entry String

Add Entry String Field
Create a new string field for the current entry.

A string field consists of a name and a value. A string field name is a generic string describing the field ("User Name", "Credit Card Number", ...). Names must be unique. An entry for example cannot have two string fields named "URL".

Name: Social Security Number

Value: 123-456-789

☒ Protect value in process memory

Help OK Cancel

6. This new field would be added.

Edit Entry
You're editing an existing entry.

Entry | Advanced | Properties | Auto-Type | History

String fields

Field Name	Field Value
Social Security Number	*****

Add
Edit
Delete
Move

File attachments

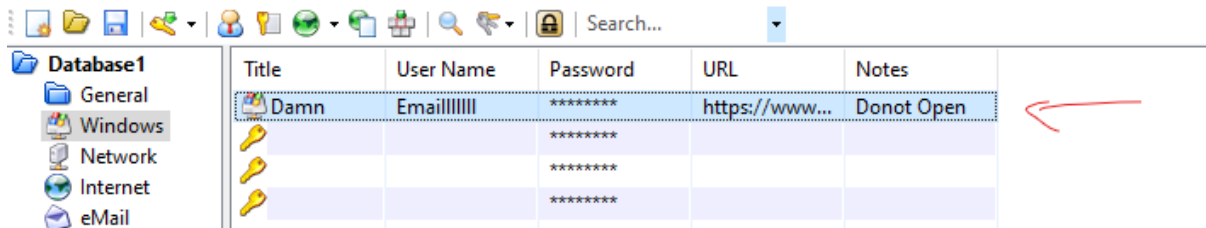
Attachments	Size
bookmarks_12_22_24.html	10 KB

Attach
Delete
Open
Save

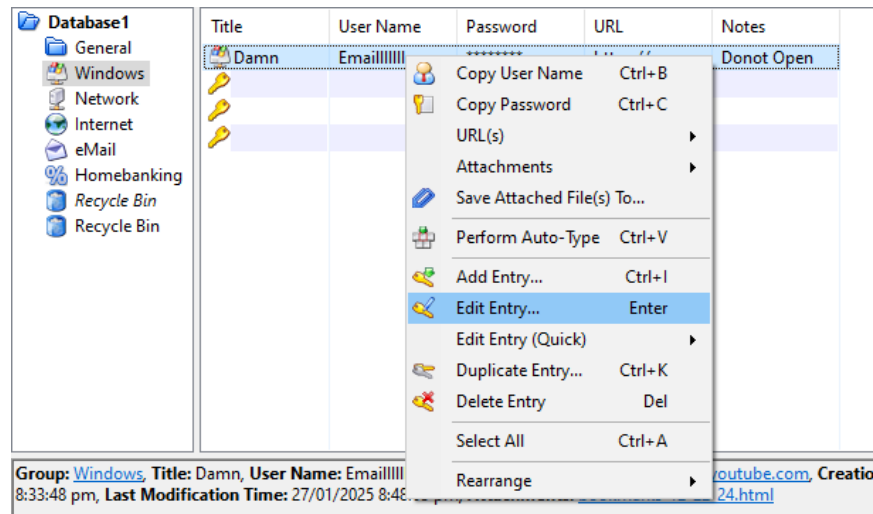
Tools OK Cancel

Activity 6: Adding URLs

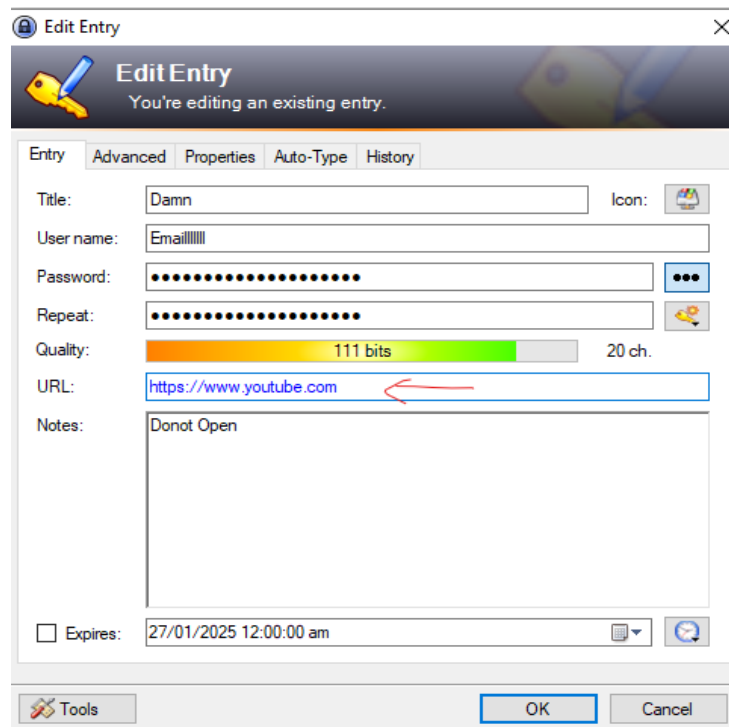
1. Click on the Email & Password where you want to store URLs.



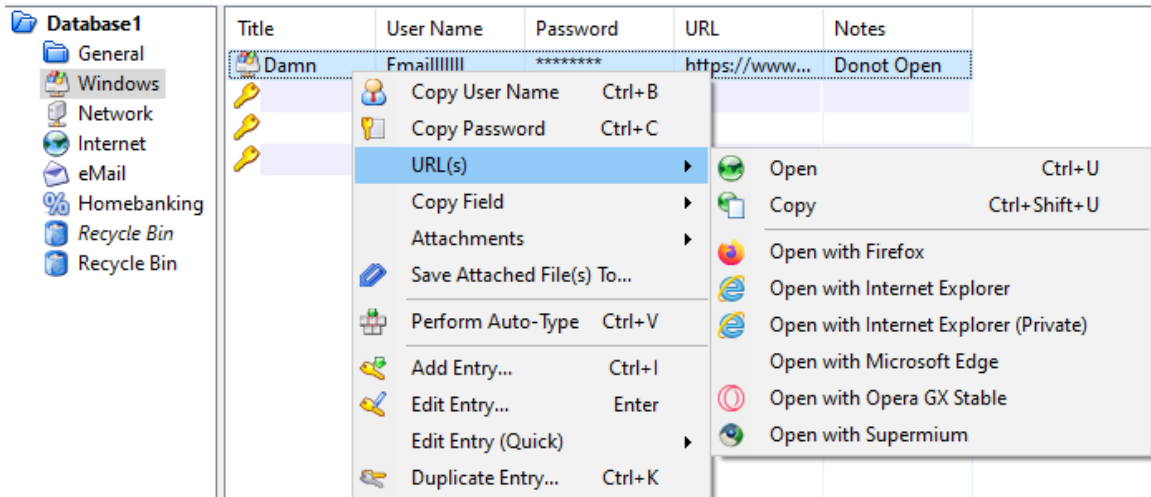
2. Right-Click on it and Click on Edit-Entry.



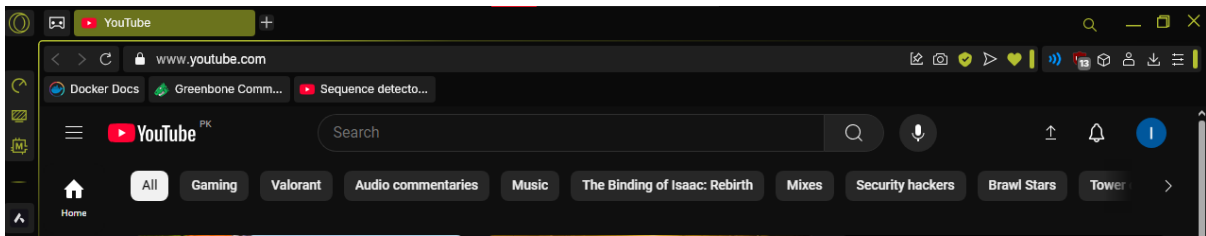
3. Now add the URL to the URL dialog box.



4. Now Click on the same email and then in URL click on Open.

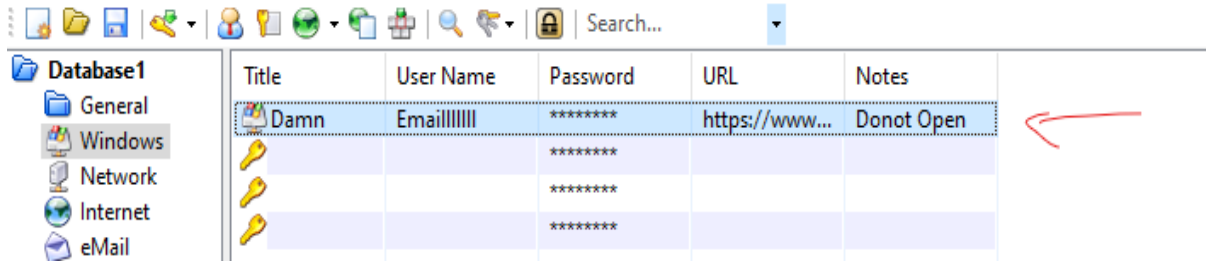


5. You will be redirected on the webpage on your default browser.

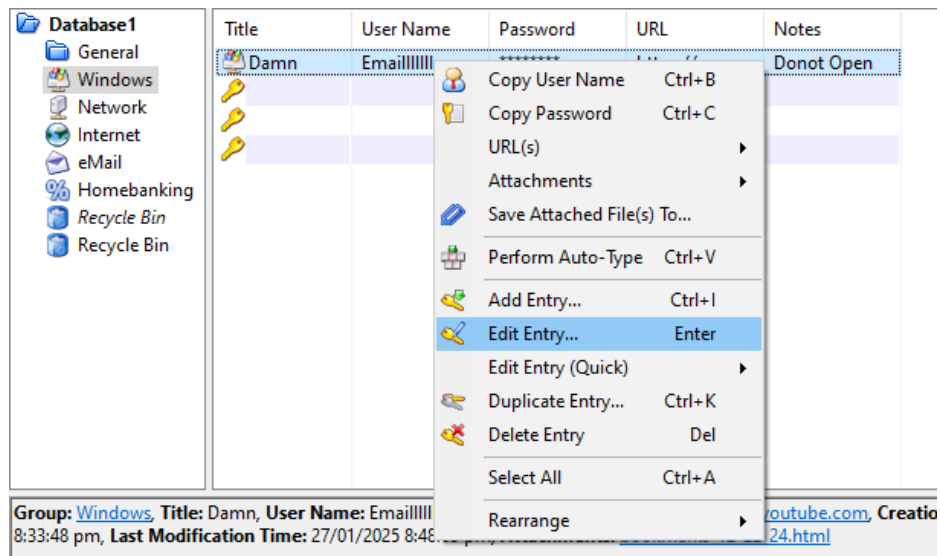


Activity 7: Attachments

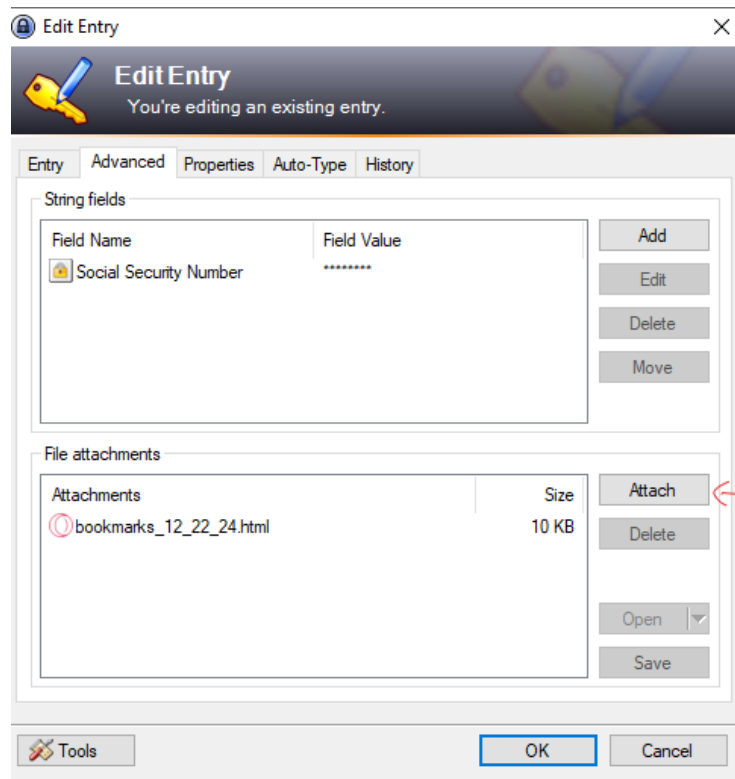
1. Click on the Email & Password where you want to store Attachments.



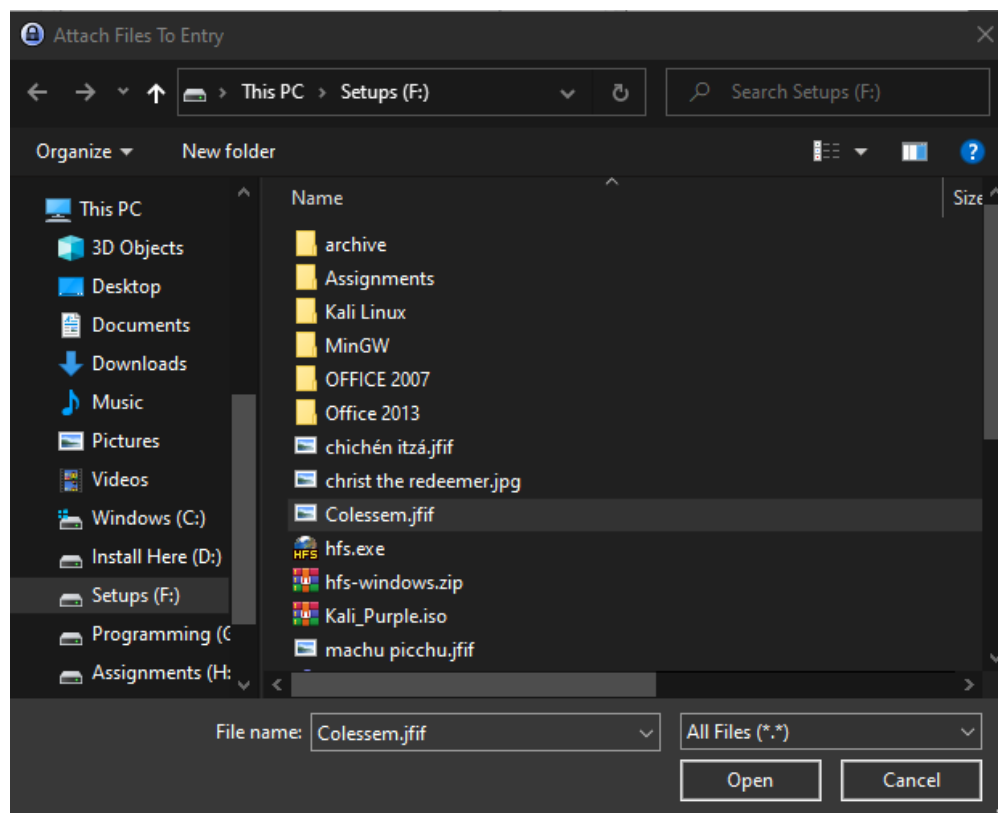
2. Right-Click on it and Click on Edit-Entry.



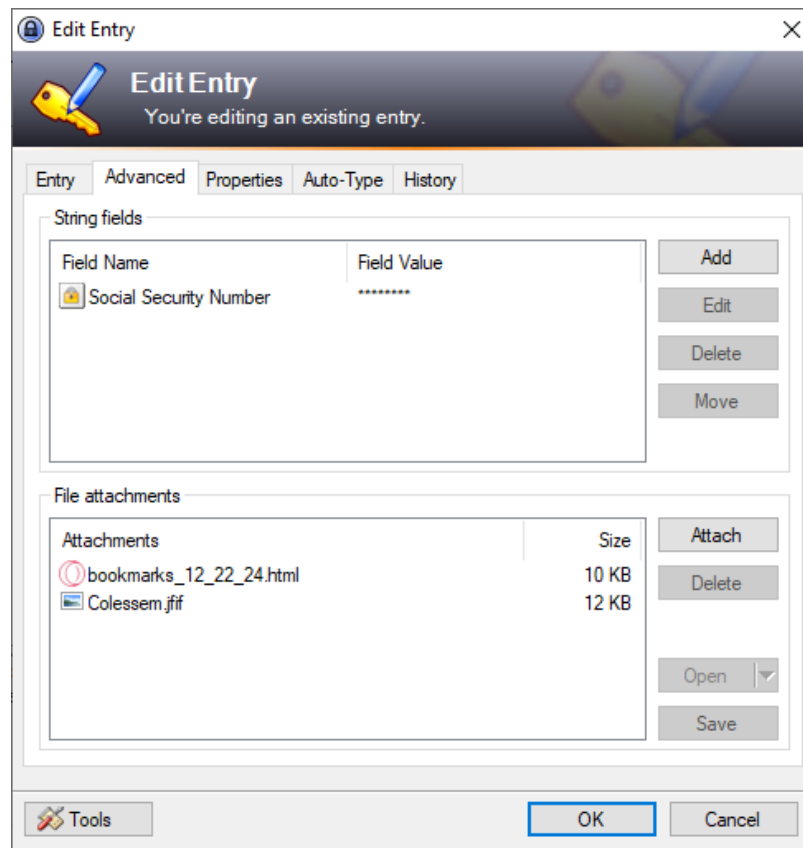
3. Now go to Advanced Tab and under Attachments, Click Attach.



4. Now attach a file of your own choice.



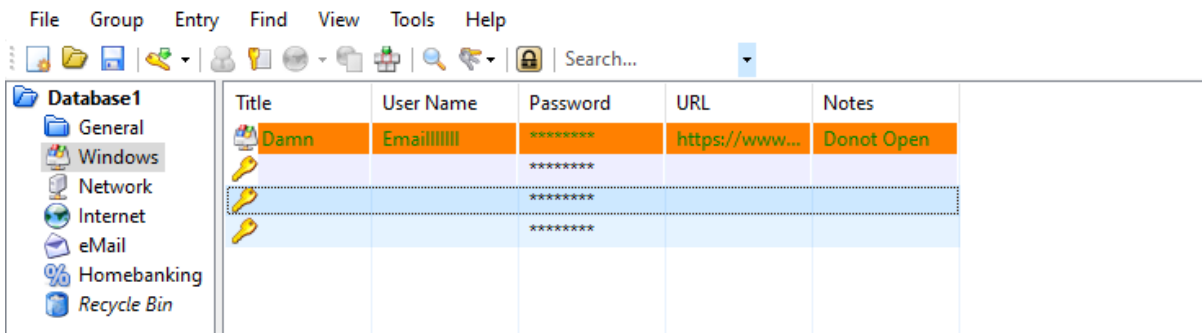
5. It will be added to it.



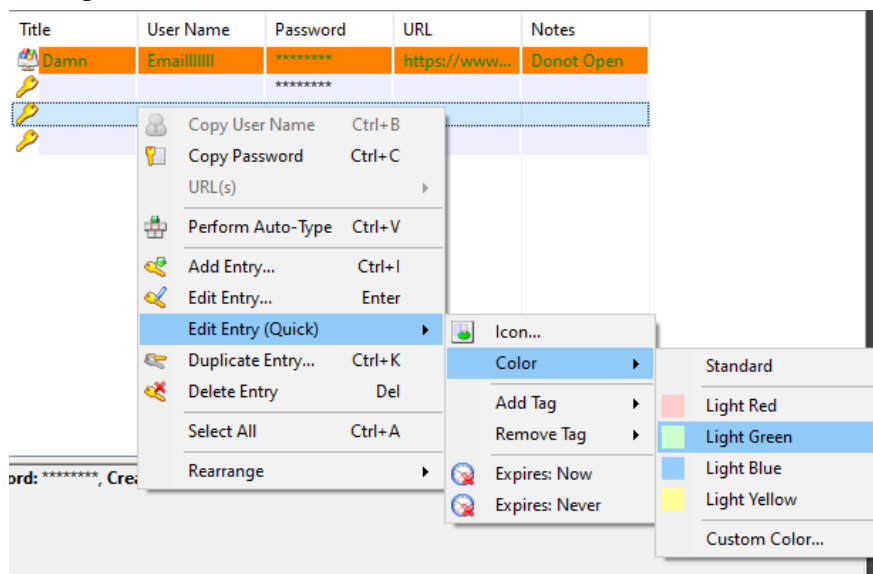
You can also view or download the attachment from the corresponding key also.

Activity 8: Coloring

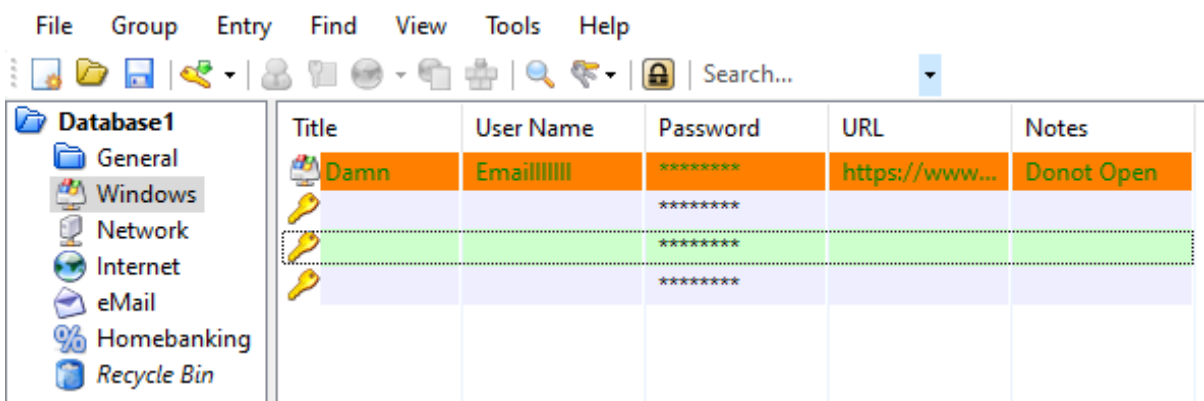
1. Select the Key that you want to Colour.



2. Right-Click on it and Go to Entry Edit (Quick) and in Colour and then choose the colour of your liking.

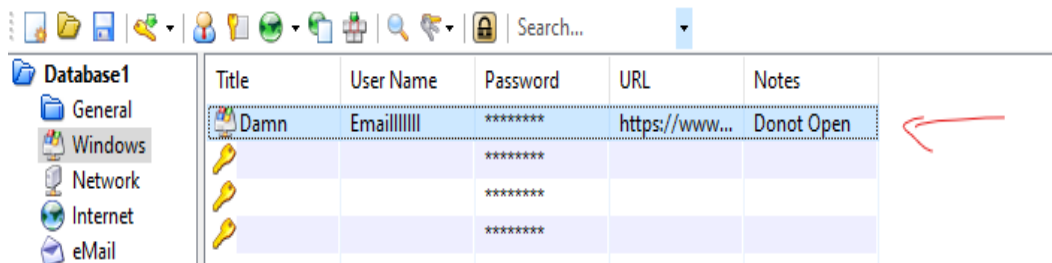


3. The Key will be coloured.

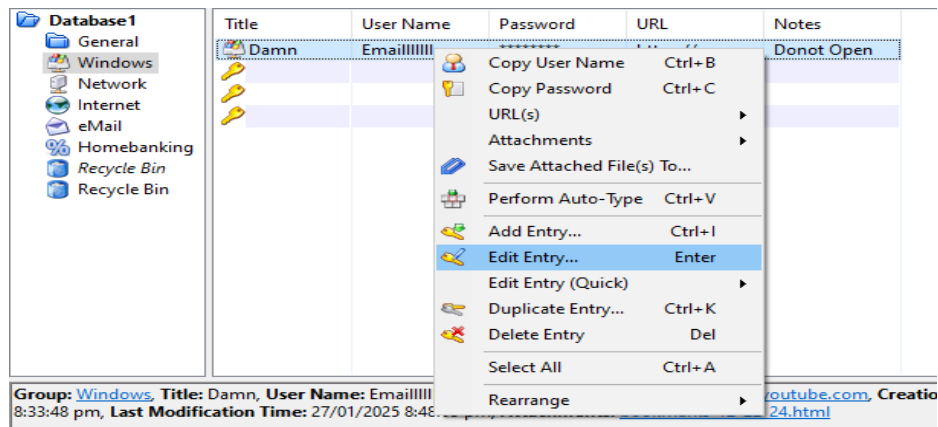


Activity 9: Check History

1. Click the Key that for which you want to check the history.



2. Now Right-Click on it and Click on Edit Entry.



3. Now go to History Tab and now you can see the occurrences this key was used.

